

IGNITE MACHINERY RENTALS & MINING LIMITED

FEASIBILITY STUDY REPORT FOR THE APPLICATION OF MINING LEASE(ML) AMENAM GOLD PROJECT

BIRIM NORTH DISTRICT, EASTERN REGION OF GHANA

Compiled and Submitted By:

Ignite Machinery & Mining Ltd

Accra-Ghana

OCTOBER, 2025

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- ☒ Application for a Mining Lease
- ☐ Application for a Restricted Mining Lease
- ☐ Existing Mining Lease
- ☐ Existing Restricted Mining Lease

LEASE APPLICANT / HOLDER DETAILS

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EXECUTIVE SUMMARY

Ignite Machinery Rentals & Mining Limited, is a private registered company in Ghana, seeking to apply for a mining lease for the Amenam concession in the Birim North District of the Eastern Region of Ghana. The concession size is approximately 9.77 sq km (47 Blocks) and can be located on Field Sheet Nos. 0601C1 / 0602D2. The company has identified both hard rock and alluvial gold zone and intends to develop the potential of gold within the Amenam property and the surrounding areas through systematic exploration methods.

The Amenam project area is situated along the western boundary of the Birim North District of the Eastern Region of Ghana. The district is bordered to the north by Kwahu West Municipal, to the west by Asante Akyem South and Adansi South Districts to the south by Akyemansa District to the east by Atiwa and Kwaebibirem Districts. The district has New Abirem as its capital.

The concession can be easily accessed along a second-class road connecting New Abirem – Nkawkaw. The concession can be accessed approximately 7.7 NW of New Abirem and approximately 43Km SW of Nkawkaw, along the main Kumasi-Accra Road. From Accra, the Amenam concession can be easily accessed using the main Kumasi-Accra Road via Nkawkaw covering approximately 181Km NW. The Birim north district is well connected to the neighboring districts and within.

The topography of the district is undulating and mountainous in nature. The district is drained mainly by the Pra River: which also serves as the boundary between the district and the Asante Akim South and Adansi South districts in the Ashanti Region. The district experiences a double maxima rainfall pattern: First rainfall season starts from late March to early July. Second season is from mid-August to late October. The district lies within the Semi-deciduous forest belt of Ghana comprising tall trees with evergreen undergrowth. It contains large species of economic trees valuable for the timber industry.

Geologically the concession is located along the north eastern part of the Ashanti Gold Belt of the Birimian Supergroup in south western Ghana. The Ashanti belt is covered by lithologies of the volcanic-sedimentary Birimian Supergroup and the overlying clastic sedimentary Tarkwaian Group which were jointly folded and metamorphosed under greenschist facies conditions during the Eburnean tectonothermal event at about 2.1 Ga. Regional foliation and subparallel shear zones hosting mesothermal gold mineralization developed during deformation coeval with metamorphism.

Hard rock mineral resource projections with reference to geological, structural, geophysical data and the Ashanti Belt historic deposit reserve indicates that, an area of 766.7K metre squared area is expected to

produce a projected tonnage of 1.76Mt, with an average grade of 1.10g/t and 62.3koz gold resource from the Amenam concession and this is a forecasted information based on technical regional data.

The evaluation of the exploration data obtained show that alluvial gold gravel beds occur in the valleys and old terraces within the concession. The buried gravel beds have an average thickness of 1.34m and have overburden with average thickness of 3.9m. The estimated reserves of gold deposits in the gravel beds in the concession are approximately 84,967.80oz.

Through intensive exploration work in the concession, the company exploration team has delineated various gold anomalous zones to operate its commercial mining activities, and aimed at extracting the approximately 84,967,80oz. of gold within the **2,872,558.0m³** volume of ore defined in its **8.30sqkm** Grid at **0.92g/m³** within Amenam grid in the **Amenam concession** for a period of **7 years** after commencement of production. The company intends to operate on **100m³/h** fixed washing plant capacity.

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TERMS OF REFERENCE

It's worth noting that Ignite Machinery Rentals and Mining Limited field exploration activities were neither managed nor conducted by Explorgeo Ltd. However, as part of their collaboration, Explorgeo received data from the company on various exploration programmes under a Non-Disclosure Agreement (NDA). This data is essential for Explorgeo to perform resource and reserve assessments, a crucial step in the preparation of the feasibility report for the Amenam Gold Project.

To ensure the accuracy and credibility of the data, Explorgeo conducted a verification process, which included field visitations and a general assessment of the delineated mineralized trends. Where possible, Explorgeo Ltd. has verified this information from independent sources or implemented methods to validate results. Explorgeo, its staff, and its directors accept no liability for any losses arising from reliance upon the information presented in this Report. These measures are essential to provide a comprehensive and reliable basis for the preparation of the feasibility report.

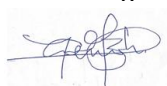
DISCLAIMER

In the disclosure of information relating to legal, technical data, title and related issues, Explorgeo has relied on information provided by Ignite Machinery Rentals and Mining Ltd. and disclaims responsibility for such information. The disclosure represents no legal opinion of Explorgeo Ltd. nor its representatives. In the disclosure of information relating to socio-political, environment and related issues, Explorgeo has relied on information obtained from public sources such as the Minerals Commission or other related departments of interest.

FORWARD-LOOKING STATEMENTS

This feasibility report was prepared by Ignite Machinery Rentals and Mining Ltd. The report includes certain “forward-looking statements”. forward- looking statements includes words to the effect that the Company or management expects a stated condition or result to occur. Forward-looking statements may be identified by such terms as “Projected,” believes”, “anticipates”, “expects”, “estimates”, “may”, “could”, “would”, “will”, or “plan”. Since forward-looking statements are based on assumptions and address future events and conditions, by their very nature they involve inherent risks and uncertainties. This list is not exhaustive of the factors that may affect any of the Company's forward-looking statements. These and other factors should be considered carefully, and readers should not place undue reliance on the Company’s forward-looking statements.

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CHAPTER 1

1.0 INTRODUCTION

Ignite Machinery Rentals & Mining Limited is applying for mining lease for the Amenam concession located in the Birim North District of the Eastern Region. The concession size is approximately 9.77 km² (47 Blocks) and can be located on Field Sheet Numbers 0601C1/0602D2. The company has delineated various alluvial and hard rock gold zones, and intends to develop the potential of gold in this property and the surrounding areas through systematic exploration methods.

The concession is located within the Ashanti belt of the Birimian. The Ashanti gold belt lie between the Kumasi basin (NW) and the Cape Coast basin (SE). The Ashanti Belt has the longest history of gold mining and is consequently the Ignite Machinery understood of Ghana's gold belts. The area was the focus for gold exploitation in pre-colonial times when small scale artisanal mining extracted gold for traditional authorities whose culture attached great significance and value to the metal. From the late 19th century, conventional mining and extractive methods were employed by British companies, leading to the development of deeper underground mines at Obuasi, Tarkwa, Prestea, Marlu, Abosso, Konongo and many other smaller shafts and adits developed on mineralization on the Ashanti Belt. New world class deposits were discovered and developed at Obuasi, Tarkwa, Iduapriem, Teberebie, Bogosu, Damang, and Akyem on the Ashanti Belt so that the Ashanti Belt now has an aggregated gold endowment (past production, resources and reserves) in excess of 125,000,000 ounces of gold. This is undoubtedly one of the world's premier gold belts.

The company has embarked on some pitting and sampling program covering the Amenam concession area to evaluate and determine the grade and size potential of the alluvial resource for the whole concession. Alluvial gold exploration is tricky due to extreme variability of gold grades along the terrace beds and nuggettisation of gold. Free gold bearing gravel beds and soils are artisanally mined by locals, hence successful mechanical mining is always a challenge. Factors such as political and industrial disruption, currency fluctuation and interest rates could have an impact on future operations, and potential revenue streams can also be affected by these factors. The majority of these factors are, and will be, beyond the control of any operating entity. Ignite Machinery Rentals and Mining Limited has adopted a detailed systematic gold investment model for profits and security.

1.1 LOCATION

The Amenam project area is situated within the Birim North District of the Eastern Region of Ghana and can be accessed close to the western boundary of the Birim North district. The District is bordered to the north by Kwahu West Municipal, to the west by Asante Akyem South and Adansi South Districts, to the south by Akyemansa District, to the east by Atiwa and Kwaebibirem Districts. The District has New Abirem as its capital.

Geologically, it is located within the north eastern parts of the Ashanti Belt of the Birimian Supergroup. The concession is approximately 9.77 km² (47 Blocks) in size and can be found on topographic field sheet numbers 0601C1/0602D2. The concession is defined by coordinates below.

Table 1.1: Amenam Concession Pillar Coordinates.

COORDINATES OF THE PILLARS						
PILLAR	LATITUDE			LONGITUDE		
	Deg	Min	Sec	Deg	Min	Sec
P1	6	25	0	1	2	45
P2	6	25	0	1	0	45
P3	6	25	15	1	0	45
P4	6	25	15	1	0	0
P5	6	25	0	1	0	0
P6	6	25	0	0	56	30
P7	6	24	45	0	56	30
P8	6	24	45	0	57	15
P9	6	24	30	0	57	15
P10	6	24	30	0	58	0
P11	6	24	15	0	58	0
P12	6	24	15	0	58	30
P13	6	24	0	0	58	30
P14	6	24	0	0	59	0
P15	6	24	45	0	59	0
P16	6	24	45	1	1	30
P17	6	24	30	1	1	30
P18	6	24	30	1	2	45



Figure 1.1: Map of Ghana showing location of Amenam concession.

**PLAN OF LAND
LARGE SCALE CONCESSION
FOR:**

Area= 9.77 SQ.KM OR 47 BLOCKS
- Shewn Edged Pink-

FS: 0601C1/0602D2

SCALE = 1:80,000

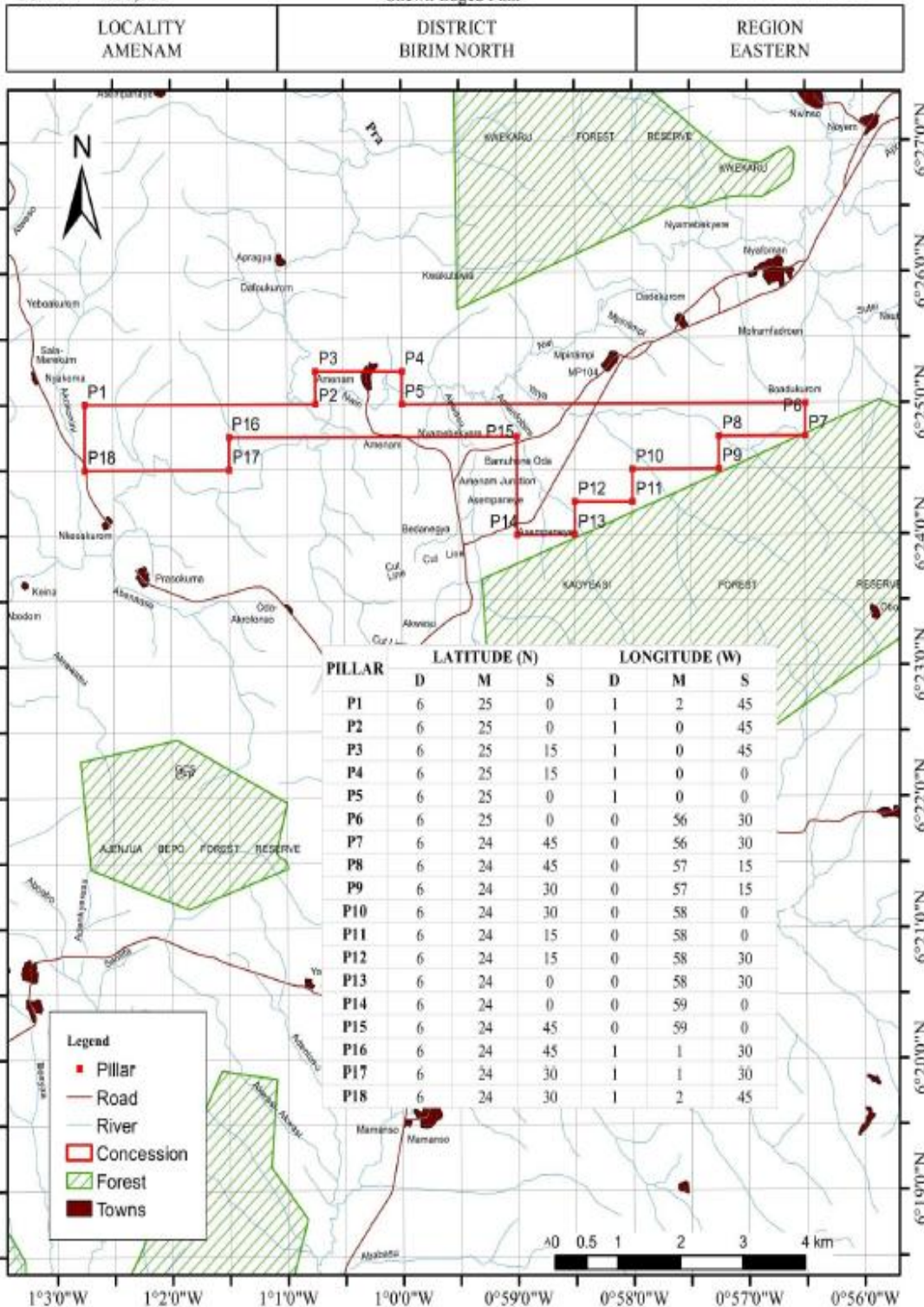


Figure 1.2: Amenam Concession Map.

1.2 GENERAL DESCRIPTION

1.2.1 Accessibility

The concession can be easily accessed along a second class road connecting New Abirem – Nkawkaw. The concession can be accessed approximately 7.7 NW of New Abirem and approximately 43Km SW of Nkawkaw, along the main Kumasi-Accra road. From Accra, the Amenam concession can be easily accessed using the main Kumasi- Accra road via Nkawkaw covering approximately 181Km NW. Accessing the concession from Kumasi is approximately 93km NW using the main Kumasi-Accra via Juaso-Obogu-Kumeso-Amenam. The Birim north district is well connected to the neighboring districts. Access within the concession is generally fair with the presence of secondary roads along the eastern, central and western parts.

1.2.2 Infrastructure

The district is investing in markets, artisan hostels, schools, and health facilities as part of its 2026–2029 development plan. These efforts aim to improve socio-economic conditions of the people and align with national development plans. The district is actively enhancing its infrastructure across multiple sectors.

The Newmont Akyem Mine is located in the district, and has contributed the socio-economic development of the district.

Some recent developments include the following:

- CHPS Compound & Nurses’ Quarters: Built in Old Abirem using mineral royalties from Newmont Akyem mines, these facilities enhance healthcare delivery in the district.
- New markets being constructed in Praso Kuma, Nyafoman, and Amoa to boost local commerce.
- A six-unit classroom block completed in Asawase, improving access to basic education and reducing overcrowding.
- Community Hostels for Artisans: Planned for New Abirem to support vocational training and economic empowerment.

1.3 GEOGRAPHIC SETTING

1.3.1 Relief

The topography is undulating and mountainous in nature.

Heights to the east of Akoasi up to the north eastern part of the district is about 244 metres above sea level. From Amuana Praso in the north to Tweapiase in the east, New Abirem and Hwiakwae to the south and Ntronang and Praso Kuma to the west is between 243-183 metres above sea level. The southern part of the district comprising the area stretching from the south of New Abirem, through Nkwateng and the surrounding areas can be described as very low lying areas; maximum height of 61 metres.

1.3.2 Drainage

The district is generally drained by the Pra River and its tributaries. The Pra River serves as the boundary between the district and the Asante Akyem south and Adansi South districts in Ashanti Region.

The Nwi, Mamang, Adechensu, Sukrang and Afotosu tributaries and several streams flow generally from the north east to the south west to join the Pra which flows southwards and enters the sea at Shama in the Western Region.

Thw Nwi River drains the concession.

1.3.3 Climate: Rainfall, Temperature and Humidity

Birim North has a hot, humid tropical climate with two rainy seasons and a dry, dusty harmattan period. Temperatures stay warm year-round, peaking in March and April, with average highs ranging from 84°F to 93°F (29°C to 34°C). Coolest months: July and August, though still warm, with highs around 84.2°F (29°C).

The district has two main rainy seasons: Major rainy season is from April to July and minor rainy season is from September to November. Dry season occur from December to March, often accompanied by harmattan winds, which bring dry and dusty condition.

Humidity levels are generally high, making the heat feel more intense. Wind chill is minimal, but the harmattan can bring cooler, dry air during the dry season.

1.3.4 Vegetation

The district lies within the Semi-deciduous forest belt of Ghana comprising tall trees with evergreen undergrowth. It contains large species of economic trees valuable for the timber industry. The Kagyeeasi Forest Reserve bounds eastern parts the concession.

CHAPTER 2

2.0 TECHNICAL INFORMATION

2.1 GEOLOGICAL SETTING

Ghana lies within the eastern margin of the West African Craton (WAC). The West African Craton stabilized in the early Proterozoic (2000 Ma) during the Eburnean Orogeny. The West African Craton (WAC) extends across western Africa, and consists of two Archean nuclei in the north-western and south-western parts. The two nuclei consist of Reguibat and Leo-Man Shields in the north and south of the craton respectively. Ghana falls within the Leo-Man Shield Archean rocks consisting of oldest banded gneiss overlain by greenstone belts with (ultra) mafic metavolcanics, banded ironstone, phyllite, greywacke and quartzite (e.g. Sula & Nimini belts in Sierra Leone) and intruded by 2.97-2.78 Ga granite (Attoh & Ekweme, 1997).

The West African Craton is the second largest region in Africa where lower Proterozoic rocks are extensively preserved. These early Proterozoic rocks comprise extensive belts of metamorphosed volcanic and sedimentary rocks exposed in Ghana, Burkina Faso, Niger and Cote d'Ivoire. In Ghana these rocks are referred to as the Birimian Supergroup System. The West African Shield represents approximately 45% of the exposed geology of Ghana, being largely restricted to the northern, western and south-western portions of the country. The shield area is confined to the southeast by a Proterozoic mobile zone and the central-eastern portion is largely veneered by late Proterozoic to early Palaeozoic sediments of the Voltaian Basin.

In general, Ghana which is located southeast of the West African Craton can be divided into several distinct terranes as:

- i) An early Proterozoic terrane (Birimian System) which hosts Igneous Mineralization of the country's mineral deposits and occupies the western and northern part of the country.
- ii) The Tarkwaian System, a distinctive sequence of clastic sediments within the Ashanti, Bui, and Bole-Navrongo Belts; iii) The Voltaian Basin, in which are preserved the late Precambrian to Paleozoic sediments that mantle the craton.
- iv) The Dahomeyan System, occupying the eastern part of Ghana.
- v) A Pan-African mobile belt, the Togo and Buem Formations, separated from the Birimian terrane by a prominent topographic feature known as the Akwapim - Togo range.
- vi) Phanerozoic sedimentary rock and
- vii) Intrusive rocks.

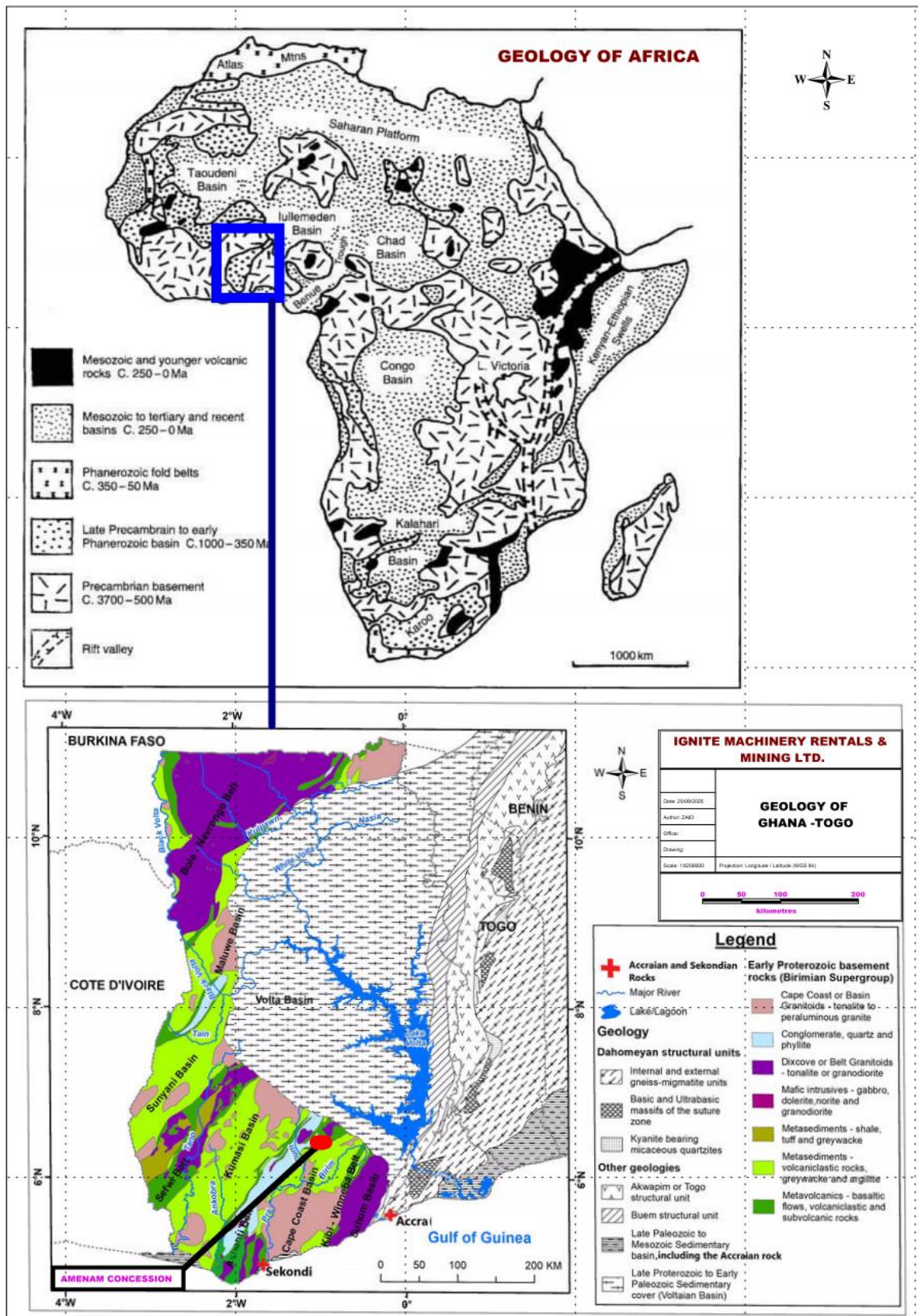


Figure 2.1: Geology of West Africa and Ghana.

2.2 REGIONAL GEOLOGY

The Concession falls within the Birimian System at the eastern margin of the West African Craton in the Leo-Man Shield formed at the southern part of the West African Craton. The Birimian group consist of five parallel approximately evenly spaced gold belts consisting of low-grade metamorphic units and tholeiitic lavas. The volcanic belts are interbedded with volcanoclastic sediments within the volcanic flows. The gold belts are separated by basins with isoclinically folded wackes, argillitic sediments and granitoids. The supra-crustal rocks are highly deformed. However, the sedimentary rocks are particularly characterized by extensive folding. The lavas are mainly of basaltic composition, though andesitic, dacitic and rhyolitic rocks are also present. Some pattern of facies distribution is shown by the Birimian sedimentary basins from the margins towards the basin centres. The transition zone between the volcanic belts and the sedimentary basins is marked by chemical facies, which has of late been found to be the site of much of gold mineralization in Ghana.

The Eburnean orogeny that stabilised the WAC and affected the birimian rocks occurred in two discrete cycles of orogenesis. An earlier cycle, the Eburnean I orogeny, was associated with eruption of the Birimian mafic volcanic rocks, intrusion of granitoids and associated metamorphism, between ca 2200 and 2150 Ma. The later, Eburnean II, orogeny involved deformation, metamorphism and intrusion of all Birimian and Tarkwaian rocks between ca 2116 Ma and 2088 Ma.

The Birimian System has been intruded by two distinctive granitoid types. The larger basin-type granitoids (and gneisses) are muscovite and/or biotite-rich and are distinctly foliated and deformed, providing a pre-tectonic appearance. The smaller belt-type (arc related) granitoids are hornblende-rich, lack the characteristic foliation of the former and are generally interpreted to be syn or post-tectonic. Despite their appearance, the belt-type granitoids are dated as being 60 to 90 million years older than the larger basin-type granitoids.

The concession is located within the Ashanti belt of the Birimian. The Ashanti gold belt lie between the Kumasi basin (NW) and the Cape Coast basin (SE). The Ashanti Belt has the longest history of gold mining and is consequently the Ignite Machineryt understood of Ghana's gold belts.

A distinctive sequence of clastic sedimentary rocks occurs in elongate troughs developed on top of the Ashanti, Oberthur (1994) and Bui, Zitzman et al. (1993 a, b) belts. These rocks host important paleoplacer gold deposits and are known as the Tarkwa System. The rocks of the Tarkwaian System represent erosional products of the Birimian and are dominated by coarse clastic sediments.

In the Ashanti Belt, the Tarkwa is made up of four units. The lowest unit, the Kawere Group, consists of immature, polymictic, matrix supported, large pebble conglomerate dominated by mafic (Birimian) pebble lithologies. The Kawere is overlain by, and is in marked contrast with, the Banket Series which consists of mature, clean, quartzite, grit, breccia and conglomerate composed in part of well sorted quartz pebble conglomerate beds known as “reefs” that host the gold mineralisation. The Banket is overlain by the Tarkwa Phyllite which consists of a transition sequence from sandstone to chloritic and sericitic phyllite. The upper Ignite Machinery Tarkwa unit is the Huni Sandstone made up of sandstone and quartzite with interbeds of phyllite.

The structure of the Tarkwaian rocks differs from belt to belt. In the Ashanti Belt, the Tarkwa System is folded into a series of northeast trending, northeast plunging antiforms and synforms. Along the northwest margin of the belt, the Tarkwaian is overturned and locally overthrust by Birimian rocks (Eisenlohr and Hirdes, 1992). The sediments have a moderate primary foliation (S1) and a strong secondary foliation (S2) near the margins of the belt.

The spatial distribution of gold mineralisation in Ghana is closely governed by north to northeast trending belts of metavolcanic rocks, which range from 15km - 40km in width. All Ignite Machinery without extension exception, the major gold deposits lie at or near close to the margin of the belt in close proximity to the strongly deformed belt-basin birimian sequences.

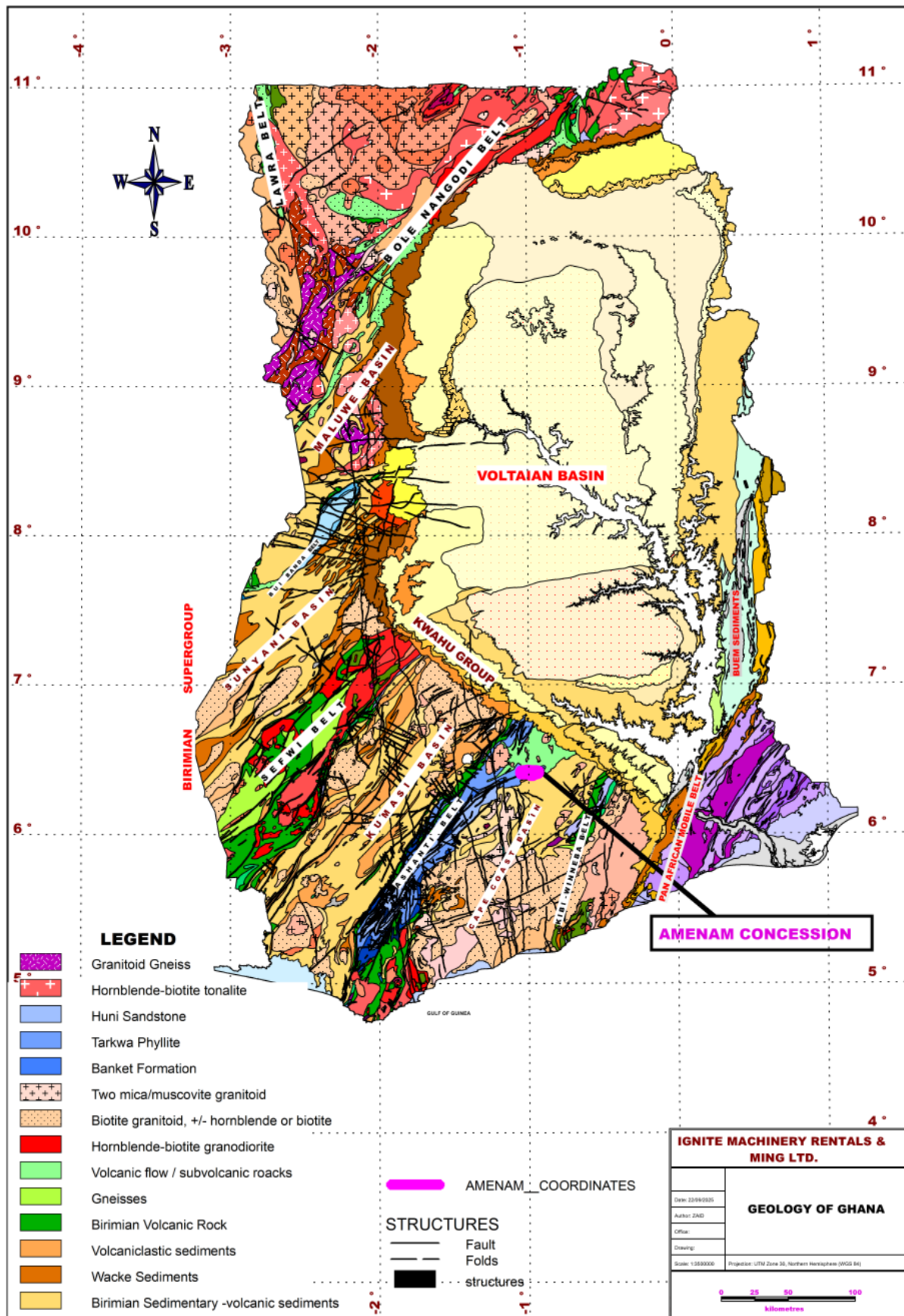


Figure 2.2: Regional Geology Map.

2.3 LOCAL GEOLOGY

The Amenam concession is located within the north eastern margin of the Ashanti Gold Belt of the Birimian Supergroup in south western Ghana.

The Ashanti belt is covered by lithologies of the volcanic-sedimentary Birimian Supergroup and the overlying clastic sedimentary Tarkwaian Group which were jointly folded and metamorphosed under greenschist facies conditions during the Eburnean tectonothermal event at about 2.1 Ga.

The concession area is largely underlain by tarkwaian sediments comprising mature, clean conglomerate, grit, breccia and quartzite belonging to the “Banket group” of the Tarkwaian System. It is the third formation in the Tarkwaian Supergroup, following the Kawere and Huni formations.

The conglomerates contain rounded pebbles of quartz and other resistant minerals. Quartzites are well-cemented and silica-rich. There are some minor dolerite intrusives that may appear as dikes or sills within the sequence.

The rocks show varying degrees of metamorphism from greenschist to amphibolite facies.

Conglomerates: greenschist facies,

Dolerites: Epidote-amphibolite facies

Quartzites: Amphibolite facies

The Banket Formation likely formed in a cratonic interior setting with recycled orogenic materials, indicating a mature sedimentary environment. It represents a humid paleoclimate with multiple episodes of sediment recycling.

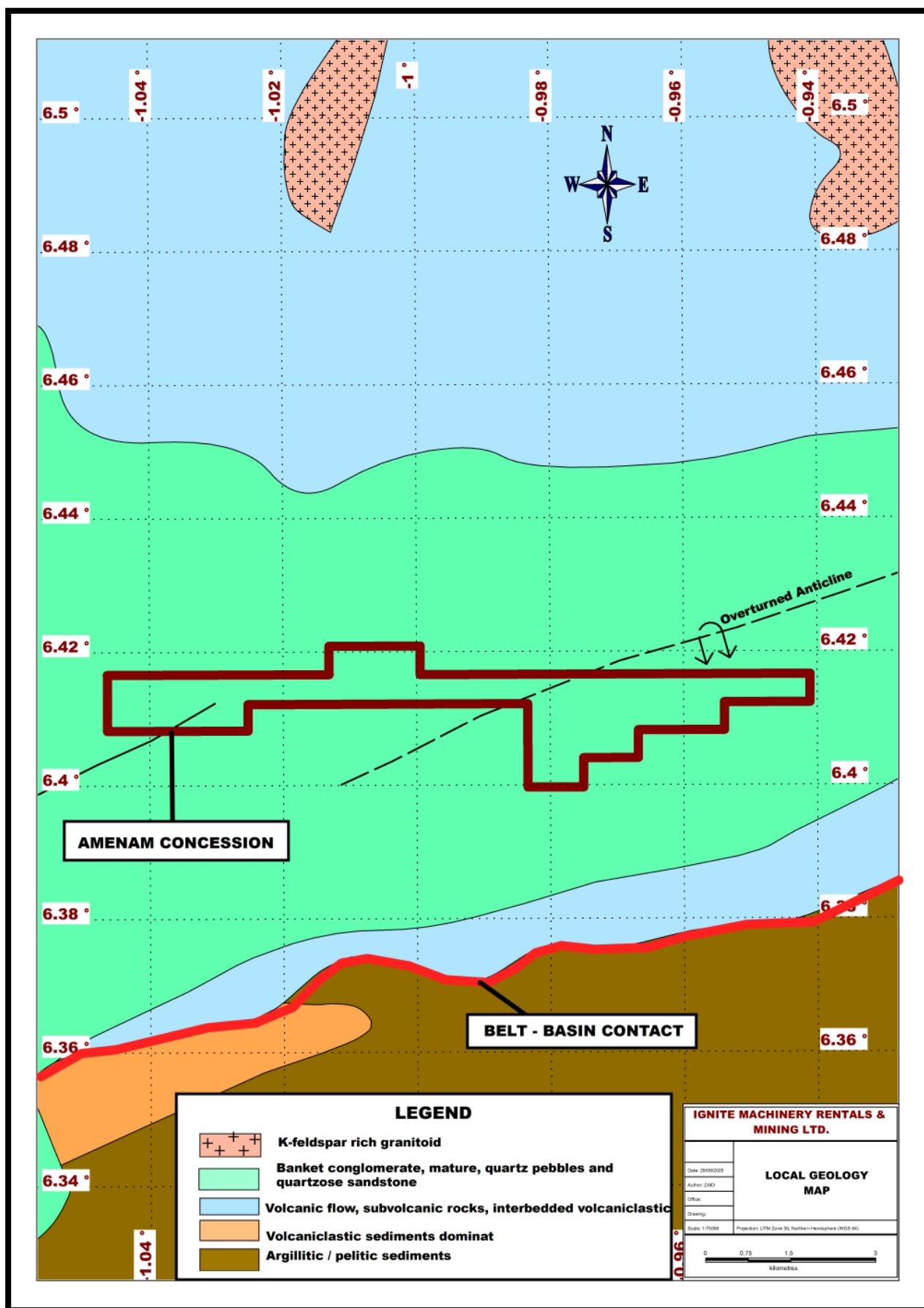


Figure 2.3: Local Geology Map.

2.4 MINERALIZATION

The Birimian rocks in Ghana are the primary source of gold mineralization trends. The primary gold occurs in quartz veins and lenticular reefs and also in some of the tuffaceous and argillaceous rocks. It is accompanied by sulphides, especially arsenopyrite but including pyrite and pyrrhotite, chalcopyrite and bornite, and a little galena and sphalerite. Along contacts zones between the host rocks and the intrusives are potential zones of gold mineralization and rare metals from the mesothermal fluids of the intrusives.

The Banket Formation is historically significant for gold mining in Ghana. Its paleoplacer nature makes it a prime target for gold exploration and extraction. The conglomerate composed in part of a well sorted quartz pebble conglomerate beds known as “reefs” host the gold mineralisation. Junner et al., (1942) named four “reefs” or conglomerate bands in the following succession: Breccia Reef - Middle Reef - Basal or Main Reef and Sub-basal Reef. The Basal or Main Reef is the Ignite Machineryt persistent conglomerate bed in the Tarkwa goldfield area and is by far the richest in gold.

The Tarkwa Goldfields deposit is also located approximately 170Km SW of Amenam concession n within the Ashanti belt.

2.5 STRUCTURE AND DEFORMATION OF THE BIRIMIAN

Ashanti Belt in Ghana is a geologically complex and economically significant region, particularly known for its rich gold deposits. Here's a breakdown of its structure and deformation history:

The belt comprises volcanic and sedimentary rocks of the Birimian Supergroup (ca. 2.2–2.1 Ga) and the overlying Tarkwaian Group (<2132 Ma).

The belt is characterized by NE-SW trending folds, shear zones, and fault systems, especially the prominent Ashanti Fault.

The Ashanti Belt has undergone multiple deformation phases, primarily during the Paleoproterozoic Eburnean orogeny (ca. 2.1 Ga). Below is a summary of the deformation phases and their characteristics.

- D1 NW-SE shortening before Tarkwaian deposition; early folding and thrusting.
- D2 Continuation of NW-SE shortening; large NE-SW folds in both Birimian and Tarkwaian rocks.

D3 WNW-ESE shortening; tighter folds with wavelengths of 100–1000 m.

D4 Sub horizontal cleavage and recumbent folds.

D5 Reactivation of earlier structures with renewed NE-SW shortening.

These deformation phases are closely linked to gold mineralization, particularly D1 through D3, which facilitated fluid flow and vein formation.

Shear zones and fold hinges are prime targets for gold exploration due to their role in channeling mineralizing fluids.

CHAPTER 3

3.1 REGOLITH AND HARD ROCK GOLD DEPOSITS

The company plans to advance the deposit sources within the concession into the regolith and hard rock. This will involve a systematic approach including geological, geochemical and geophysical methods from field mapping to drilling stage.

Three broad exploration stages include:

- Area Selection
- Target Definition
- Ore deposit Delineation

Area selection typically involves using broad geological concepts, regional geological mapping and syntheses, airborne geophysical surveys, and regional geochemistry (such as systematic sampling of sediments in drainage channels, or sampling weathering crusts). Favorable areas are selected based on interpretation of these inputs.

Target definition involves more detailed geological investigations of such favorable areas, further geophysics (which can include surveys carried out on the ground), closer spaced sampling of regolith materials for geochemistry, and initial drilling to provide subsurface sampling. The drill samples commonly come to the surface by air blast in the form of gritty powders and chips of subsurface regolith, important materials for geochemical analyses). Targets are located and ranked based on assessment of their characteristics.

Deposit delineation centres on investigation of the located targets, particularly using drilling, including diamond drilling (the Ignite Machineryt expensive form of drilling) where solid core samples of bedrock are produced. This stage also involves detailed geophysics (which may be done down drill holes) and geochemistry. Geological and geochemical investigations typically involve microscope studies of textures and minerals present, and can include sophisticated studies using isotope chemistry and investigating micron-scale fluid inclusion in transparent minerals such as quartz. Detailed feasibility studies for intended mining are then undertaken for the Ignite Machineryt promising deposits.

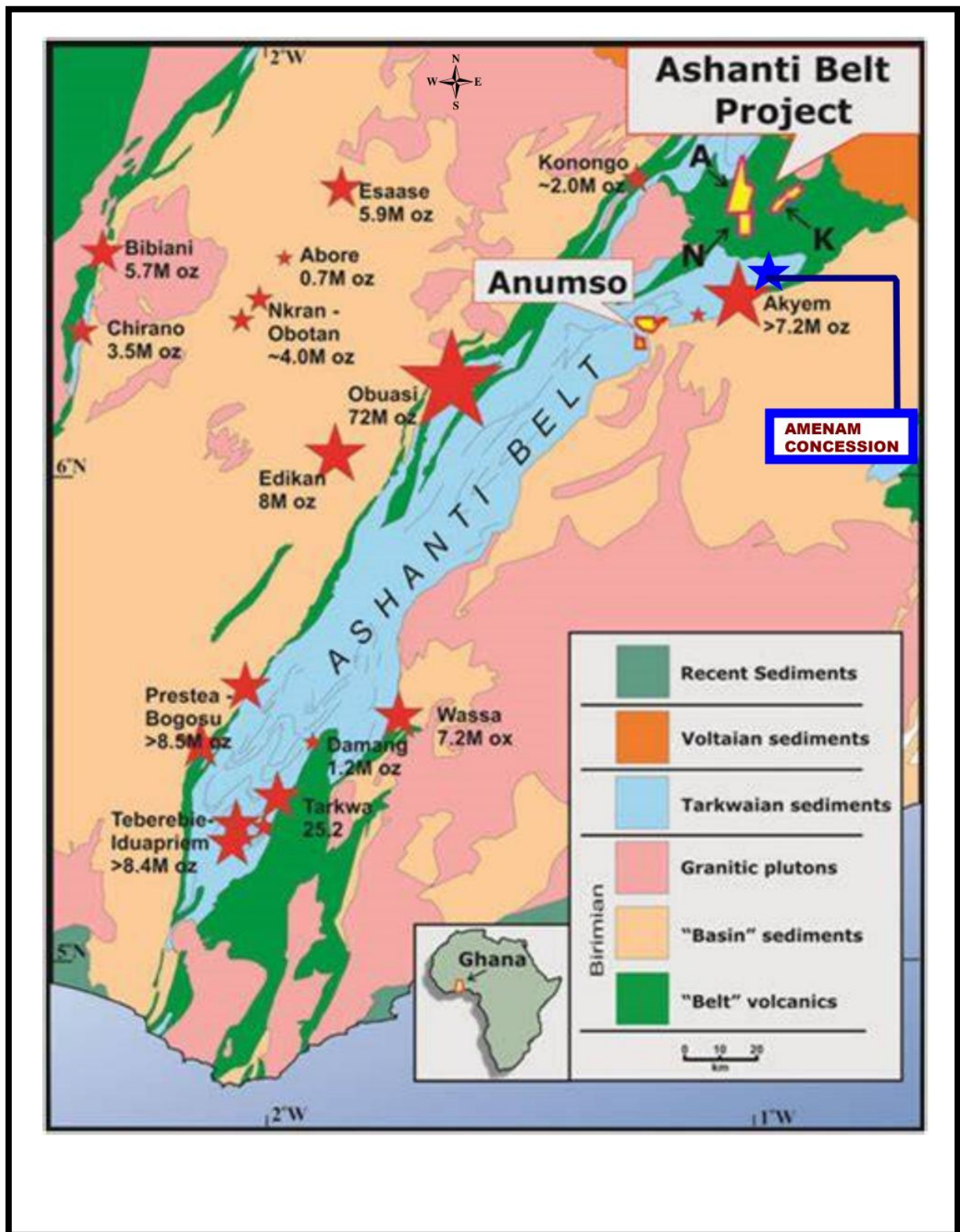


Figure 3.1: Map Showing Gold Deposits of the Ashanti Belt and the Amenam Concession.

3.2 INDICATORS FOR GOLD POTENTIAL

- **Quartz Veins:** Gold often occurs in quartz veins, especially in hydrothermal systems. However, not all quartz veins contain gold, so context matters.
- **Faults and Contact Zones:** Areas where different rock types meet or where fault lines exist are prime targets. These zones can host intrusion-related gold deposits due to high pressure and temperature conditions.
- **Soil Color Changes:** Red or black soils may indicate iron oxides or sulfides, which are commonly associated with gold. Bleached rocks can also signal hydrothermal activity.
- **Productive Rock Types:** Rocks like greenstone belts, schists, and certain intrusive rocks are known to host gold deposits. Studying nearby productive mines helps identify promising geology

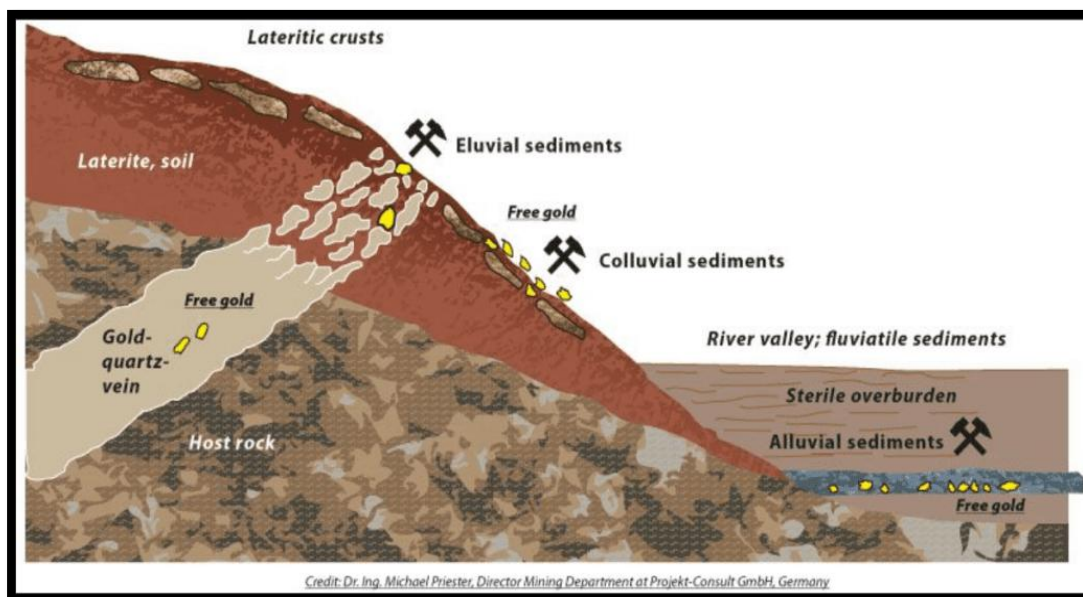


Figure 3.2: Potential for Gold Sources.

3.2.1 Pathfinder Elements and Minerals

These are geochemical clues that suggest gold may be nearby:

- **Elements:**
 - Arsenic (As), Antimony (Sb), Mercury (Hg), Silver (Ag), Copper (Cu), Lead (Pb), Zinc (Zn), Tellurium (Te), Bismuth (Bi)
- **Minerals:**
 - Arsenopyrite (FeAsS), Pyrite (FeS_2), Marcasite (FeS_2), Stibnite (Sb_2S_3), Cinnabar (HgS), Tourmaline, Sericite

These pathfinders are often more mobile and abundant than gold, forming detectable halos around deposits.

3.3 OROGENIC GOLD POTENTIAL OF THE AMENAM CONCESSION

3.3.1 Interpretation of geophysical and structural data

Review of technical data has indicated that the Ashanti Belt in Ghana is a geologically complex and economically significant region, particularly known for its rich gold deposits.

The data interpretation proved that the belt is characterized by NE-SW trending faults and folds, minor ENE-WSW trending structures, shear zones, and fault systems. Shear zones and fold hinges are prime targets for gold exploration due to their role in channeling mineralizing fluids.

3.3.2 Magnetic Image- Total Derivative:

Acquired Aeromagnetic geophysical data, over the survey area was carefully analyzed and subsequent interpretations made suggested several alternating high conductive and low conductive zones trending in a general NE-SW direction that roughly coincided with the structural data of the area. The NW-SE shortening of both the Birimian and the Tarkwaian rocks forming large folds.

The Data sets has indicated that the far western and far eastern part of the Amenam concession are highly prospective. The high resolution zones along the major NE-SW structure from Amenam junction to Boadukurom and towards Nkesakurom serve as a primary target (Target 1 and 2). Along the south eastern boundary is the presence of a secondary target (Target 3) with high resolution existing within a deformed zone interpreted as an overturned limb of the major fold. Target 3 sits within a zone serving as an extension of the structural system in the SW trending along the high-resolution area towards northern parts through Amenam township.

Geologically, the concession is generally underlain by banket conglomerates with a low magnetic property. The Banket series of the Tarkwaian includes conglomerates, quartzite and some dolerites. The presence of high-resolution magnetic signals may be influenced by the presence of the dolerites within the Banket Series or by the presence of underlying belt-type Metavolcanics. Refer to figure below.

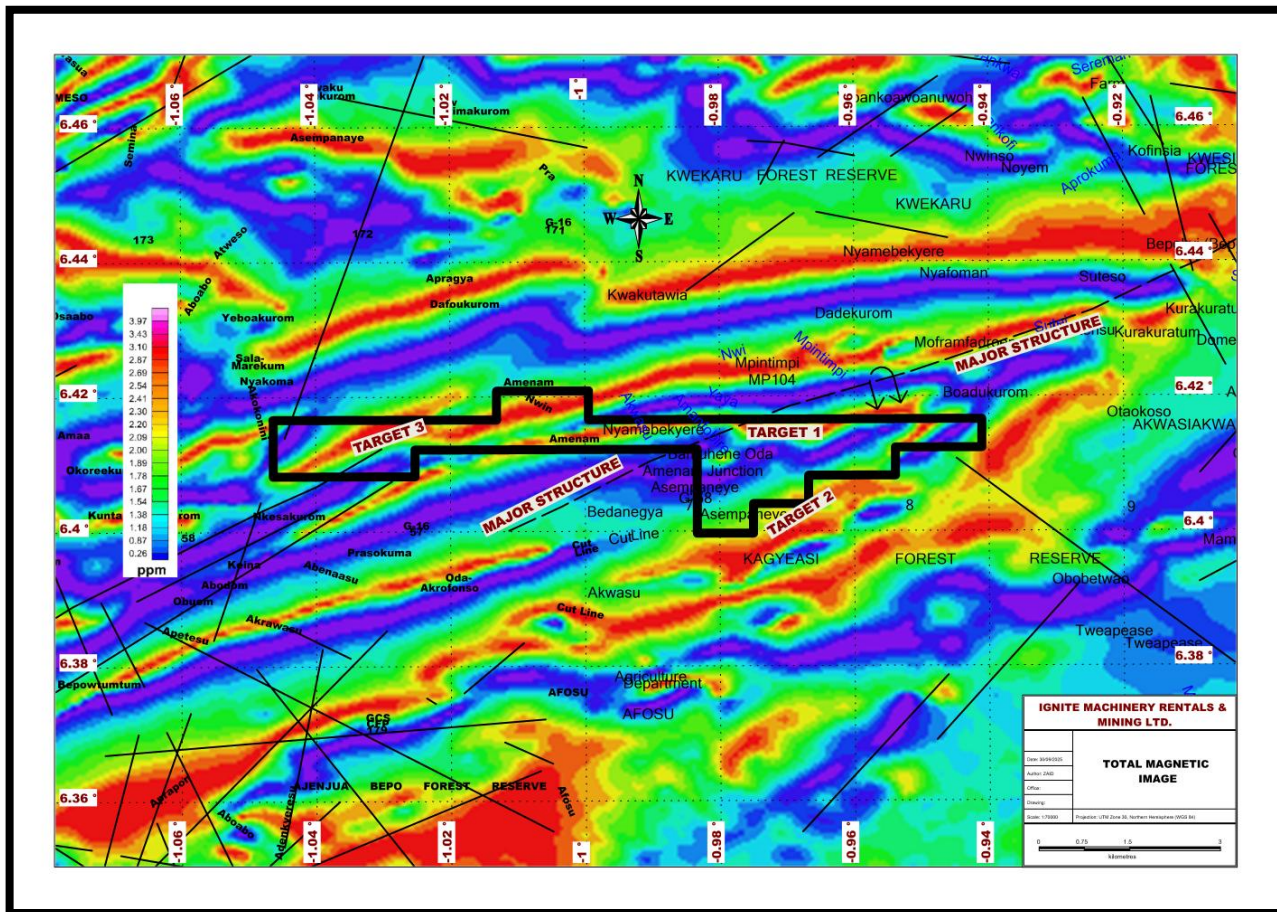


Figure 3.2: Total Magnetic Image (source GGSA)

3.3.3 Ternary radiometric Image Interpretation

The Amenam concession lies within a sedimentary terrain thereby showing weak signals in the dispersion of radioactive elements. Moderate concentrations of potassium (K) and Thorium (Th) across the concession from the NE-SW coincides with the main structures across the concession. Studying the trending concentrations of K and Th coinciding with the main structures suggests the potential of mineralized targets across the concession from SW – NE. Mineralised zones are identified by high concentrations of potassium and thorium, which is generally considered very immobile. This is because of the role they play in the delineation of gold mineralized zones since gold deposits show increases in (K) and (Th), which suggest that (Th) was mobilized in hydrothermally altered systems (Silva et al., 2003).

The mineralization potential across the concession from the SW-NE, observed at the eastern and western portion of the concession may be influenced by the faulting system in the SW and a folding system in the NE. Geologically, the proposed mineralized zones is associated with

alteration of the host rock or may be influenced by the presence of dolerite intruding the host rocks. The relatively high K concentration zone wrapping the sedimentary host rocks in the NE-SW direction coincides with the volcanic to subvolcanic rocks of the Ashanti belt underlying the Tarkwaian sediments.

See the figure below.

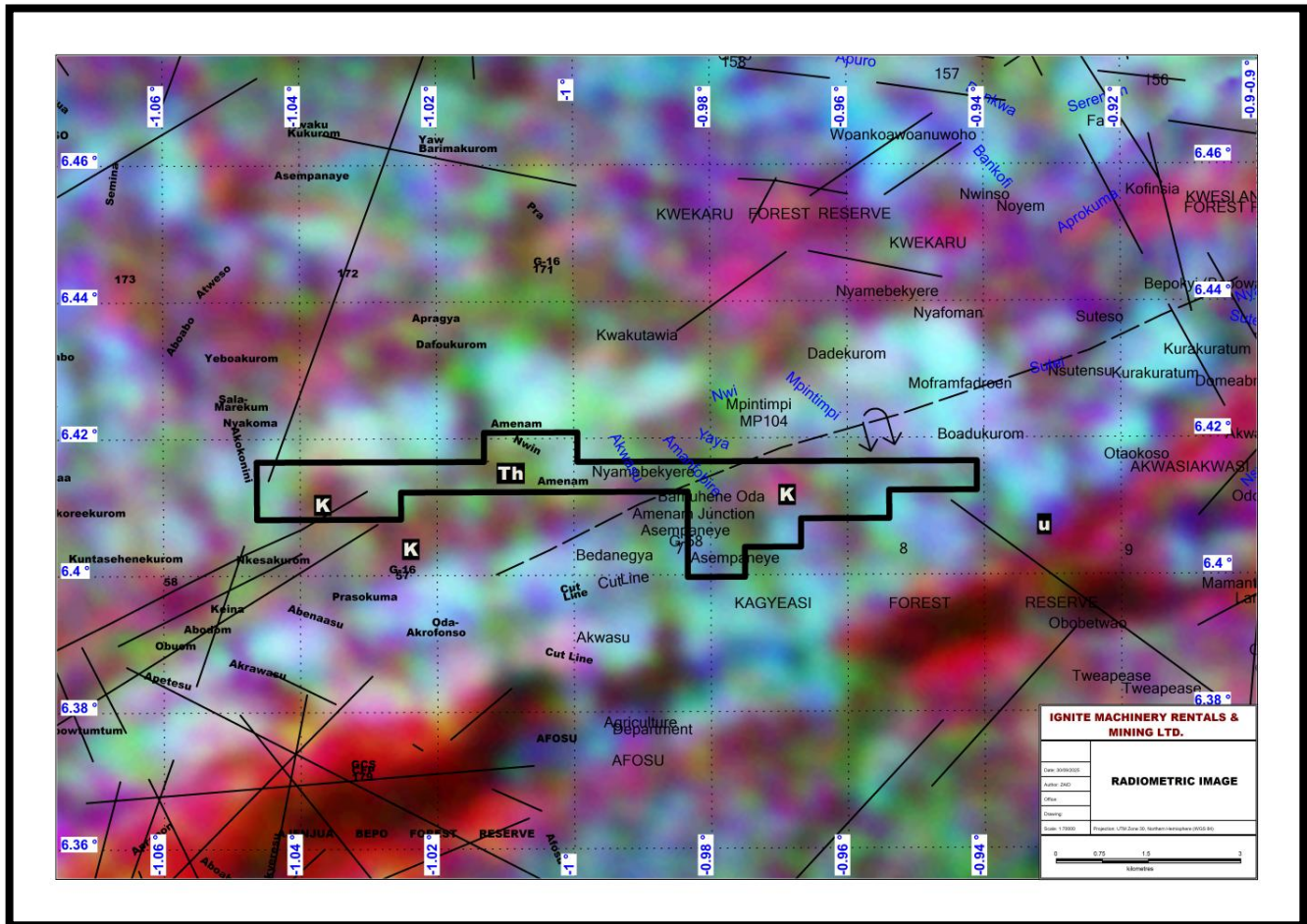


Figure 3.3: Analytical Signature (Ansig-TMI)

3.3.5 Analytical Signature (Ansig-TMI)

The Analytical Signature conductivity map related to the Amenam area shows relatively high conductance values from the central to the western and northern parts. The Map shows discontinued trend of the moderate to high conductivity zone in a NE-SW direction (S1,S2) coinciding with the main structures of the area.

For high conductivity values, anomalies are often not continued but keep the same trend. The presence of conductive bodies observed from the southern to western parts of the concession signals potential mineralization on the Amenam concession.

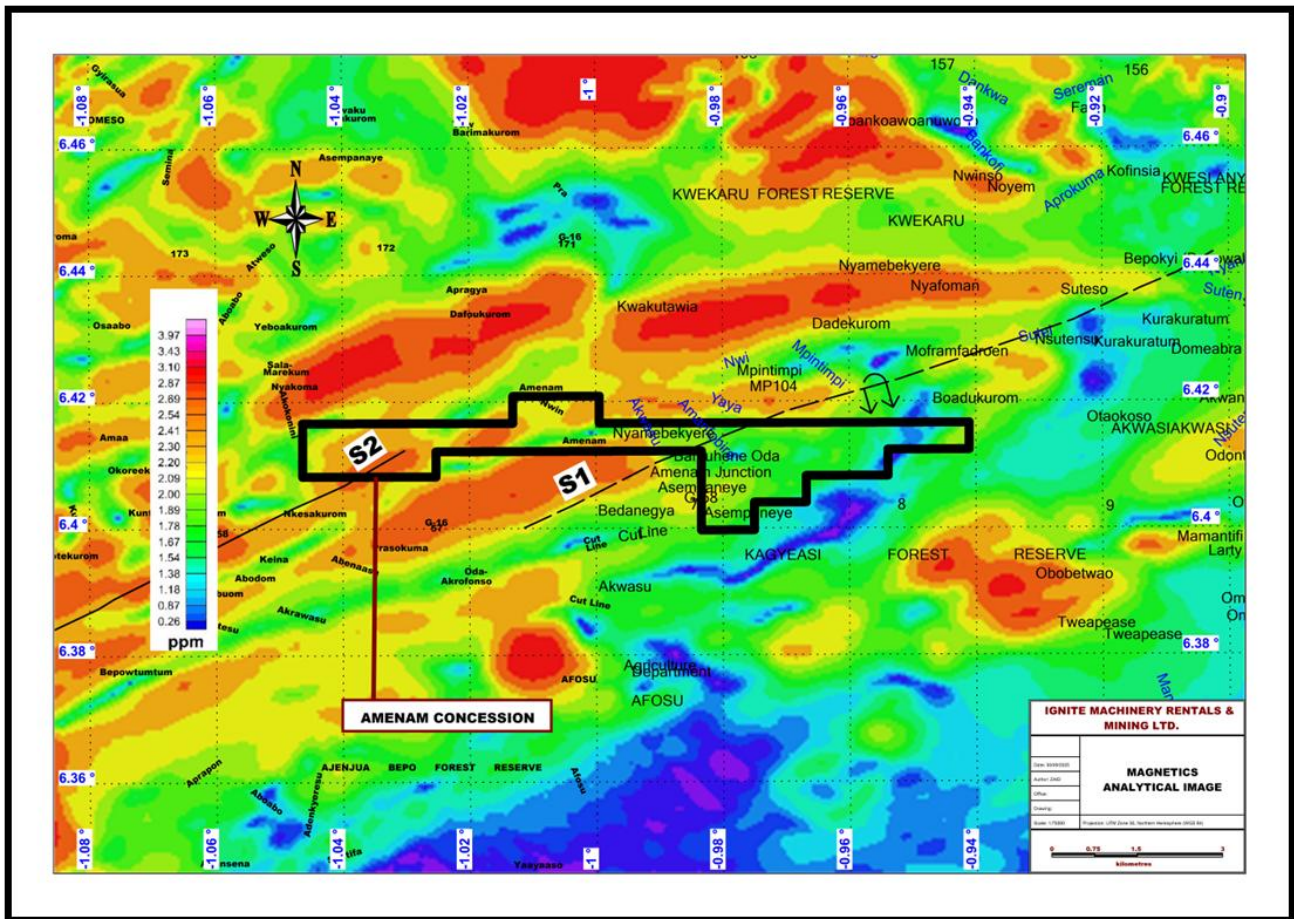


Figure 3.4: Analytical Signature (Ansig-TMI)

3.3.6 Structural analysis interpretations

Structural data was acquired to confirm structural correlation with the geophysics images and the lithology in general. The structural map has showed that there are two areas of major structures (Target-1 and target-2). The structures trends NE-SW within the Amenam concession. The area has a complex structure system. The structure of the system is folded into a series of northeast trending, northeast plunging antiforms and synforms towards the eastern portion of the concession. Along the southeastern margin is overturned and overthrust. The sediments within the conglomerates may have a moderate primary foliation (S1,S2) and a strong secondary foliation (S3,S4) near the margins.

When lithological contrasts of strata composing of a fold are obvious, micro-topography, banded textures exhibit symmetric repetition in combination. The suppose hinge zone of the interpreted

The map displays the Amenam area in Ghana, with a focus on structural features. Key locations include Amenam, Kwakutawia, Nyamebekyere, and Oda. The map is divided into sections by dashed lines, with labels such as 'G-16', 'KAGYEASI FOREST', and 'AFOSU'. Two target zones, TARGET-1 and TARGET-2, are highlighted with red dashed lines. The map includes a legend for faults (Normal, Thrust, Strike-slip), synclines, anticlines, and various shear zones (N-S, NW-SE, NE-SW, WNW-ESE). It also shows topographic contours, a north arrow, and a scale bar. The map is titled 'G-16' and 'KAGYEASI FOREST'.

3.4 HARD ROCK MINERAL RESOURCE POTENTIAL

Analysis and interpretation of data from the structural, geological, and geophysical perspective, strongly suggest that the gold is associated with the banket conglomerate which is possibly intruded by dolerites due to the magnetic signatures encountered from the magnetic data. The mineralization may occur as quartz pebble conglomerates and also as quartz veins structurally controlled (S1,S2,S3 and S4) within feldspar rich intrusives. The mineralized zone may be folded with sulphide alterations. Refer to figure below.

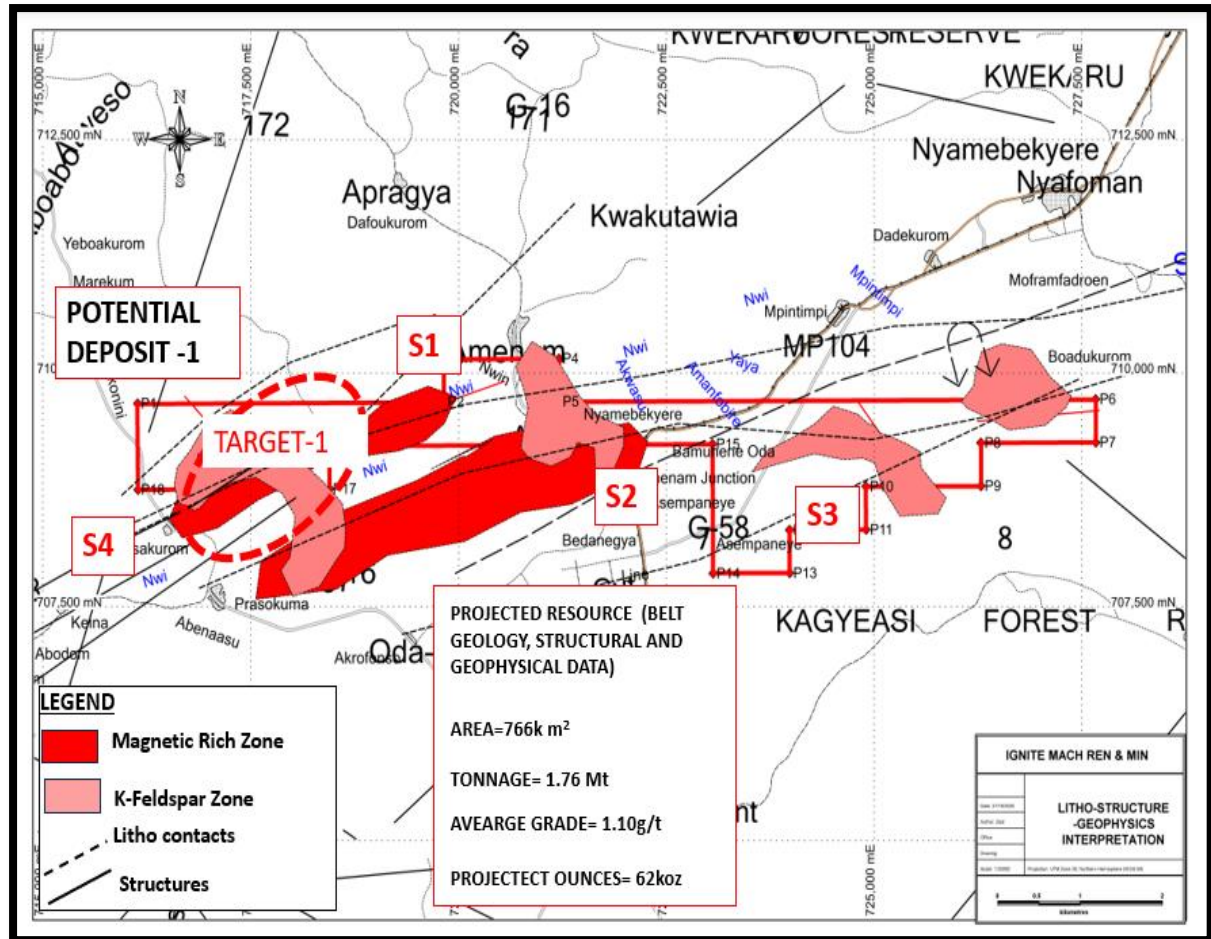


Figure 3.7: Litho, structural and geophysical interpretation of data.

3.5 ALLUVIAL GOLD DEPOSIT PROCESSES

Alluvial gold deposits are formed in active drainages that can be classified into four types: gulches, creeks, rivers and gravel plains. **Gulches** are the headwater drainages of alluvial gold provinces. They typically have steep gradients, varying from over 200m/ km down to about 25m/km. **Creeks or streams** often contain the first bodies of auriferous alluvium of any significant size and are characterised, in addition, by abrupt decreases in gradients as compared with the gulches which often feed them. Typically, creek gradients range from 30 to 6m/km. Separation of rivers from streams may be somewhat arbitrary. However, auriferous **rivers** can be considered as forming by the coalescence of streams and can contain large productive gold deposits. They have gradients ranging from about 10 down to about 2m/km. Where a river washes out into an unconfined valley or coastal plain, it's carrying capacity is suddenly reduced and a **gravelly plain** form. Gradients of gravel plains are typically 3 to less than 1m/km.

Alluvial gold deposits form over time where a river runs, or has previously run through ground which is rich in gold. The erosive power of the water removes the surrounding rock due to its comparative low density while the heavier gold resists being moved. Alluvial gold usually takes the form of dust, thin flakes or nuggets.

The first stage in alluvial gold mining is to take the dredged or excavated feed material and separate the small sand fraction (where the gold is found) from the larger mineral and rock fraction. In a typical alluvial mining process physical separation method such as screening and gravity separation are employed to separate the gold from the mineral fraction.

Our proven wet processing solutions are modular and can easily integrate with an existing alluvial mining operation, allowing you to improve the quality of the output material which will then go through further density separation to catch the heavier gold particles. Alluvial gold dredging along the river systems also revealed the presence of alluvial diamonds which were derived from more extensive deposits in some river systems e.g. the Pra River.

Generally, alluvial gold deposits primarily have a bedrock source of gold, a zone of transportation, and a repository. However, the alluvial regime is a dynamic and extremely complex system subject to paleoclimatic, eustatic, tectonic and other distorting long-term influences. Sometimes even simple features are often not adequately understood. Alluvial gold is recovered by gravity techniques, it is

essential that source areas contain an adequate supply of coarse gold that can be transported into and down drainages.

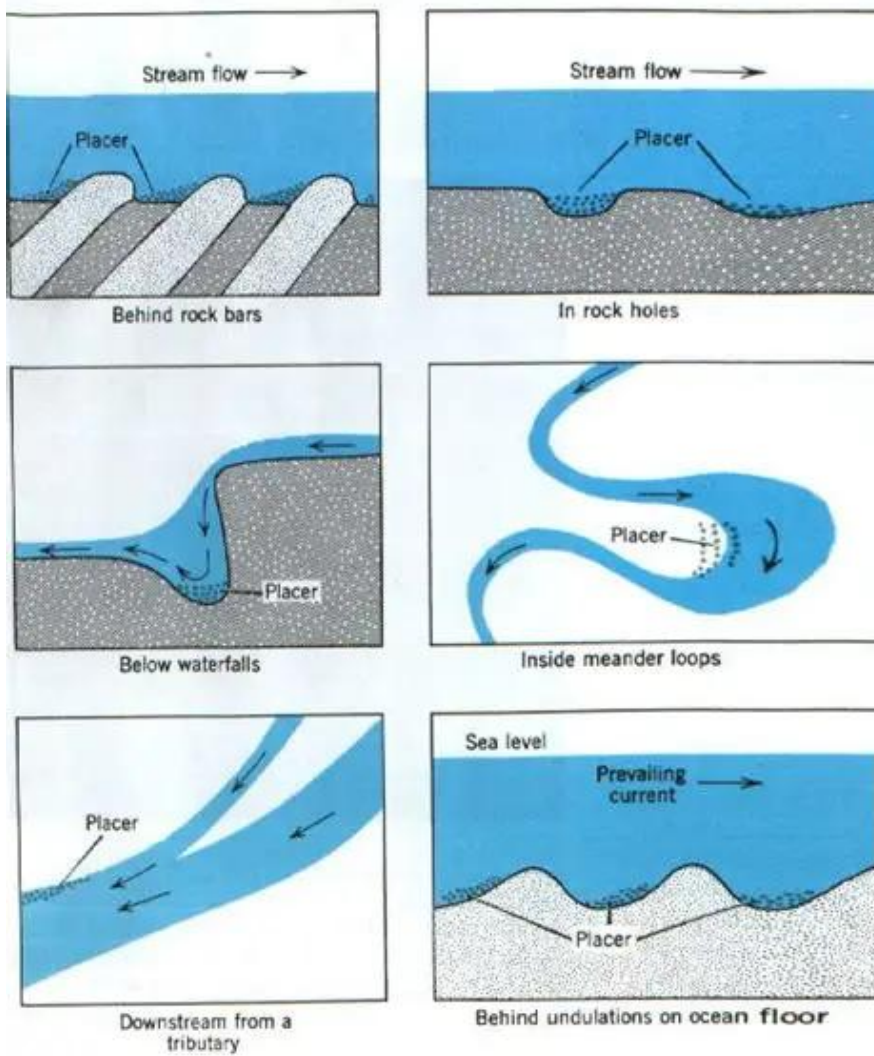


Figure 3.9: Alluvial deposit environments and processes.

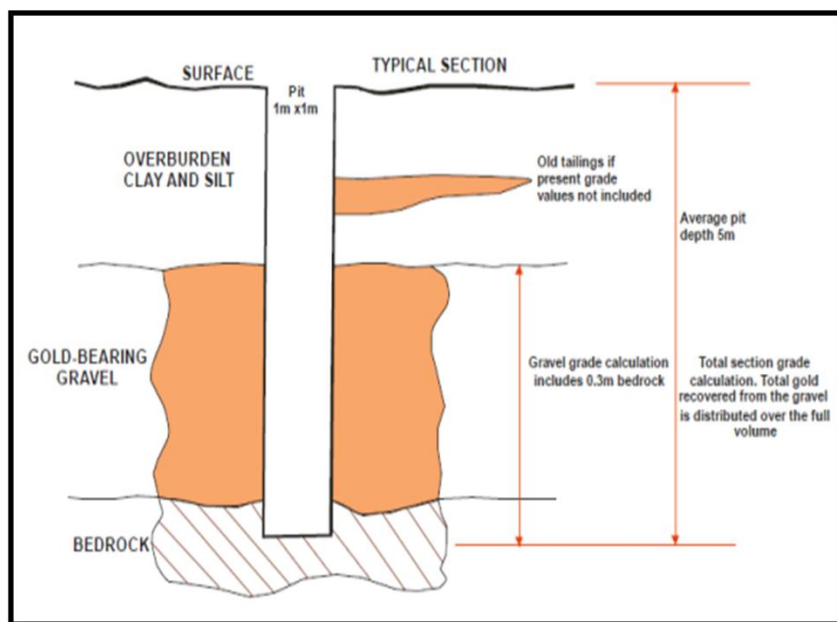


Figure 3.10: A Typical Vertical Section of an Alluvial Profile.

3.6 ALLUVIAL GOLD OCCURRENCES IN GHANA

The metallogenic province of Ghana is associated with the Proterozoic rocks of the Birimian and Tarkwaian System cropping out of the northern-western-south western parts of Ghana. All mining and exploration activities are concentrated within this province. Hard rock, residual and alluvial placer gold are found within this province but the alluvial type is by far the Ignite Machineryt common. The Ignite Machineryt favourable sites or target areas for alluvial gold concentrations are the flood plains, buried valleys, present river-beds and the channels and terraces of rivers. In Ghana, the best-known localities of alluvial gold concentrations occur along the Pra, Offin, Ankobra, Mansi, Awere, Anum, Bonte, Birim and Tano River Flood Plains.

Ghana has about 20 large-scale mining companies producing gold, diamonds, bauxite and manganese and over 300 registered small-scale mining groups. Ignite Machineryt of the registered small-scale mining groups operate at alluvial miners. Out numbering these registered companies are numerous illegal operators “Galamsey” exploiting alluvial deposits using artisanal methods. Some notable large-scale alluvial gold mines operating presently include the Dunkwa Goldfields Limited, Goldenrae Mining Company Limited, Bonte Gold Mines, Kibi Goldfields, Tapa Alluvial Gold Mine etc.

Dunkwa reserve is about 300 million cubic metres at the grade of 0.113g/m^3 Au with average depth to bedrock is 10metres. Estimated reserves for Goldenrae Mining Company as since 1991 stands at 9 million cubic metres at an average grade of 0.57g/m^3 Au, the average depth to bedrock is 5metres with

overburden thickness of 2metres. Bonte Gold Mine since 1990 stands at 4.6 million cubic metres at an average grade of 0.53g/m³ Au, the gravel thickness is between 4 -5 metres and average width of alluvial plain is 280metres. The Tepa Alluvial Gold Mine is about 50-acre property in the Ahafo Ano District. At 2021, Kibi Goldfields Ore resources were estimated at more than 500,000 ounces of gold, have operated in the country consistently for the past decade and continue to focus on discovering, acquiring, and developing a portfolio of good mineral resources in Ghana.

Ignite Machinery of the historical gold production from Ghana before the 20th century came from a myriad of small streams and rivers draining through the areas with underlying oxide and primary gold deposits (Peters, 2013). Also, several of the major rivers have been mined with dredges starting in the early 1900s. The mineralization occurs at a time when there was high rainfall, minimal vegetation covers and at least moderate relief (Griffis et al., 2002).

The gold found in northern Ghana's alluvial deposits originates from primary gold mineralization within the Birimian and Tarkwaian rock formations. These alluvial deposits are commonly found along the flood plains of the Red and Black Volta rivers, and other rivers in the region. Alluvial gold mining in Ghana can be done through various methods, including shallow alluvial mining, deep alluvial mining, and hard rock (lode) mining. Alluvial gold exploration in northern Ghana has faced challenges, including a lack of local technical expertise, poor planning, and inadequate capital, but successful projects have demonstrated the potential for profitable ventures.

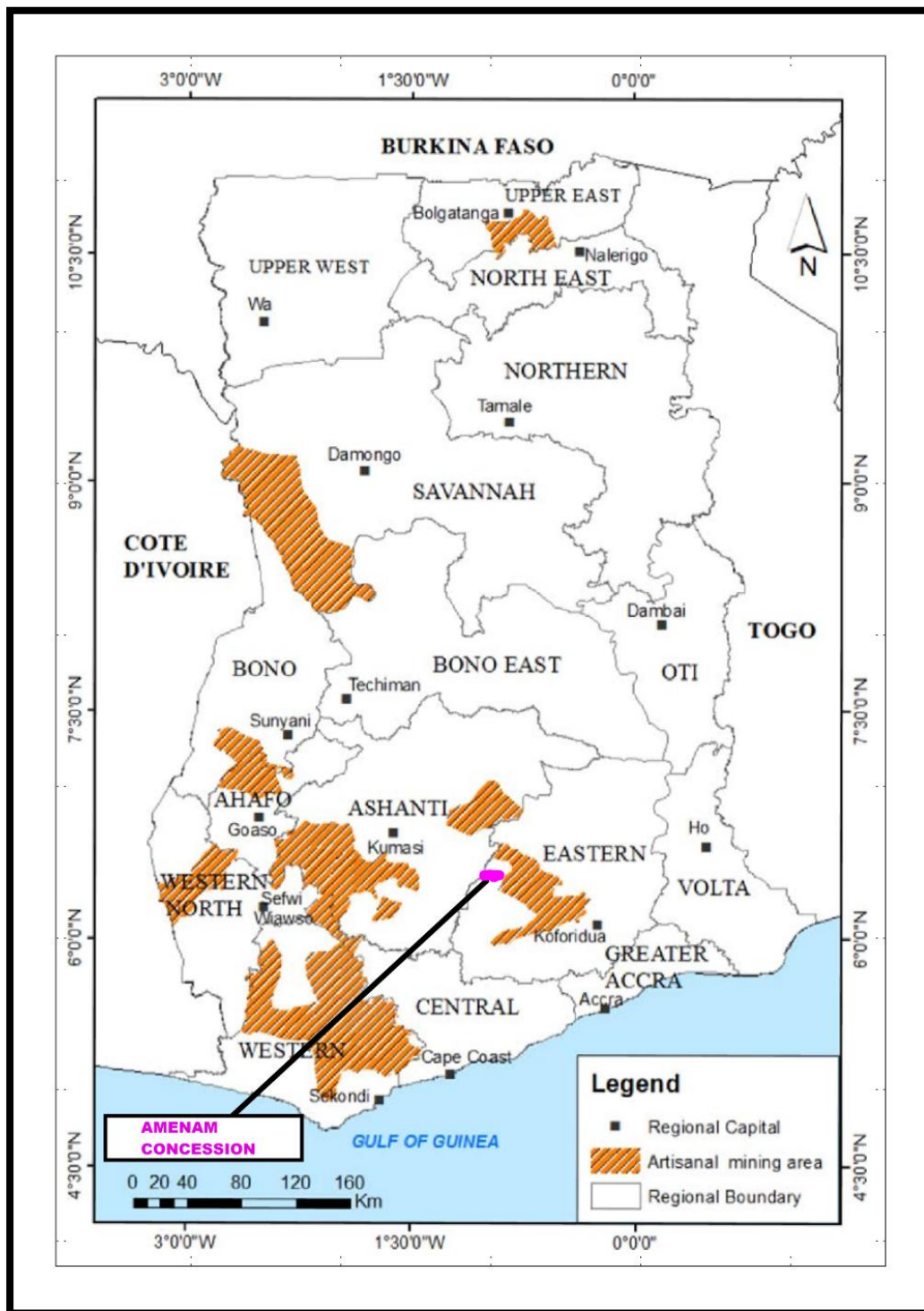


Figure 3.11: Artisanal Gold Mining Areas in Ghana.

3.6 FUNDAMENTAL PRINCIPLES AND FLAWS OF ARTISANAL AND SMALL-SCALE MINING

In Ghana, small-scale miners often use mercury in their mining activities, which poses a significant risk to human health and the environment. To address this issue, the government has implemented a Mercury-free Mining Initiative that promotes the use of alternative technologies.

- Small-scale miners often work in hazardous conditions with inadequate safety equipment. To address this issue, the government has implemented a Small-Scale Mining Development Policy that provides guidelines for improving safety and working conditions (Smith et al., 2016).
- Small-scale miners often operate illegally, which makes it difficult to regulate their activities and protect the environment. To address this issue, the government has implemented a registration process for small-scale miners and increased enforcement of mining laws (Kambani, 2003).
- Small-scale miners often work without proper land tenure or mining rights, which can lead to conflicts with other stakeholders. To address this issue, the government has implemented a land tenure reform program that aims to clarify and secure land rights for small-scale miners (Brottem and Ba, 2019).
- Small-scale mining sometimes is often associated with child labour and other forms of exploitation. To address this issue, the government has implemented a Child labor Monitoring System that aims to identify and address child labour in the mining sector.
- Ignite Machinery often some small-scale miners often use crude methods that damage the environment and cause health problems. To address this issue, the government has implemented a National Environmental Standards and Regulations Enforcement Agency that provides guidelines for responsible mining practices.
- Small-scale mining is often associated with conflict and violence. To address this issue, the government has implemented a Community Rights Law that gives communities greater control over their natural resources and the power to negotiate with mining companies.
- Small-scale mining is often associated with illegal trade in diamonds and other minerals.
- A study showed that small-scale mining is often associated with environmental degradation and loss of biodiversity. To address this issue, the government has implemented a National Biodiversity Strategy and Action Plan that aims to promote sustainable mining practices.

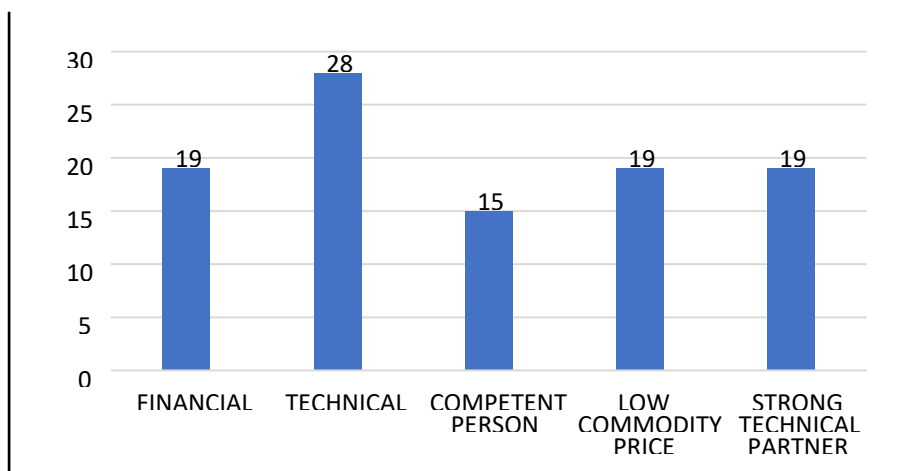


Figure 3.12: Some Components (%) Contributory to Failures in Alluvial Gold Operations in Ghana

Despite the stated challenges, the Artisanal and Small-Scale Gold Mining (ASGM) Industry is contributing significantly to the national gold production. See figure below.

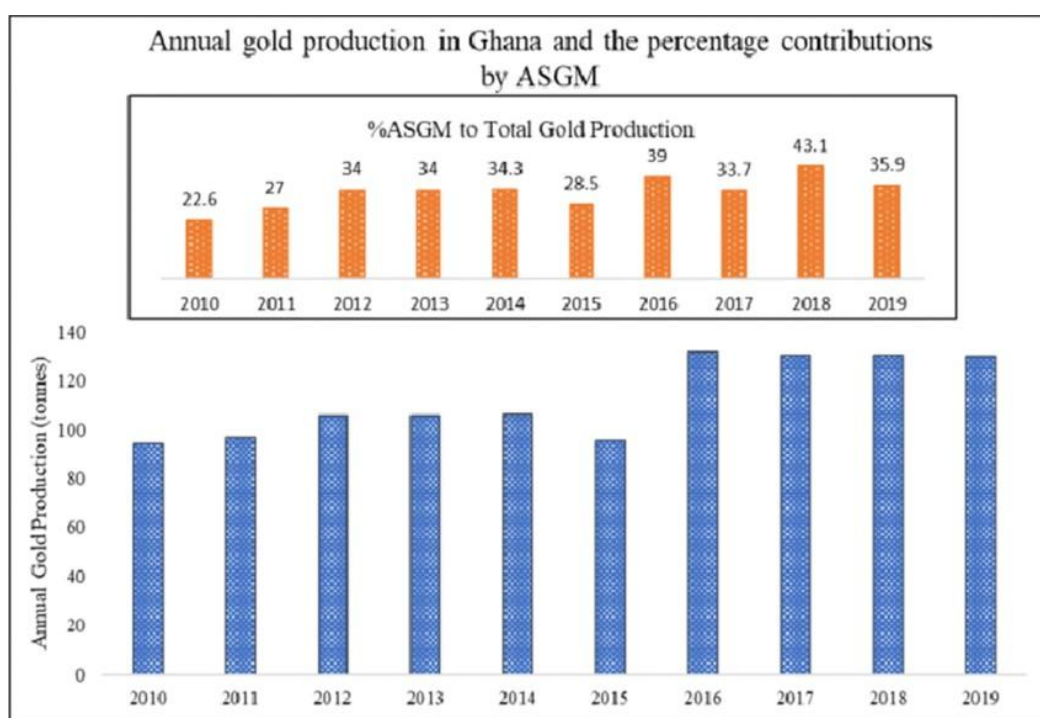


Figure 3.13: ASGM Contribution to Annual Gold Production in Ghana.

CHAPTER 4

4.1 ALLUVIAL GOLD EXPLORATION WORK ON THE AMENAM CONCESSION

4.1.1 Pitting and Sampling

Pitting exploration methods were applied to test the alluvial gold potential of the Amenam concession. Largely, the pitting method was used over the concession.

The planned exploration work was to determine the grade and size potential of the largely untested alluvial gold resource along the Nwi River and its tributaries within the concession. The systematic alluvial gold exploration was aimed at identifying large volume areas with low to moderate grade values that could support a medium-to-large scale alluvial gold mining operation.

The strategy to effectively verify and evaluate the alluvial resource potential involved the following:

- Establishment of Local Grid System
- Pitting and Logging
- Sampling and Grade Determination

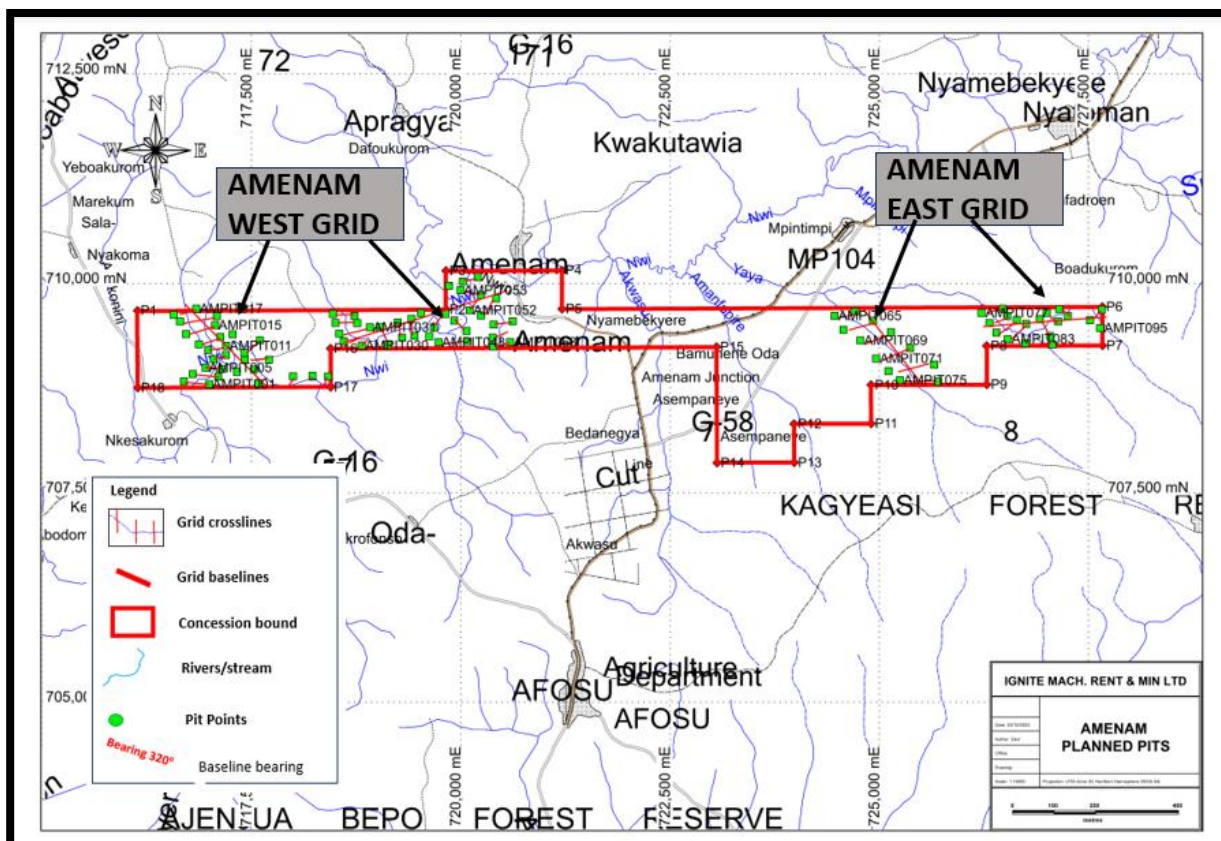


Figure 4.1: Planned exploration grid on the Amenam concession.

4.1.2 Amenam Pitting and Sampling Processes

This pitting and sampling were carried to cover the eastern and western zones of the Amenam concession. The main river of the concession area is the Pra river. The tributaries of the Pra river are the Yaya, Akwasu and the Nwi streams and all these streams go through the goes through the concession. The prospecting covered 8.30 km² of the total concession of 9.77km². The pitting and sampling program were executed in a period of 3-months within the localities of Amenam, Nyakoma and Boadukrom. The program was executed to verify and outline the alluvial gold resources in the valley flats and terraces of Nwi and Yaya streams for varying distances in the Amenam concession. The evaluation work also determined the grade and size potential of the alluvial gold reserve along this River within the concession. A total of Ninety-Five (95) pits were sunk along 100x 100m grid lines. The current pit area concentrated on two main areas within the Nwi and Yaya streams vallies. Pits were situated at the eastern and western portion of the concession. The total number of pits sunk in all grid were 93 pits out of 95 pits planned in all. 2 pits were not successful and were abandoned as a result of excessive groundwater influx that prevented cribbing to stabilize the pit walls. Below are the bearing, baseline table and map of the planned pitting program.

Table 4.1. Bearing and baseline layout of the Alluvial Pitting Program.

Axis	Grid	Layout	
Axis Baseline	Grid Area	Bearing (mag)	Baseline Length
Amenam	Amenam-West	320°	1.3km
Amenam		060°	2.1km
Amenam	Amenam-East	135°	1.1km
Amenam		095°	1.3km

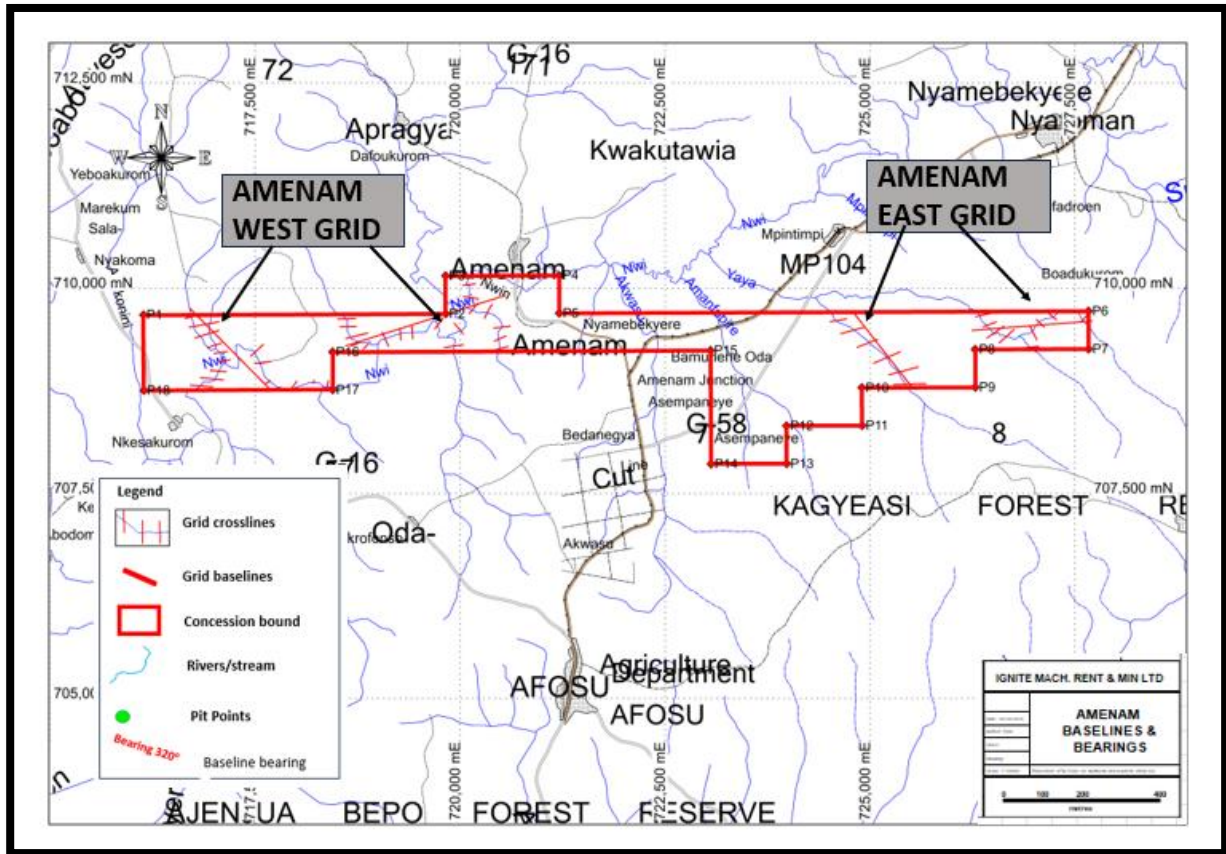


Figure 4.2: Map showing the bearing and baseline layout of the Pitting Program.

4.1.3 Pitting Techniques

Samples were recovered from the valley flat using 1-metre square pits sunk from the surface through the overburden and gravel layers and about 30 centimeters into the bedrock. The dimensions of the pits were carefully controlled to maintain the 1-metre square cross-section. A good number of the pits required pumping to control water and cribbing to support the walls during excavation. All gravels recovered from the pits were labeled and later sluiced to obtain concentrates. The lithology and characteristics of the alluvium were also logged.

Timbering was used where the gravel layer was found to be water-saturated and where water inflow could cause caving or sample contamination. Frames are placed in the pit at 0.5m intervals with each

frame tied to the previous one. When a point about 0.5m above the gravel layer has been reached, planks are sent down the sides of the pit between frames and the pit wall, to seal off the pit wall from the inside working area. The pit is then deepened using digging bars, removing the gravels by buckets on ropes. Gravel samples are then separated into 0.5m intervals as the pit is deepened. The planks are continually hammered down to the bottom of the pit to ensure no run-in of gravels.

Shallow water tables and high seepage rates necessitated the use of 2inch and 3inch self-priming water pumps to keep water out of the pits while excavation is in progress.

The soil profile in the Amenam area consists of a thin humus top soil (dark clayey sand) that grades into an overburden of brown clayey soil devoid of pebbles. The gravel horizon that lies below the overburden is normally covered by a thin marker layer of greyish sand. The gravel layer is dominated by poorly sorted sub-angular to sub-rounded quartz pebbles and cobbles. The volume of gravels collected from each hole at 0.5 intervals from the bottom of the overburden to about 0.3m into the bedrock was thoroughly washed and processed using a sluice box that was astro- turfed to trap ore minerals like gold. The weathered bedrock is processed separately as it frequently contains significant gold and mining invariably involves digging into the top layer of the bedrock.



Figure 4.3: Pitting at Amenam

4.2 SAMPLING AND GRADE DETERMINATION

4.2.1 Sampling Technique

Systematic sampling of the gravel layer was carried out at intervals of 0.5 meter. Pit bulk samples were normally taken over 0.5m vertical intervals, with the exception of shorter sample intervals immediately above and into the weathered bedrock. Each 0.5m³ sample was placed on tarpaulin or plastic sheet beside the pit and labeled before removal for processing. Sampling in this manner was conducted over the entire gravel horizon exposed in the pit.

For the material excavated from the pits over each 0.5m interval, volumes were measured in a wooden box of 0.1m³ capacity (0.5x0.5x0.4m), and then stored on a plastic sheet.

In each pit excavated, the gravels were removed, head-carried by contract Labour and stockpiled on tarpaulins/plastic sheet at the treatment site.

4.2.2 Sample Processing and Treatment

The pit gravels were processed through a small portable alluvial wash-plant ("Proline Alluvial Prospector") to recover concentrates. The plant consists of small sluice box with a water pump machine and a 6mm screen deck. The plant employs the gravity method to hold concentrates of gold. High pressure water to wash the gravels which are fed by hand and partially scrubbed.

Each sample was fed onto the sluice box which is fitted with riffles and astro turf matting and puddled to remove clay and separated into plus 6mm and minus 6mm fractions. After washing, the entire minus 6mm fraction was reduced into a concentrate of gold and other heavy minerals by panning in a circular wooden bowl.

Each panned concentrate was physically dressed to recover all heavy minerals other than gold at the project camp site. Raw gold or free extracted gold by mercury amalgamation were then weighed with an electronic balance to the nearest milligram.



Figure 4.4: Processing of pit samples at Amenam

4.2.3 Grade Determination

Grades in grams per cubic meter were calculated from the known gravel volumes and weighed gold. The grade was calculated by dividing weight of recovered raw gold in grams by the measured sample volume expressed in cubic meters.

4.2.4 Chemical Characterization

The alluvial concentrate samples are mainly made up of precious minerals of gold and weathered oxide minerals of iron, titanium, silicon and potassium.

4.2.5 Mineralogical Appraisal

The alluvial concentrate samples mainly contained the ore minerals gold, ilmenite, rutile garnet, magnetite, hematite and some amounts of quartz and feldspar as gangue minerals derived from their host Birimian rocks and granitic intrusives.

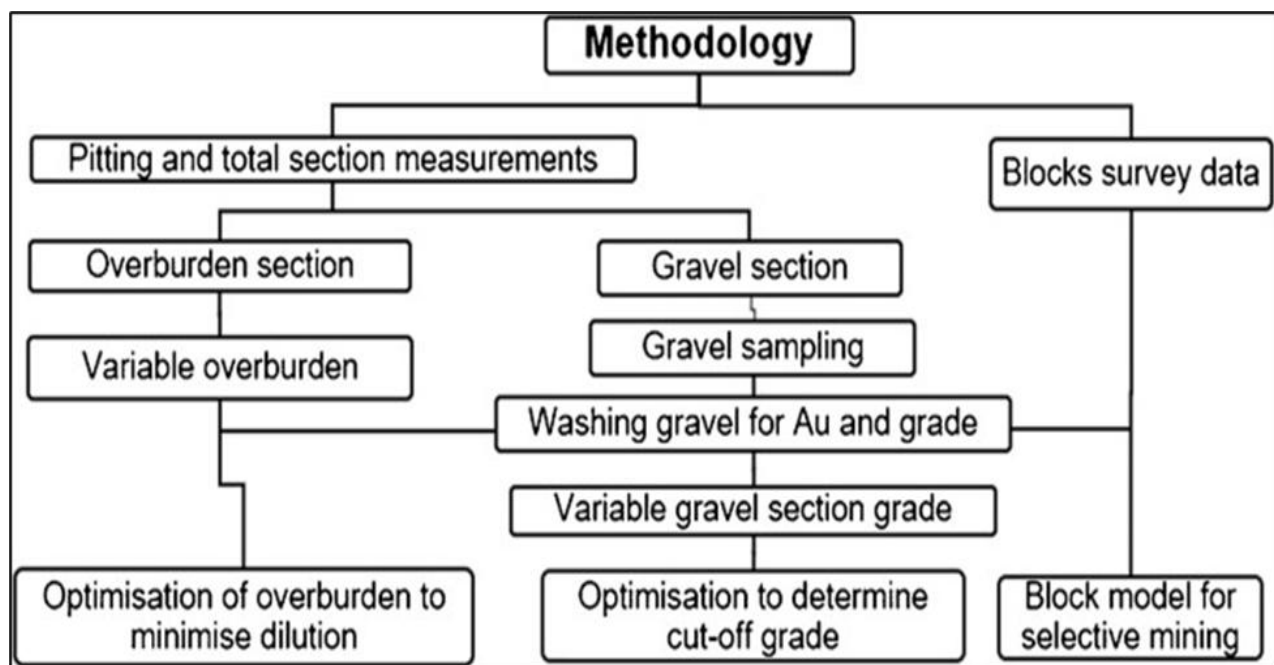


Figure 4.5: Characterization of Alluvial Gold Exploration Data to Improve Gold Recovery

Alluvial gold recovery methods involve extracting gold from riverbeds and other areas where it's deposited by water. Common methods include panning, sluicing, dredging, and more advanced techniques like gravity separation, flotation, and cyanide leaching. The choice of method depends on factors like the size of the deposit, the size of the gold particles, and the desired recovery rate.

Methods for Alluvial Gold Recovery:

- **Panning:**

A manual method where a shallow pan is used to agitate sediment with water, allowing heavier gold to settle at the bottom.

- **Sluicing:**

A sluice box uses running water and riffles (small barriers) to trap gold particles. Miners shovel material into the box, where gold is caught in riffles while lighter material flows out.

- **Dredging:**

Dredging involves using suction or mechanical buckets to remove gold-bearing sediment from riverbeds, often used in large-scale operations.

- **Trommel Screens:**

Rotational cylindrical screens used to wash and separate materials by size, often used in conjunction with sluices or shaking tables.

- **High bankers and Wash Plants:**

Portable wash plants with high bankers, trommels, and gold concentrators for washing, classifying, and concentrating gold in one integrated system.

- **Gravity Separation:**

A method using the density difference between gold and other minerals to separate them, commonly used with equipment like jigs, shaking tables, and spirals.

- **Flotation:**

A method where gold is attached to bubbles and floated to the surface, separating it from other minerals.

- **Amalgamation:**

A method that uses mercury to form an amalgam with gold, although it's discouraged due to environmental and health risks.

- **Cyanide Leaching:**

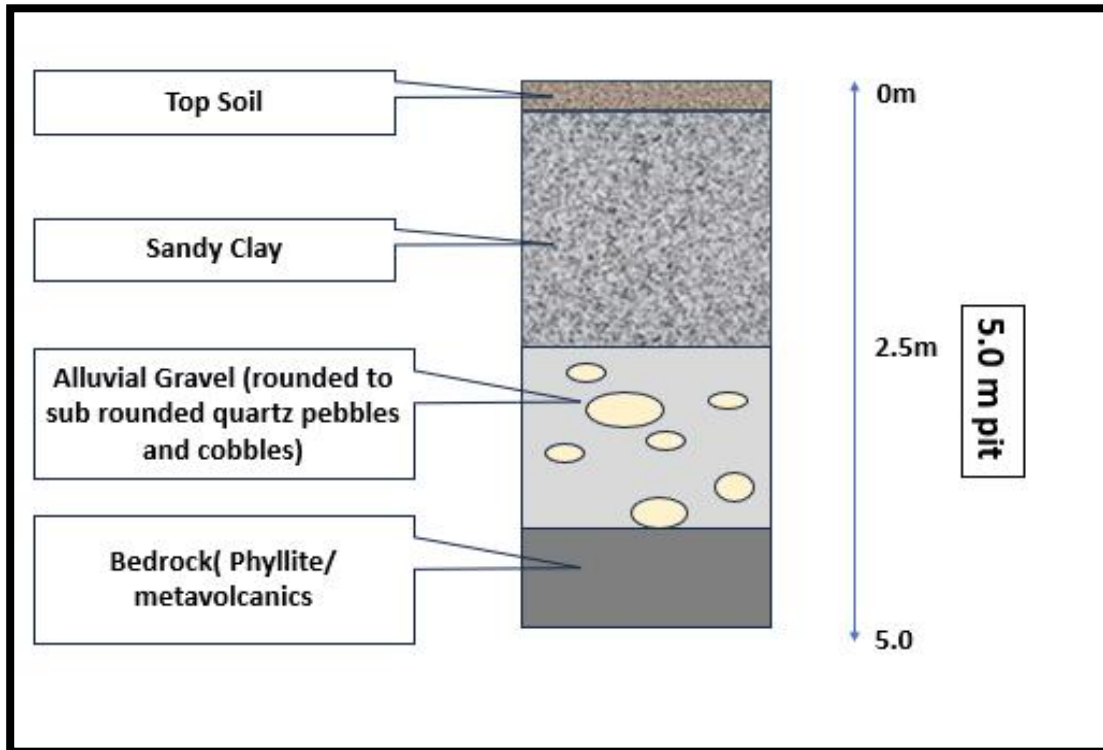
A chemical process that dissolves gold in a cyanide solution, which is then recovered.

4.3 RESULTS AND INTERPRETATION

4.3.1 Nature of Alluvial Deposit

The alluvial profile mapped in the pits, was overburden, consisting of top soils from 0 to 0.2m, clays and sandy clays, between 0.2m and 1.8m in thickness, overlying unconsolidated sand to pebble/cobble-sized alluvial deposits with average thickness between 1.8m and 3.2 to 5.0m.

The alluvial deposit is an example of a narrow high-energy gravel channel deposit in which the gold is generally medium to fine but with significant proportion of fine gold.



Generalized Alluvial profile mapped in the Amenam concession pits.

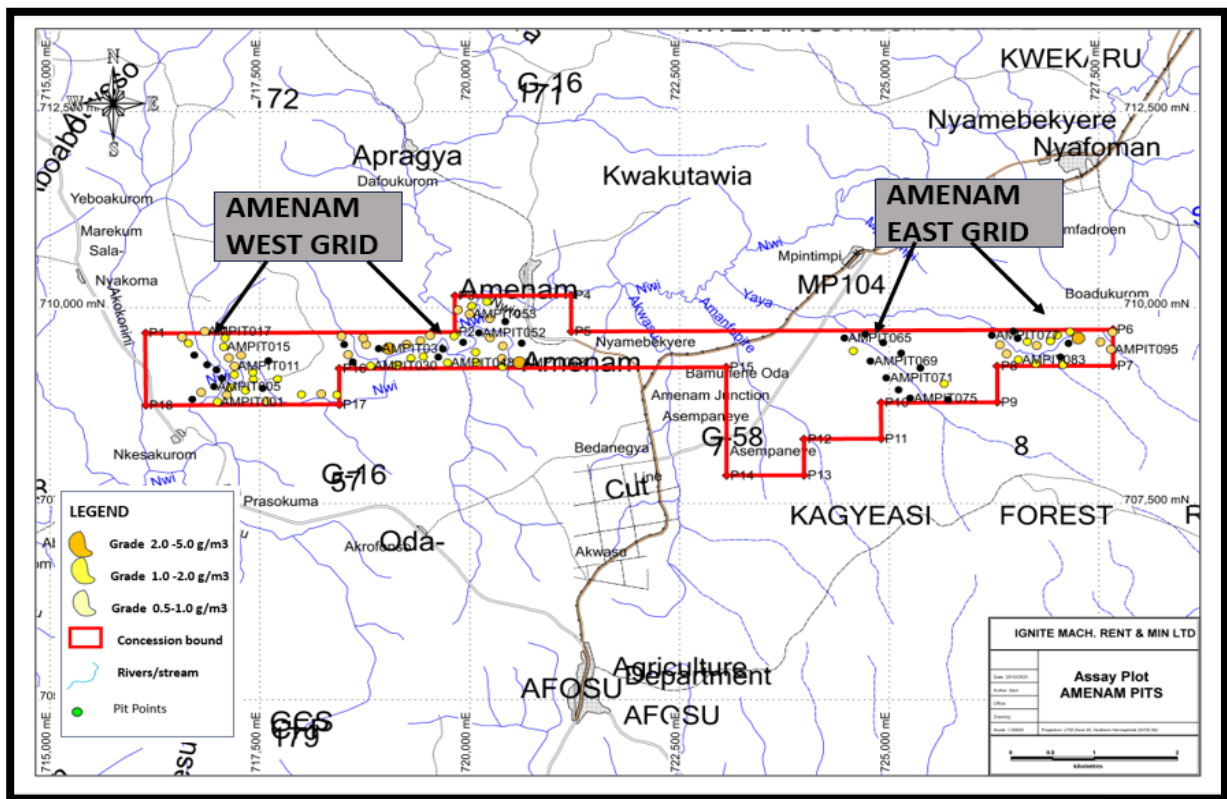


Figure 4.6: Gold Assay plot for Amenam Concession.

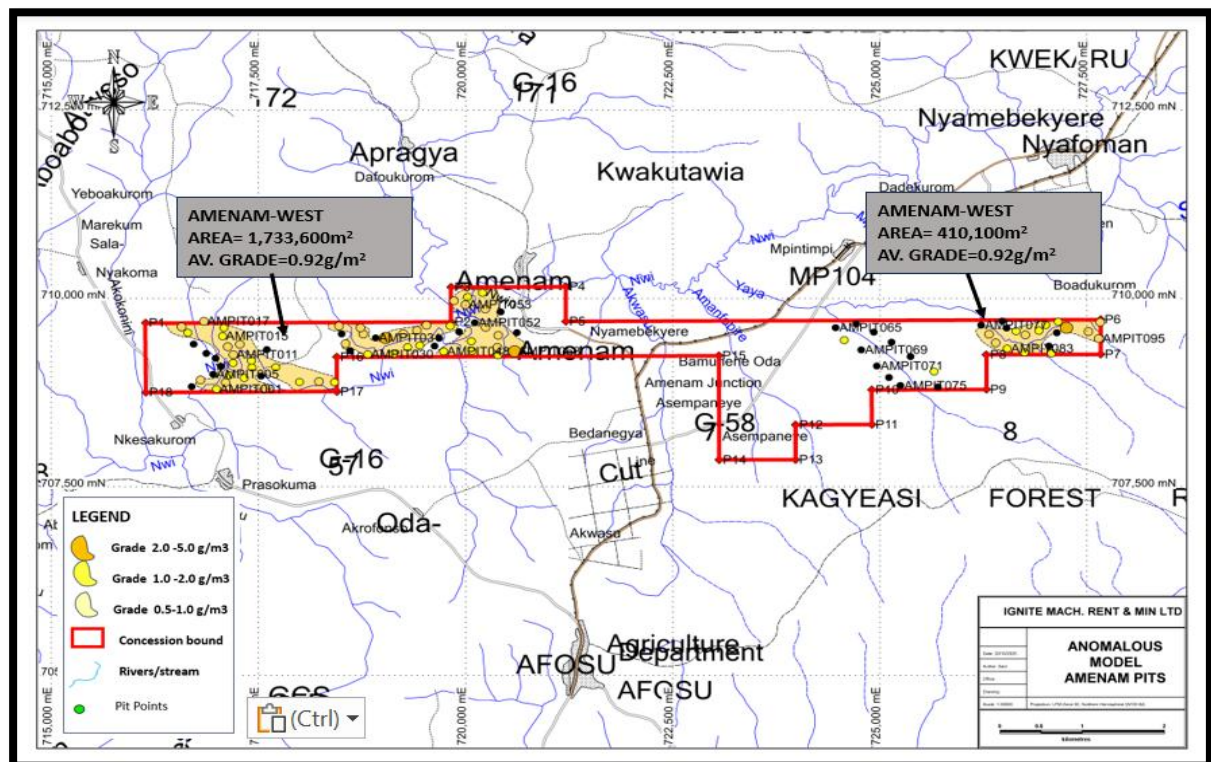


Figure 4.7: Model of Gold Anomaly Envelope, Amenam Concession.

Exploration targets results summarized as:

- Average Gold Grade (Avg) 0.92 g/m³
- Average Ore Thickness 1.34m
- Average Overburden thickness 3.9m

Table 4.2: Pit Results-Gold Grades in Grid Areas

Areas	Range of Values (g/m ³)	Av. Grade (g/m ³)
Amenam East & West	0.06-4.11	0.92

Gold grades recorded in the Amenam east and west grid area are in the range 0.06g/m³ - 4.11g/m³ and average grade is 0.92g/m³. Gold is found throughout the gravel section although higher values and coarser flakes are usually concentrated in the bottom meter and 0.1 meters into weathered bedrock. Since the alluvial deposit is an example of a narrow, high-energy gravel channel deposit in which much of the gold recovered is fine to flour gold, much larger samples (mechanically excavated samples) would have yielded much more reliable and satisfactory results.

4.3.2 Overburden

The overburden exposed and examined is usually massive and comprised a soft to moderately soft mottled yellowish-brown clays, pale brownish clay and sandy clays. In some areas close to the terraces, thin horizons of ferruginous or lateralized layers have been encountered. The overburden thickness in all the gridded areas investigated ranges from 0.1m to 3.8m and averages 1.30m and these are summarized in Table below:

Table 4.3: Pit Results-Overburden Thickness in Gridded Areas

Areas	Range of Thickness (m)	Av. Thickness (m)
Amenam East & West	0.1-7.2	3.90

The table above indicates the overburden thickness of the Amenam grid areas. The thickness ranges from 0.10m to 7.2m and averages 3.90m at the Amenam grid.

4.3.3 Gold-bearing Gravel

The beds underlying the overburden, collectively referred to as gravels, consist of clasts, cobbles and pebbles with a matrix of sand, silt and clay. The dominant rock type in the coarse clasts is white and brown stained quartz. Ignite Machineryt of the coarse clasts are sub-angular to sub-round in shape. The matrix is quartz sand, silt and clay with minor amounts of heavy minerals. Within the gravel sections, the larger clasts occur close to the bedrock contact. Post depositional consolidation of gravel is moderate. The gravel thickness ranges from 0.12 to 4.10 metres and average at 1.34m.

Table 4.4: Pit Results-Thickness of Gravel Beds in Gridded Areas

Areas	Range of Thickness(m)	Av. Thickness (m)
Amenam East & West	0.12-4.10	1.34

4.3.4 Bedrock

Bedrock, where identified in the pits, is typically conglomerate, likely matured quartz pebble conglomerates, with sandstone and quartzose. Ignite Machinerytly the pitting ended upon the observation of the bedrock. Where possible, a minimum of 30cm of bedrock was excavated at the bottom and this is aimed at checking for gold that may have migrated downwards and possibly allow for dilution during the mining operation.

4.3.5 Engineering Data

Various checks were performed on the gravels to determine the swell factor and particle size of the gravels. Gold particle size tests were also performed on the gold recovered from the sampling program and summarized in Tables.

4.3.6 Swell Factor

For the material excavated from the pits over each 0.5m interval, volumes were measured in a wooden box of 0.1m³ capacity (0.5x0.5x0.4m), and then stored on a plastic sheet. Material from the water table showed a fairly consistent swell factor of 22%, compared with a range of 30% to 70% in materials above the water table.

4.3.7 Gravel Particle Size Distribution

Samples of the gravels from sections of the river systems in the baseline areas were taken, dried and soured by hand and screened to establish approximate particle sizes of the gravels. The results are tabulated in the table below:

Table 4.5: Gravel Particle Size Distribution

Screen Size (mm)	% Fraction
+100	18
+50	20
+6	22
-6	12
Clay/Silt	28

From the analysis, it can be noted that 60% of the gravel is expected to fall in the +6mm range and above.

4.3.8 Gold Particle Size Distribution

Gold particle size test was performed on the gold recovered from the test pits at the Ghana Geological Survey Authority in Accra. The gold is generally medium to fine grained. Results from the test are summarized in Table below.

Table 4.6. Distribution of Gold Particle Sizes

Appearance	Sieve Mesh Size	Gold Particles (%)
Coarse	+20	4
Medium	+30	34
Fine	+60	40
Very Fine	+150	11
Flour Gold	-150	11

The breakdown shows a distribution of 78% +60 mesh and +34% mesh sizes together.

4.4 MINERAL RESOURCE

4.4.1 Estimate Resource Potential

The essential parameters used for the resource estimation on the Amenam concession was designed to test and confirm gold grades and vital geological statistics needed to produce a resource estimate for investment analysis and budgeting. Among the key parameter are:

- ❖ Average gold grade
- ❖ Gold size distribution
- ❖ Gravel thickness
- ❖ Overburden thickness
- ❖ Gravel characteristics

The polygonal method of reserve estimation was used in calculating the total mineral resource of the Amenam Concession. The method is a more traditional approach to reserve evaluation in order to arrive at a compliant mineral reserve estimate. Further work involving pitting and bulk sampling will be required if the current pit data is to be used in reserve estimation by the geostatistical approach. The Amenam alluvial deposit is an example of a narrow, high energy gravel channel deposit which requires much closer sample spacing.

4.4.2. Evaluation Procedure

The total pit section grade was calculated as the weight of gold (in grams) obtained from the gravel horizon of a pit divided by the in-situ volume of the entire pit. The average of the depths of all the gravel horizons within the pay zone was then computed. The pay-zones for this estimation were grades greater than or equal to 0.5g/m^3 , however other areas with grades above 0.3g/m^3 may be mined in future when the initial reserve is exhausted. The volume of gravel within the pay zone was calculated as the product of the average gravel horizon thickness computed and the surface area of the delineated pay zone. The average gravel grade of the pay deposit was estimated as simple arithmetic average of all the pit gravel section grades within the pay zone. The quantity of gold in grams was then computed as the product of the average gravel grade of the pay zone and the volume of gravel estimated within the pay zone.

Table 4.7: Resource Table for Amenam Concession

PARAMETERS	QUANTITY	UNITS
Average Gold Grade (Avg)	0.92	g/m ³
Ore Thickness (gravel)	1.34	m
Average Overburden thickness	3.90	m
Total area of deposition	2,143,700.00	m ²
Total volume of ore (TVg)	2,872,558.00	m ³
Gold content (Potential Reserve)	2,642,753.36	g/m ³
Potential Reserve(oz)	84,967.80	oz
Gross Present value of ore (Base rate \$2,500)	212,419,490.08	USD

The estimation of the Amenam resource is based on a 100x100m spaced pitting program. Since much of the gold is fine, much larger samples are required to yield satisfactory results. These two factors favor mechanical pitting (excavator pits), which yields much larger samples. Based on the information and data obtained during the pitting program, the volume estimate of the potential reserves of gold- bearing gravels along the channel and valley flats of the Amenam grid, approximately **2,872,558m³**. This was estimated while considering a lower cut-off of 0.5g/m³. The total gold content is **2,642,753.36g/m³** and the total gold content in ounces is **84,967,80oz**. The cut-off gives room to expand the area of interest if a much lower cut-off is considered as shown in the table below.

Table 4.8 Resource Table for Gravel and Gold Content at the Amenam Alluvial Prospecting Area

Grid	Average Gravel Thickness (m)	Block Area in (Sqm)	Block Gravel Volume(m ³)	Average Grade ((g/m ³)	Gold Content (g)	Gold Content (oz)
Amenam East & West	1.34	2,143,700	2,872,558	0.92	2,642,753.36	84,967.80

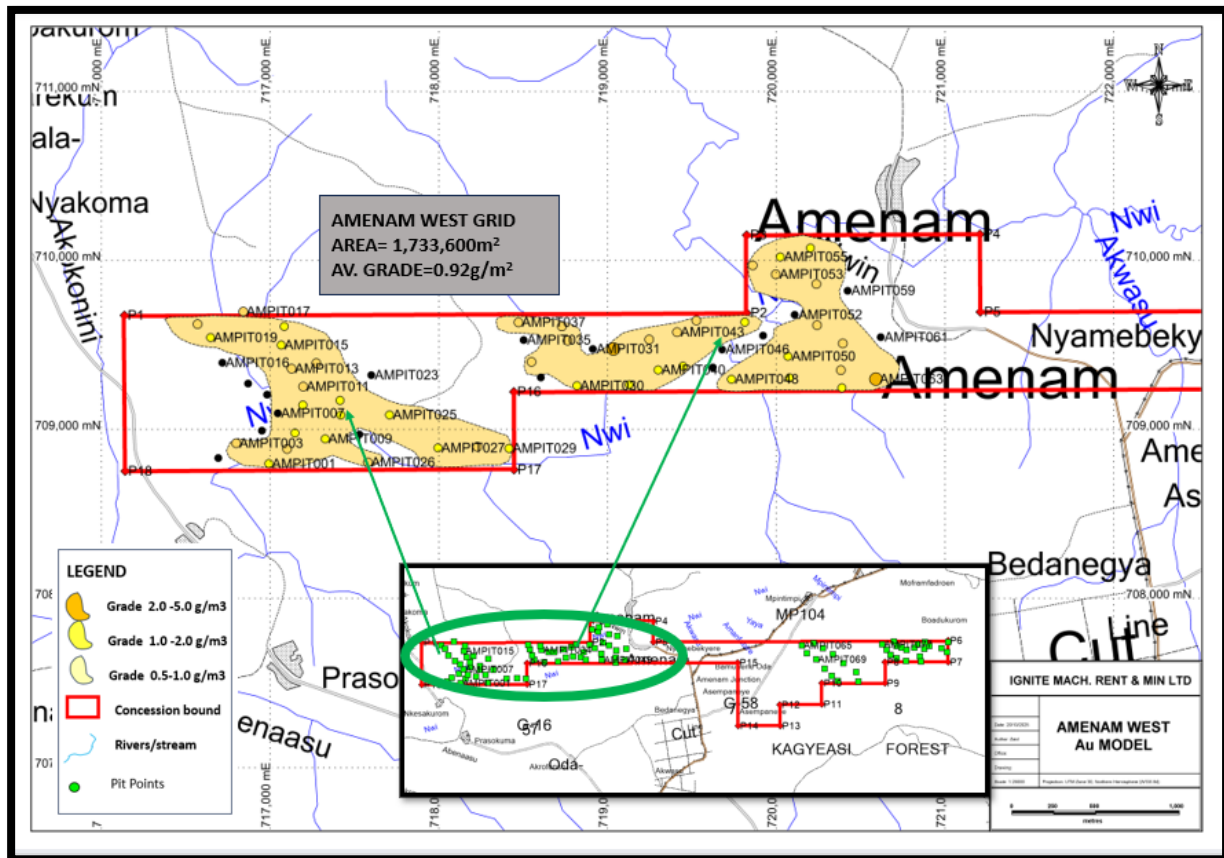


Figure 4.8: Model of Gold Anomaly Envelope-Amenam-West Grid

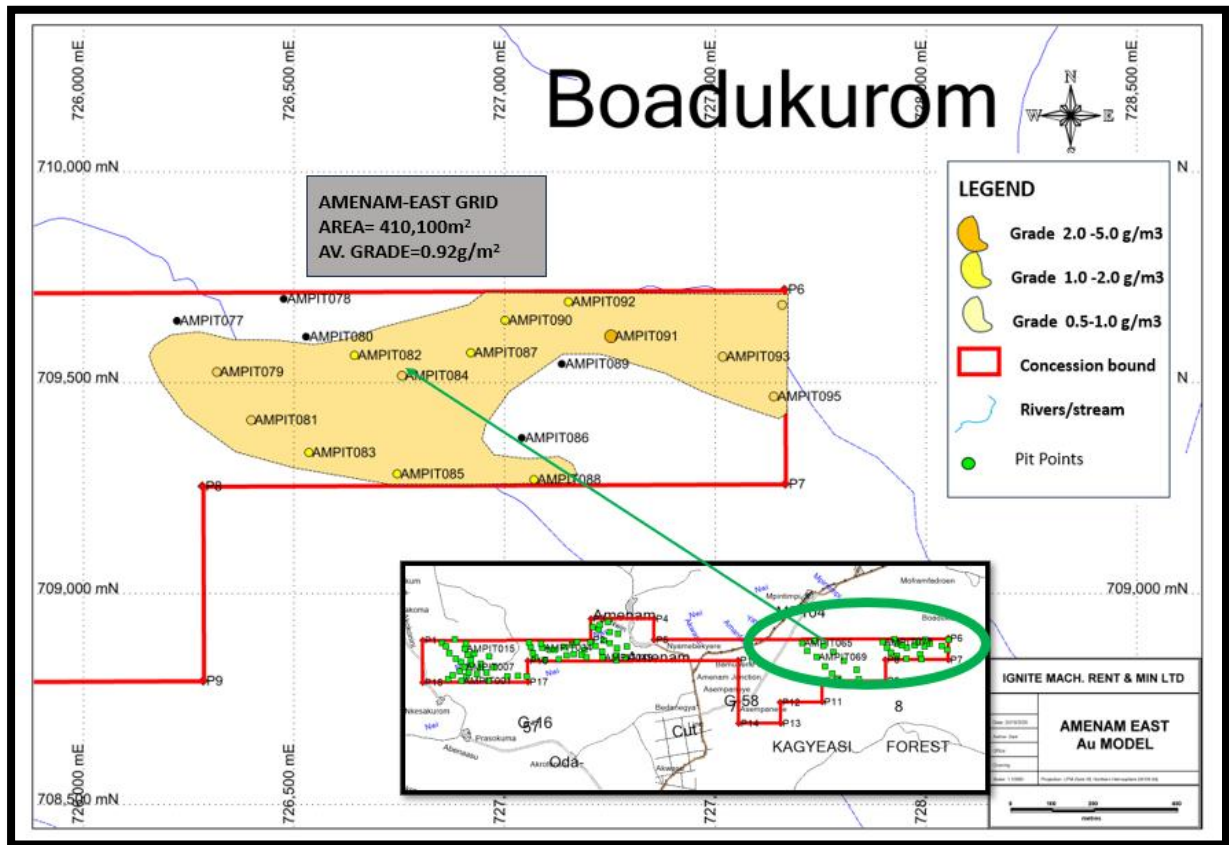


Figure 4.8: Model of Gold Anomaly Envelope-Amenam-West Grid

The alluvial gold evaluation for the entire project area, established the existence of gold- bearing gravels with an average gold grade of 0.92g/m^3 and an average gravel thickness estimated at 1.34m in pay-zone areas with a total area of $2,143,700\text{m}^2$. Based on these results, the potential gold-bearing gravel volume in the valley flats and channels of the Concession are estimated to be in excess of $2,872,558.0\text{m}^3$. The in-situ gold content in the gravel block is estimated to be in excess of $2,642,753.36\text{g/m}^3$ thus estimated in ounces as 84,967.80oz. The estimation was based on field data pitting projects carried out at the Amenam east and west grid on the Amenam concession.

4.5 ESTIMATION OF MINE LIFE

Total Volume of gold bearing gravel estimated in the concession = 2,872,558.00m³

Capacity of Trommel = 100m³/hr.

For 8 hours shift for a day, the total volume of gravel = 100m³/hr. x 8hrs = 8,00m³

For a period of six working days per week, 800m³ x 6 days = 4,800m³

For a period of four weeks per month, the total volume of gravel, 4,800 x 4wks = 19,200m³

Total volume of gravel for one year (11 months), 19,200m³ x 11 = 211,200m³

NB: Management of the company intends to operate a day and night shift

The total volume of ore processed for two shifts (16 hours per day) in a year is

$$211,200 \times 2 = 422,400.0\text{m}^3$$

Hence the mine life of the main plant unit of the Amenam Mine is

$$2,872,558\text{m}^3 \div 422,400.0 \text{ m}^3 = 6.8\text{years.} = 7\text{years}$$

Therefore, the mine life for Amenam property is approximately 7 years.

Table 4.9: Summary of Mineral Reserved for IGNITE MACHINERY LTD

Grid	Average Gravel Thickness (m)	Block Area in (Sqm)	Block Gravel Volume(m ³)	Average Grade (g/m ³)	Gold Content (g)	Gold Content (oz)	Gross present value of ore (\$) 2,200/oz
Total	1.34	2,143,700	2,872,558	0.92	2,642,753.36	84,967.80	212,419,490.08

Table 4.10: Summary of Mineral Reserved, Plant Equipment Setup and Mine Life for the Amenam Property

SUMMARY OF MINERAL RESERVED, PLANT EQUIPMENT SETUP AND MINE LIFE		
(PARAMETERS)	(QUANTITY)	UNITS
Total Volume of Gravel (m ³)	2,872,558.00	m ³
Plant Capacity	100.00	m ³ /H
Number of Plants	1.00	
Total Volume (M ³) of Gravel Washed/Shift-Day (8hrs)	800.00	m ³
Total Volume (m ³) of Gravel Washed Per Week (6 Working Days)	4,800.00	m ³
Total Volume (m ³) of Gravel Washed Per Month	19,200.00	m ³
Total Gravel Washed Per Year (Single Shift-11 months)	211,200.00	m ³
Total Gravel Washed Per Year (Day & Night Shift)	422,400.00	m ³
Mine Life (total volume/gravel washed per year)	6.80	years
Mine Life (Years)-(LOM)	Approx. 7 years	LOM

4.6 RESOURCE AND RESERVE POTENTIAL

The Potential Mineral Reserve of the entire Amenam concession is estimated to be 84,967.80oz.

The estimate was defined based on exploration results obtained from the alluvial pitting and sampling programme on the entire concession. The assessment involved the calculation of average thicknesses of gold-bearing gravels for the resource areas in order to establish the volume of gold-bearing gravels in Amenam concession.

The alluvial gold pitting and sampling program in the project area, established the existence of gold-bearing gravels with an average gold grade of 0.92g/m³. Based on these results, the potential gold bearing gravel volume in the entire concession is estimated to be in excess of 2,872,556.0m³.

CHAPTER 5

5.0 MINING PROJECT

The Amenam Project will adopt the Ignite Machinery practical method of exploiting the measured reserves, which is strip-mining, utilizing excavators for both removal of overburden and the excavation of the gold bearing gravels. The gold resources on the Amenam concession are primarily concentrated in the gravel horizon, with very little occurring in the saprolite of the bedrock. Before removing the waste, management of the Company intends to conduct rigorous grade control and sampling within the boundaries of the delineated ore zones during the mining operation.

An excavator will remove the lateritic clay overburden up to the top gravel contact, which is often indicated by a horizon of white marker sand. The removed overburden material will be stockpiled in anticipation of its eventual use for reclamation purposes. The primary mining machinery to be used are the bulldozer, excavator, and dump trucks.

In selecting a mining technique for the project, the following have been considered:

- The mining method should be technically feasible and economical.
- The mining method should maintain a high standard of safety for the workers and inhabitants.
- The mining should have minimal impact on the environment.

The grade of the deposit necessitates a high-capacity, low-cost mining method. All the above requirements are met by the mining method chosen. The Amenam Project will employ the Ignite Machinery practical method of exploiting the proven (measured) reserves, which is strip-mining with excavators for both overburden removal and excavation of the gold bearing gravels. The conditions for dry stripping techniques are ideal, allowing for a quick and cost-effective extraction rate.

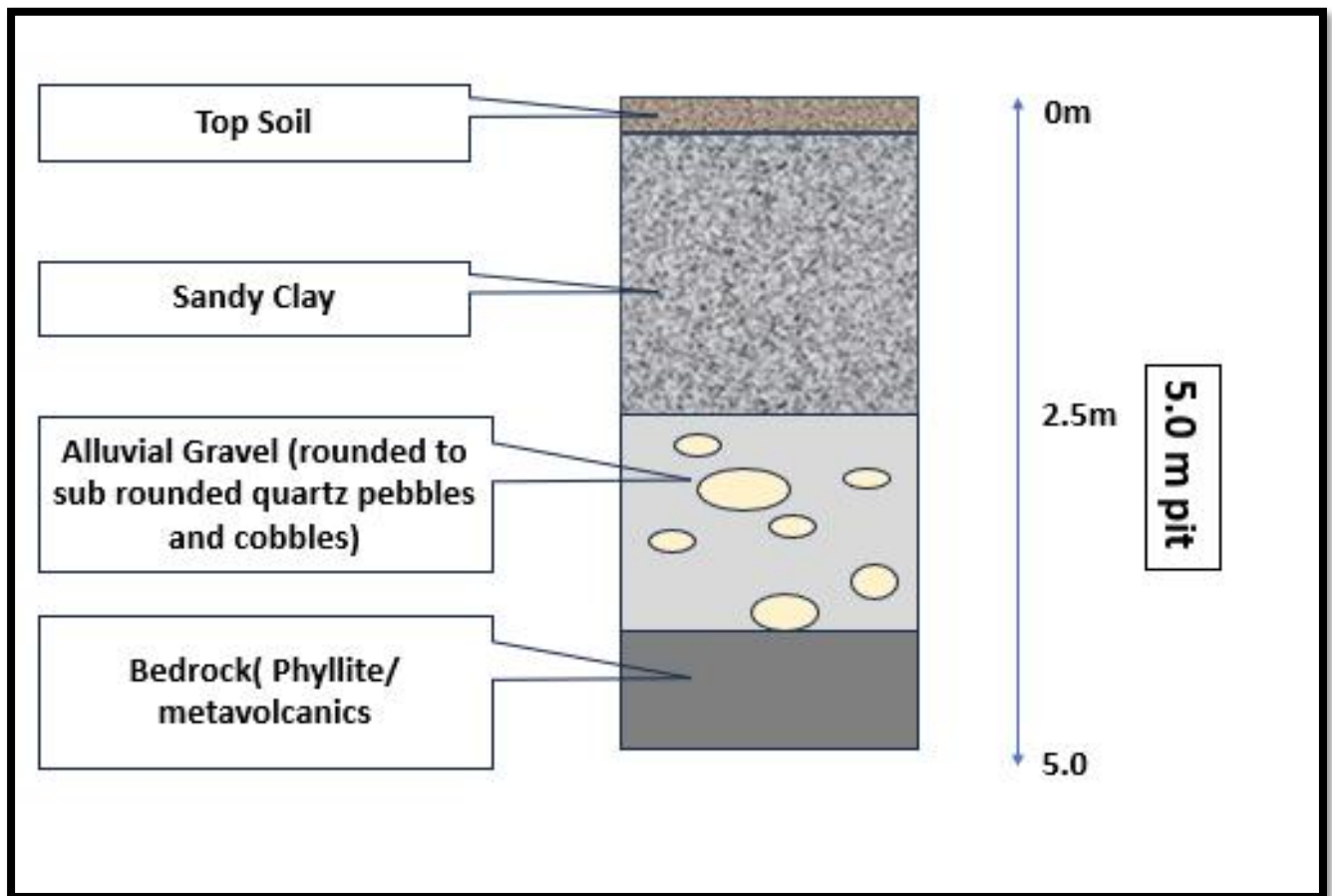


Figure 5.1. Generalized Alluvial profile mapped in the Amenam concession pits.

5.1 STRIP MINING CONCEPT

The Strip-mining approach is grounded in a tried-and-tested technique used for mining placer deposits that share similarities with the Amenam deposit. This method has demonstrated success in various locations in Ghana and other regions of West Africa over the years. The proposed strategy for extracting the confirmed (measured) reserves at the Amenam concession involves a phased approach, including:

- Surface Preparation
- Overburden Removal
- Excavation and loading of the ore
- Transportation of the ore to the processing plant
- Processing of the ore
- Restoration

5.1.1 Surface Preparation

In anticipation of subsequent mining activities, the initial phase involves the meticulous preparation of the mining site. To clear the area, a chain saw machine will be employed to fell trees. During this process, valuable timber species will be selectively collected for utilization in civil works within the mining area. The remaining timber will be cut into manageable pieces and dried, providing a sustainable source of firewood for local residents.

Furthermore, the surface preparation will entail the systematic removal and stockpiling of topsoil. This nutrient-rich layer, which serves as the foundation for the existing vegetation, will be strategically preserved for future re-vegetation efforts. The use of a Caterpillar D6 or D7, is deemed suitable for the efficient execution of this task. This method ensures a careful and environmentally conscious approach to site preparation, emphasizing the importance of sustainability in the mining process.

5.1.2 Overburden Removal

Overburden, the non-ore material overlaying the valuable ore deposits, must be systematically cleared to facilitate access to the desired resources. In our mining approach, the overburden will not be transported; instead, it will be strategically piled alongside the excavation for future utilization during the reclamation process.

The backhoe excavator is considered for the efficient removal of the overburden. This decision is based on a cost-effective approach. By opting for a backhoe and bulldozer combination, we ensure a balanced blend of efficiency and cost-effectiveness in managing the overburden removal process. This strategic choice aligns with our commitment to optimizing resource allocation while maintaining environmental stewardship throughout the mining operation.

5.1.3 Excavation and Loading of Ore

Following the successful removal of overburden, the ore is now exposed, marking the beginning of the excavation, and loading phase. Notably, some ore deposits may exhibit compact characteristics, demanding a substantial amount of force for effective excavation. Considering this, the selection of appropriate equipment becomes paramount.

Given the need for robust force in ore excavation and considering pit conditions, the backhoe excavator emerges as the preferred choice. This decision aligns with our commitment to operational

efficiency and cost-effectiveness. The backhoe excavator's versatility and higher breakout force make it well-suited for the challenges posed by compact ore deposits, ensuring a seamless and effective excavation and loading process.

5.1.4 Transportation of Ore

Transporting the gold-bearing gravel from the mining site to the processing plant is a crucial aspect of our operational plan. The haul distances may vary, spanning from a few hundred meters to several kilometers. To efficiently carry out this task, heavy-duty dump trucks have been strategically chosen for their robust capabilities and load-bearing capacity.

The selected heavy-duty dump trucks are designed to navigate diverse terrains, ensuring the secure and reliable transportation of the ore material. Their suitability for varying distances within our operational range positions them as the optimal choice for this crucial phase. This decision underscores our commitment to a well-planned logistics strategy, enabling the seamless and timely delivery of the gold-bearing gravel to the processing plant.

5.1.5 Processing of Ore

The processing of the extracted ore will be carried out through a systematic series of operations within a dedicated plant. This facility is equipped with essential components to ensure efficient recovery and extraction. The key elements of the processing plant include; Vibrating Grizzly, Scrubber cum Screen Trommel, Knelson Concentrators and a Gemini Shaking Table. This integrated approach to ore processing underscores our commitment to employing advanced technologies for optimal recovery and resource utilization.

5.1.6 Restoration

In an integrated approach to resource management, the processing plant tailings, composed of washed boulders, cobbles, pebbles, and sand, will serve a dual purpose. These tailings will be systematically conveyed to the mined-out areas, contributing to backfilling operations. Importantly, this backfilling process will run concurrently with the mining operations, ensuring a comprehensive and sustainable utilization of extracted materials. The logistical efficiency of this process is maximized by utilizing the same dump trucks responsible for transporting ore to the processing plant. On their return journey to the mining face, these trucks will be loaded with washed gravel or tailings, minimizing transportation-related overhead.

To facilitate this multifaceted operation, specialized equipment will be strategically positioned at the tailings site. This machinery not only loads dump trucks efficiently but also plays a crucial role in de-silting the sandpit. Considering the diverse responsibilities, an excavator with an extended boom and a light to medium weight profile emerges as the ideal choice for this task. This tailored approach aligns with our commitment to optimizing operational efficiency and environmental stewardship.



5.2 EQUIPMENT AND MATERIAL REQUIREMENTS

5.2.1 Mining Excavators

As previously outlined, a single backhoe excavator, complemented by a bulldozer, will play a pivotal role in both overburden removal and ore excavation and loading. The choice of excavator size is directly correlated with our planned production rate, a crucial factor in ensuring operational efficiency.

The treatment plant designated for use boasts a capacity of 100m³/h. Consequently, the selected excavator must possess the capability to strip overburden and load more than 100m³ of gravel within a one-hour time frame. Given a stripping ratio of 1:2, where stripping precedes mining to expose ore for extraction, an additional excavator, supported by a dozer, will be employed specifically for the stripping process ahead of mining operations.

To meet our production goals and create a sufficient ore stockpile for the treatment plant, the ore mining crew aims for a production rate of 100m³/h. The choice of the ore mining excavator will be made to achieve this targeted mining rate. The productivity of backhoe excavators, particularly under medium digging conditions, is outlined in Table 5.1, guiding our selection process for optimal performance and output. This meticulous approach to excavator selection aligns with our commitment to precision and efficiency in meeting production targets for a seamless mining operation.

Table 5.1 Productivity of Backhoe Excavators

Bucket Size (m ³)	Cycle Time T(sec)	Production Rate Q(m ³ /h)
0.9	17	79.9
1.0	17	87.8
1.2	18	98.7
1.4	19	107.7
1.6	19	121.8
1.7	20	129.9
2.3	21	162.8
2.8	22	186.7
3.7	24	227.2
4.6	26	262.5
9.1	27	205.5

For the required production rate, an excavator with a bucket size of 1.4 - 4.6 m³ will suffice. In this case, a CAT 345 excavator with 1.5 - 4.6 m³ bucket capacity or equivalent will be used.

5.2.2 Feeding Excavator

To efficiently feed the processing plant with a capacity of 100m³/h, an excavator with a minimum capacity of 2.5m³/min is required. Therefore, it is recommended to employ a CAT 345 excavator or an equivalent model for this essential function.

The selection of this specific excavator ensures not only the necessary capacity for plant feeding but also takes into consideration the operational efficiency and reliability associated with the CAT 345 or its equivalent. This strategic choice aligns with our commitment to using equipment that meets the demands of the processing plant while maintaining high standards of performance.

5.2.3 Tailings Handling Excavator

In adherence to our design considerations, a single excavator will be assigned the crucial tasks of de-silting the sandpit and loading all tailings material onto trucks. Effective desilting requires the excavator to have a relatively long boom for optimal reach and performance.

However, it is essential to strike a balance between functionality and weight to avoid sinking while cleaning the sandpit. In consideration of these requirements, both the CAT 320 and Daewoo 220 excavators are deemed suitable for this dual-purpose operation. Their design features, including a long boom for effective desilting, coupled with an optimal weight profile, make them well-suited for the demands of the task at hand.

This strategic choice ensures that our equipment is not only capable of efficient sandpit maintenance but also minimizes the risk of environmental impact, aligning with our commitment to responsible and sustainable mining practices.

5.3 TRUCKS

The choice of trucks for specific tasks is directly influenced by the size of the loader used in the excavation process. In this project, the proposed mining excavator is designed with a 2.5m³ bucket size. The volume estimation for a 2.5m³ excavator operating under optimum conditions is detailed in Table.

Table 5.2 Volume Estimation of Truck Loaders

No. of Passes	Volume (m ³)
6	15
7	17.5
8	20
9	22.5

From Table above, it is determined that 22.5 m³ CAT articulated trucks, requiring nine passes of the excavator per truckload, are a suitable choice for efficient material transportation.

The trucks will carry ore to the treatment plant and carry tailings on their return trip. Cycle time studies of similar trucks working under similar conditions in a similar mine in the country have an average cycle time of about 18 minutes. An average cycle time of 18min is therefore used to determine the number of trucks required to meet the production target.

Truck output formula is given as:

$$P = 60 * C * \frac{E}{T}$$

Where P = Truck Output (m³/h)
C= Truck Capacity (22.5 m³)
E= Truck Efficiency
T = Truck Total Cycle Time (18mins)

Assuming an Efficiency of 95%

$$P = 60 * 22.5 * \frac{0.95}{18}$$

$$P = 49m^3/h$$

$$Number\ of\ Trucks = \frac{\text{required production}}{\text{truck output}} = \frac{100m^3/h}{49m^3/h} = 2.04 \approx 2$$

Therefore, one trucks will be sufficient. However, to take care of maintenance schedules an additional truck will be required. Two (2) trucks will therefore be needed for the project. Table shows the machinery requirements for the operation.

Table shows the machinery requirements for the operation.

Table 5.3 Summary of Selected Equipment

Project Item	Quantity
CAT 14G Grader	1
CAT 345 Excavator	1
CAT 966F Front End Loader	1
CAT D6 Bulldozer	1
Dewatering Pump	2
Dump Trucks	2
Water Tanker	1
Generator	4
Gold Trommel Wash Plant	1
Nissan Bus	2
Office Equipment	1
Office furniture & fittings	1
Pick Up (4 X 4)	4
Station Wagon	4
Tractor Loader Backhoe	1
Utility Truck	2

5.4 MINE SAFETY

5.4.1 Geotechnical Considerations

The alluvial gold is embedded within loose gravel, predominantly composed of quartz pebbles and cobbles cemented in a clayey sand matrix. The extraction of these unconsolidated gravels for treatment in the washing plant does not involve hard rock drilling or blasting with dynamite. To uphold the highest standards in geotechnical aspects, the company has implemented comprehensive procedures and protocols in collaboration with a private geotechnical consultant. This ensures the effective management of alluvial surface mining operations.

Monitoring and Deformation: Rigorous geotechnical monitoring is conducted on pit wall faces, upper slopes, and accessible benches to assess their rates of deformation. This proactive approach helps in identifying and addressing potential stability issues promptly.

Settling Pond Construction: The sites chosen for the construction of settling ponds are underlain by weathered sedimentary rocks covered by overburden. To guarantee the stability and longevity of the settling ponds, a thorough foundation investigation will be carried out. This investigation aims to identify and analyze key geotechnical factors that need consideration during the construction phase.

Key Considerations:

1. **Slope Stability:** Ensuring the stability of the side walls of the ponds is of paramount importance. Rigorous analysis and reinforcement measures will be implemented to mitigate the risk of slope instability.
2. **Water Seepage Prevention:** Preventing potential water seepage through buffer zones during heavy rainfall is a critical factor. Robust engineering solutions will be employed to safeguard against any adverse effects of water seepage.
3. By integrating these geotechnical considerations into the mining and construction processes, the project aims to ensure the safety, environmental and long-term viability of the project.

5.4.2 Dewatering and Surface Run-off Management

Processed water will be recycled as required to meet discharge permit regulations and support mining operations during low precipitation periods, ensuring a zero-discharge operation at the project site. Collected water includes precipitation pumped from the mine, direct precipitation on the dust control pond, Bulk Storage Reservoir, off-channel storage reservoir, and dewatering well yields.

To manage inflows to the Tailings Storage Facility (TSF), tailings water and runoff will be directed to the Bulk Storage Reservoir, minimizing water storage in the TSF. Losses include water retained in tailings, evaporation, and seepage. The remaining water is recirculated to the processing plant via the Bulk Storage Reservoir, with recirculation from the tailing's thickener. Because the tailings will be non-segregating thickened tailings, the volume of water available from tailings storage for return to the process plant is significantly reduced in the later years of production.

A portion of this flow will meet processing plant freshwater needs, and excess water from wells can be used for process water or dust control. Permission from the Water Resources Commission (WRC) will be sought by the company if drawing from water bodies is necessary to maintain operational flow balance.

Mine-pumped precipitation is stored in the dust control pond. Any deficit for dust control is compensated with water from dewatering wells or the off-channel reservoir. In a wet year, excess runoff can be held in pit sumps, and unused water can be directed to the process plant.

The main sources of water inflow into the pit will be ground water filtration and surface run-off. Oversize tailings from the plant as well as overburden from the next cut will be dumped into the previous cut. This will be done immediately after mining to ensure minimum ground water seepage into the pit. However, during heavy downpours or mining at areas with high seepage of water, all water will be directed in the lowest point created by deepening the first bite from which the excess water will be pumped into the settling ponds. Surface run-off will be prevented from entering the pit by constructing small berms using overburden material around the perimeter of the pit to prevent excessive water entering the pits during rainfalls.

Alternatively, the preferred and least expensive option of dewatering a pit is to divert the water off the pit to a safe place downstream. Management of the company considers the construction of temporary drainage system 5-15 m behind the high wall to intersect storm water from hills and groundwater shedding from swamps into the pit. The water is then collected by gravity into a sump and discharged safely to a downstream water source. The company has conducted an estimate for sump dewatering based on the pump capacities and the effective pumping hours

5.5 PRE-STRIPPING

Topsoil, a precious resource considering its scarcity, should be extracted separately, stockpiled, and subsequently replaced as a surface covering during reclamation and restoration efforts whenever feasible. Placing the stockpile as close to the mining block as possible, without impeding subsequent operations, is crucial. It should remain in its designated location until the land is fully reclaimed. To preserve the quality and fertility of the soil source, minimizing topsoil stockpiling is essential to prevent compaction and reduced fertility. Soil handling, removal, and re-spreading should only be carried out in dry conditions.

Caution must be exercised during gravel excavation to minimize dilution. Rigorous monitoring and continuous sampling are imperative to keep under/over mining to a minimum, especially in regions with undulating bedrock. Expect higher gold values within the base gravel horizon, making it necessary to include a minimal amount of bedrock with the bottom gravels.

5.6 MINING (OVERBURDEN REMOVAL AND GRAVEL EXTRACTION)

The excavator is preferred due to its adaptability and resilience in handling challenging ground conditions. Its short cycle time and versatility extend its utility for general tasks across the mining site. The selected CAT 345, equipped with a 4.6m³ bucket, aligns perfectly with these operational needs. For the efficient transportation of gravel to the processing plant, the recommendation is for all-wheel drive articulated trucks. It's envisioned that these trucks will double their utility by transporting tailings back to the mined-out areas on their return trip. To maintain continual operations, four trucks are initially proposed, with a prudent addition of an extra unit to account for breakdowns and maintenance considerations.

The bulldozer takes on a pivotal role, primarily involved in clearing, grubbing, constructing access roads, and waste management. Anticipating wet conditions, the recommendation is for a Caterpillar D6 or its equivalent, featuring low ground pressure (LGP) undercarriage and extensive width tracks. This dozer also plays a complementary role, working alongside the excavator in the crucial task of stripping ahead of mining operations.

The machinery selection, including the excavator, trucks, and bulldozer, is meticulously tailored to ensure operational efficiency and adaptability to varying mining conditions. The inclusion of an extra truck as a backup underscores the commitment to uninterrupted operations, accounting for potential breakdowns and necessary maintenance.

5.7 LIFE OF MINE

Operational Parameters

Guided by process test work, the gravity separation processing plant will operate at a feed rate of 100m³/h to ensure a buffer against equipment breakdowns and weather uncertainties.

Total Volume of gold bearing gravel estimated in the concession = 2,872,558.00m³

Capacity of Trommel = 100m³/hr.

For 8 hours shift for a day, the total volume of gravel = 100m³/hr. x 8hrs = 8,00m³

For a period of six working days per week, 800m³ x 6 days = 4,800m³

For a period of four weeks per month, the total volume of gravel, 4,800 x 4wks = 19,200m³

Total volume of gravel for one year (11 months), 19,200m³ x 11 = 211,200m³

NB: Management of the company intends to operate a day and night shift

The total volume of ore processed for two shifts (16 hours per day) in a year is

$$211,200 \times 2 = 422,400.0\text{m}^3$$

Hence the mine life of the main plant unit of the Amenam Mine is

$$2,872,558\text{m}^3 \div 422,400.0 \text{ m}^3 = 6.8\text{years.} = 7\text{years}$$

Therefore, the mine life for Amenam property is approximately 7 years.

Life of Mine Calculation:

$$\text{Life of Mine(LOM)} = \frac{\text{Reserve}}{\text{Plant Annual Throughput}} = \frac{2,872,558\text{m}^3}{422,400 \text{ m}^3} = 7\text{years}$$

As common industry practice, ongoing intensified exploration of the property is anticipated. Any successes in exploration efforts have the potential to extend the Life of Mine beyond the initial estimate. The Company's strategic planning considers operational parameters, maintenance schedules, and the potential impact of ongoing exploration activities, demonstrating a proactive approach to project longevity and adaptability to industry trends.

CHAPTER 6

6.0 MINERAL PROCESSING

6.1 GRAVEL PARTICLE SIZE DISTRIBUTION

Samples of gravel sourced from specific sections of the river systems in the baseline areas underwent a meticulous process. The process included drying, manual sorting, and screening to establish approximate particle sizes of the gravels. The results are presented in Table below.

For the material extracted from the pits at each 0.5m interval, volumes were measured using a wooden box with a capacity of 0.1m³ (0.5x0.5x0.4m). Subsequently, the measured material was carefully stored on a plastic sheet. Materials excavated from the water table consistently exhibited a swell factor of approximately 20%. In contrast, materials above the water table demonstrated a broader range, spanning from 30% to 70%.

These findings provide valuable insights into the characteristics of the extracted gravels, crucial for effective mineral processing and subsequent gold recovery efforts.

Table 6.1 Distribution of Gold Bearing Gravel

Screen Size (mm)	% Fraction
+100	11
+50	23
+6	20
-6	18
Clay/Silt	28

From the analysis, it can be noted that 60% of the gravel is expected to fall in the +6mm range. The clay content was found to be sandy clay matrix.

6.1.1 Gold Particle Size Distribution

Gold particle size test was performed on the gold recovered from the test pits at the Ghana Geological Survey Authority in Accra. The gold is generally medium to fine grained. Results from the test are summarized in Table below.

Table 6.2 Distribution of Gold Particle Sizes

Appearance	Sieve Mesh Size	Gold Particles (%)
Coarse	+20	4
Medium	+30	30
Fine	+60	46
Very Fine	+150	8
Flour Gold	-150	12

The breakdown shows a distribution of 80% plus 60 mesh and plus 30 mesh sizes together.

6.1.2 Gold Fineness Determination

The gold specks were digested with aqua-regia and the subsequent determination of the grade was achieved through Atomic Absorption Spectroscopy (AAS). The outcome revealed a gold purity of 90%, equivalent to 21.72 carats. Notably, silver emerged as the predominant impurity, constituting 4% of the composition. It's imperative to acknowledge that without conducting broader and more comprehensive testing, the exact percentage of minerals present in the saleable concentrate remains unknown, underscoring the importance of future analyses for a more detailed understanding.

6.1.3 Gold Liberation Characteristics

Within the confines of the Amenam concession, the gold particles stand as liberated entities, a state often referred to as free gold. Notably, the predominant presence of small quartz inclusions nestled within larger nuggets serves as the primary source of locked gangue particles. Encouragingly, the gangue encapsulated within these larger nugget particles can be readily separated and removed during the subsequent smelting process. This liberation characteristic bodes well for the efficient extraction and processing of gold from the geological matrix.

6.1.4 Pre-Concentration Amenability of Ore

The gravel within the Amenam concession exhibits favorable characteristics for pre-concentration. A comprehensive gravel particle size analysis reveals that 60% of the gravel falls within the +6mm range. The pre-concentration process will be seamlessly integrated into the washing and screening operations. This strategic approach involves the removal of all particles larger than the anticipated

gold size, effectively enhancing the grade of the remaining material. The incorporation of pre-concentration techniques is a pivotal step, ensuring the optimization of ore processing and the extraction of valuable gold resources.

6.1.5 Gravel Washing Characteristics (Gravel Washability)

The washability of gravel serves as a critical indicator of its susceptibility to washing. Generally, the higher the clay content in the gravel, the more challenging it becomes to wash. On the Amenam concession, the presence of a sandy clay matrix with minimal clay content renders it free washing.

Recognizing the tendency of clay to entrap gold particles, a proactive approach is taken to prevent such entrapment. A clay blaster, an integral component of the washing plant, will be employed. This specialized equipment ensures thorough washing and subsequent separation of the ore into its constituent minerals. To enhance the washing process, lifter bars will be incorporated into the Trommel/Scrubber, generating agitation and a cascading effect.

Efficient washing is further guaranteed by the application of a high volume of spray water in a counterflow manner to the screen surface. The residence time within the Trommel ensures the complete liberation of particles, setting the stage for the ensuing gravity separation process. This multifaceted approach underscores the commitment to effective gravel washing, a pivotal step in optimizing gold recovery.

6.2 PLANT DESIGN

The design of alluvial gold treatment plant must take many factors into consideration in order to achieve an optimum recovery of gold. Among the factors taking into consideration are:

- Characteristics of ore to be processed.
- Size distribution of ore and gold particles. Fractions of the gravel reporting to the various sections of the plant.
- Volume of ore/gravel to be processed per day to recover gold to maintain economic viability for the project.
- The effective maximum and minimum sizes of the liberated gold particles and their distribution in the ore.
- Optimum operating conditions of the gravity separating equipment.
- Infrastructure requirement
- Robustness of the final design regarding operating conditions.

6.2.1 Selection of Processing Plant

The gravel and gold within the Amenam concession exhibit distinct characteristics, including the presence of free washing and normal washing gravel, as well as a gold particle size spectrum ranging from fine to coarse grains. Based on these specific attributes, the chosen method for treatment involves the utilization of sluice boxes, complemented by further concentration through a 7.5” Knelson Concentrator.

The treatment process begins with the initial washing and screening of the gravel before directing it to the sluice boxes for gravity concentration. Sluice boxes stand out as a prevalent gravity separation device in the industry, valued for their simplicity and effectiveness in heavy mineral recovery. They prove efficient in recovering gold particles ranging from 25mm to 100 microns. However, the efficacy of gold recovery varies based on operational practices, with well-operated sluices achieving recovery rates between 80 to 100%.

The chosen PROLINE sluice box design has been successfully employed in similar projects with comparable gravel compositions within the country. Noteworthy advantages of the PROLINE design include:

- Fast Waste Flush: Efficient removal of sands and gravels.
- High Retention: Excellent retention of all heavier minerals.
- Variable Flow Adjustment: Allows for adjustment between fast and slow waterways.
- Water Stop Feature: Facilitates clean-up or quick assessment of current gold levels in the sluice.
- Selective Capture Zones: Captures coarse gold in lower flow zones, while fine gold recovery is enhanced by adjusting the flow rate.

The PROLINE design operates by creating a continuously lowering water pressure zone, enabling the sequential settling of gold particles of varying sizes and weights at different intervals. This approach ensures a constant capture system throughout the entire length of the sluice box, optimizing the recovery of gold from the diverse particle sizes present in the gravel.

6.2.2 Processing Plant Flow Sheet

The flow sheet for the Amenam project has been designed to ensure the optimum recovery of gold from the treatment plant. This comprehensive design incorporates insights gained from on-site gravel assessments, logistical considerations in the operational area, and valuable input from seasoned

professionals with extensive practical experience in plant design. The culmination of these factors ensures a robust flow sheet aligned with the specific characteristics of the ore.

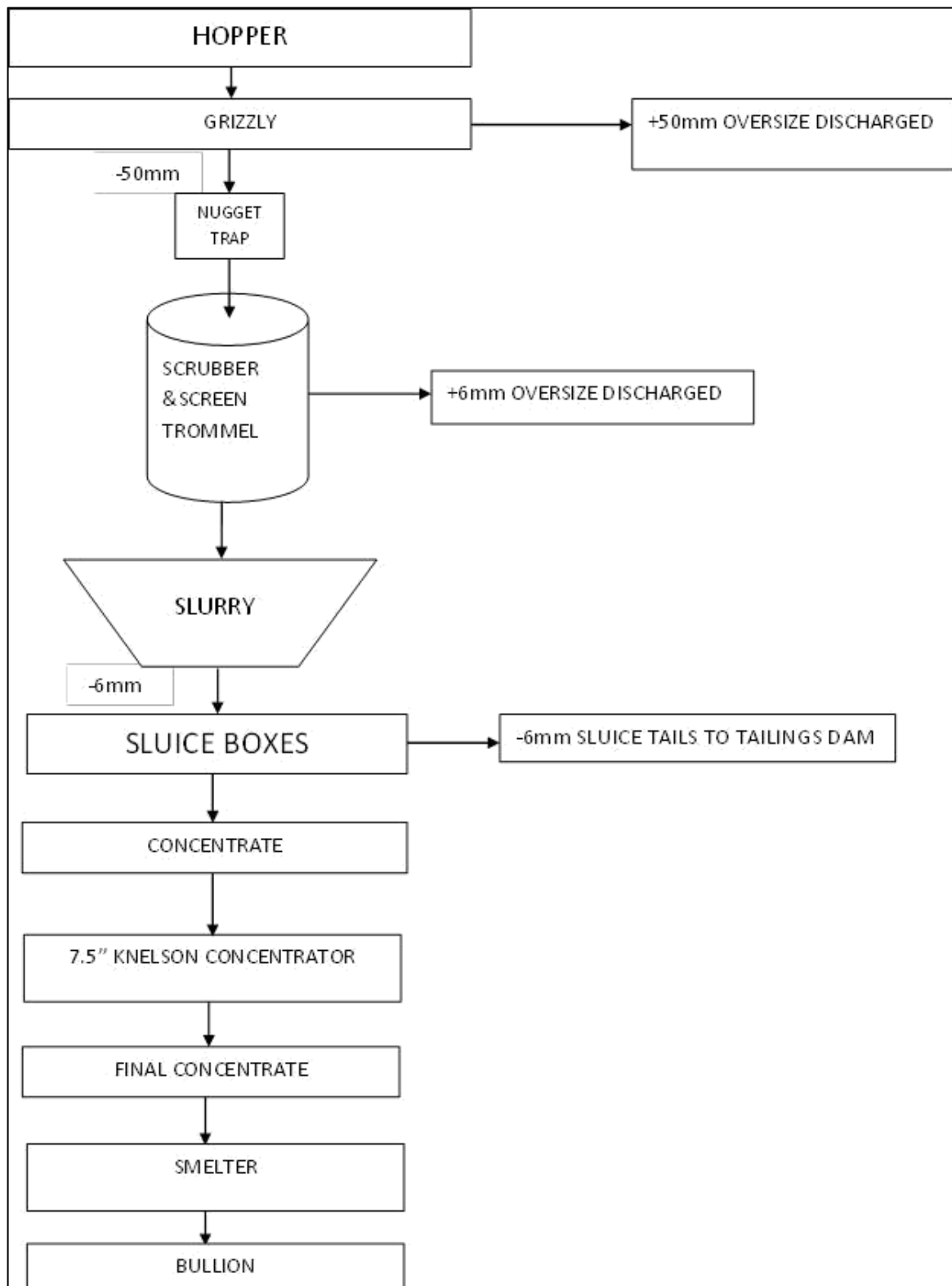


Figure 6.1: Proposed Plant Flow Sheet

6.2.3 Processing Plant Flow Sheet Description

The intricately designed flow sheet for the Amenam Project Processing Plant, illustrated in Figure above, outlines a systematic approach for optimal gold recovery. The plant's ore intake involves direct dumping from 22.5m³ trucks at the mining face into the feed hopper or, alternatively, feeding from a stockpile using a Front-End-Loader. To minimize truck idle time, an ore stockpile located close to the feed hopper ensures a seamless operational flow.

To regulate the ore flow onto the grizzly and prevent uncontrolled spillage, chains are integrated into the hopper. An 8" giant water monitor facilitates an even feed rate of 120m³/hr. onto the grizzly, with water from the monitor serving as added moisture at a rate of 750 gal/min. Positioned at the grizzly's edge, a clay blaster tackles clayey ore. The +50mm fraction is conveyed to an oversize stockpile, while the undersize -50mm progresses through the scrubbing trommel and into the screen trommel.

A nugget trap strategically placed at the trommel entrance captures gold nuggets. The +6mm fraction from the screen trommel is directed to a conveyor and subsequently discharged to the oversize stockpile. The +50mm and +6mm oversize material are stored for transport back to the mined-out pits on return trips of trucks. The -6mm fraction (undersize of the screen trommel) gravitates into a slurry tank, flowing through 18" pipes to three sluice box units.

From the 18" pipe, the material is divided into three 14" pipes, each leading to a sluice box. Control valves on the 14" pipes enable selective shut-off during sluice box cleaning intervals (twice per shift, i.e., every 4 hours). Sluice tails follow gravity to a three-stage settling pond. The concentrate extracted from the sluice boxes undergoes an upgrade process through a 7.5" Falcon Concentrator. Subsequently, the concentrated material is smelted to produce gold bars. Power, lighting, and other energy requirements for the operations are supplied by a 550 KVA diesel generator.

6.2.4 Process Water

The treatment plant's process water requirements will be sourced from the four-stage tailings settling ponds post-decanting, ensuring an eco-friendly approach with minimal discharge into the environment.

The initial supply of process water will be facilitated by an 8” diesel pump drawing from an off-channel reservoir, strategically constructed from one of the tributaries within the nearby Pamu river drainage system, secured with the necessary permits from the Water Resources Commission. Subsequent process water needs will predominantly be met by the settling pond. Water loss is anticipated due to evaporation and spillage at the plant. Supplemental makeup water will be drawn from the nearby river, tributary/streams as required.

The tailings and recycle ponds will be ingeniously constructed from the initial excavation of mining pits, serving as reservoirs for slurry tailings, pit water, and runoff during mining operations. This susBirim Northable water management approach aligns with environmental best practices, ensuring responsible use and conservation of water resources.

6.2.5 Tailings Management System

The processing plant at the Amenam Project will generate two types of waste: oversize material (derived from the grizzly and screen trommel) and tailings from the sluice boxes. Oversize material will be conveyed and stacked for reclaiming mined-out pits and constructing haul roads. The tailings from the sluice boxes will undergo gravity discharge into a three-stage settling pond system, a design integral to the initial excavation of mining pits. The pond walls will be constructed with sufficient width to serve as functional haul roads.

The three-stage settling pond arrangement is meticulously designed to facilitate the settling of coarse particles in the first pond, where coarse sand pebbles will settle near the sluice discharge area. This material will be periodically excavated for reclamation purposes. The second pond, connected via a weir, allows for continued settling of suspended particles, ensuring fines settle during the gravity flow through the arrangement. Clean water, separated from suspended particles, will flow to the third pond through another weir.

The third pond will be interconnected with the off-channel reservoir. Utilizing gravity, clean water from the third pond will be directed to the reservoir. Subsequently, this water will be pumped back to the treatment plant for efficient reuse, emphasizing a sustainable approach to water resource management within the project.

6.3 INFRASTRUCTURE

The Amenam township has not seen much infrastructure in terms of schools, hospitals, tarred roads, police station etc. but some fundamental facilities including potable drinking water, electricity, and a good mobile communication and internet services are available. During the construction phase of the mining site, essential physical structures pivotal for the seamless implementation of the project will be established. The key facilities to be erected on-site include: Operating plant (Washing plant site with equipment)

1. Operating Plant (Washing Plant Site with Equipment)
2. Maintenance Workshop and Warehouse
3. Fuel Storage Facilities
4. Water Storage Facility
5. Office Accommodation
6. Washroom and Toilet Facilities
7. Changing Room
8. Canteen
9. Site Clinic

6.3.1 Plant Area Buildings

The warehouse will be built of blocks and used for the storage of spare parts, lubricants, tagged instruments, paints and general supplies that will be needed for welding works, mechanical, electrical works, civil works etc. It will also store supply equipment such as generators, water pumps, compressors, welding machines, conveyor parts etc. Flammable materials such as lubricants, thinner and paints will be stored in accordance with their environmental and Material Safety Data Sheet (MSDS) information. As a safety measure, no hazardous material may come to the project site without the Company securing MSDS information and ensuring that the required storage and handling techniques are adhered to.

The facility will have an office complex which consists of offices for administrative work, a washroom, workers' canteen and first aid facilities for workers.

6.3.2 Energy Supply

- a. Power Supply Requirements: The power supply for the mining operations will be obtained from the National Grid for which the necessary arrangements for the connection will be made with the Electricity Company of Ghana. The main consumer of electric power at the mine site will be the gravel treatment plant. Other power outlets will be offices, stores,

security lights at strategic places like the open pits, fuel and lubricant storage depots and the main gate to the site.

- b. Power Supply Option: Apart from the high- tension power supply to the mine site, stand-by generator sets (2) will be procured and installed at the site for use during external power failures. Power consumption specifications are as follows:
Main High-Tension Power = 340 KW
Maximum Operating Power = 280 KW
Average operating Power = 240KW
- c. Slimes Disposal: A power generating set will be provided for the pumping of slimes for safe disposal at constructed dam sites away from water bodies. The generating set will be rated at 200 KVA.
Connected power supply = 130 KW
Average Operating Power = 110 KW
- d. Lighting of Mine Area: Generator rating = 7.9 KVA
- e. Dewatering of Mine Pits: Pump engine rating = 11 KW

6.3.3 Fuel Storage Facilities

Diesel fuel will be the main fuel to be consumed on site for fueling the standby generator and other trucks. Diesel storage tank will be provided on site. The fuel storage area will be constructed with concrete base to prevent leakage into the ground in the event of a spill during fueling.

6.3.4 Water Supply

- a. Operational Water Supply: Water supply to the treatment plant will be obtained from an intake point located on the banks of the streams upon attaining water use permit from the Water Resources Commission in accordance with the laws of Ghana. A mobile generator will be placed at the intake point for the pumping of water into the settling ponds from where other pumps will work to deliver water to the treatment plant for gravel washing. Dust control on accesses, haulage roads and aggregate areas will be catered for by pumping water from the streams into tankers for suppression purposes. The pumps are rated at 200KVA.
Connected power supply = 130 KW
Average Operating Power = 110 KW
- b. Raw Water Supply: The streams and rivers within the concession will be the main source of raw water supply for the gravel washing plant of the proposed Amenam Mine.

- c. Potable Water: Generally, IGNITE MACHINERY Ltd will source potable water from Ghana water company into polytanks for use at the site. Additional boreholes will be constructed on the mining site as alternative source of water to the site office buildings for drinking and cooking in the canteen.
- d. Sewerage System: Plumbing works will be done to ensure efficient discharge of sewerage from the washrooms to the manholes to be constructed at the site. Waste water from the workers' canteen and dirty oil and lubricants will appropriately be discharged through drains into a special container for collection and disposal in an environmentally safe manner. All these measures will be taken to prevent the pollution of water bodies in the area.

6.3.5 Tailings Storage Facility (TSF)

Provision has been made for the construction of a Tailings Storage Facility at the site. Waste gravel with differently sized material will be stored for future uses in back-filling the mined-out pits. Fine tailings will also be stockpiled separately to be used in the future land reclamation programme. The TSF type will be shallow open pits to be constructed at the mine site. The tailings collected from the processing plant will be stockpiled here for use.

6.3.6 Communications

The company will make available a two-way radio system for the mining, plant and security departments. Other essential departments will also be provided with the two-way radios. General communication will be via cellular telephone to all staff. Vodafone, MTN, Airtel-Tigo are available in the area. Credits for cellular phones will be provided for management staff, supervisors and other essential staff. Modems of the various mobile phone companies will also be used for internet connections.

6.3.7 Accommodation

The Company plans to construct modest houses to provide residential accommodation for and local senior staff of the mining project. Since the Company will largely employ its labour force from the catchment area, it will assist the junior workers to rent accommodation from the nearby towns and villages to save costs. Transportation arrangement will be made in such instances.

6.3.8 Security

The security of the mining operations will be ensured through putting in place adequate safety and protective measures. A security company with trained personnel will be engaged to oversee the security arrangements of the mining activities. A security post will be constructed at the main entrance to the mine site and manned for 24 hours on rotational basis. The washing plant and office block areas will be protected with a wire mesh fence. The key places at the site that will be kept under security surveillance always include: the gold room and the open pits.

CHAPTER 7

7.0 HUMAN RESOURCE REQUIREMENTS

The human requirement for this project is as follows in Table below. Ignite Machinery of the employees apart from the key technical personnel will be sourced from the local communities. If there are technical personnel from local communities who meet the competence requirements, they will be given priority during employment.

Table 7.1: Human Resource Requirements

Resource Item	Number-Ghanaian	Number-Other National
Managing Director	1	-
Administration and Finance Manager	1	-
Secretary	2	-
Mine manager	1	-
Project Geologist	1	-
Accounts Officer	2	-
Pit Foreman	2	-
Administration Officer	1	-
Metallurgist	1	-
Metallurgist Assistant	1	-
Mechanical Engineer		
Environment, Health & Safety Officer	1	-
Chief Security	6	-
Mine surveyor	1	-
Transport Officer	1	-
Electrical Officer	1	-
Plant Labour	2	-
Operators	6	-
Auto Technician	2	-
Cook	2	-
Mining Labour	20	-
TOTAL	55	

ORGANIZATIONAL STRUCTURE FOR MINING OPTIMIZATION SOULUTIONS LIMITED

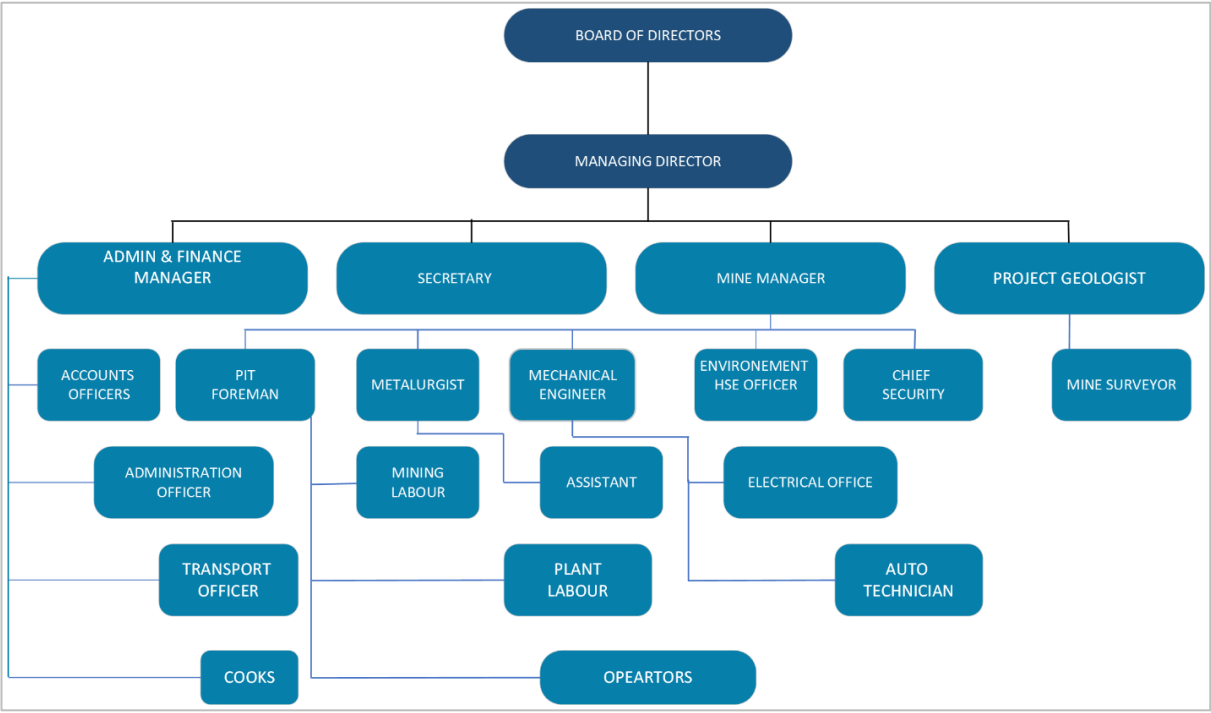


Figure 7.1: Organizational Structure



CHAPTER 8

8.0 ENVIRONMENT

An Environmental Impact Assessment (EIA) for developments, projects, or undertakings has been a requirement in Ghana since 1989. In June 1995, the Ghana Environmental Protection Agency (EPA) established new procedures for EIAs involving gradual phases depending upon the nature, complexity, and location of the undertaking (Ghana Environmental Impact Assessment Procedures, 1995). Between 1995 and 1999, the EPA reviewed and revised these procedures. In June 1999, the revised procedures were adopted and passed by Parliament as Legislative Instrument 1652 Environmental Assessment Regulations (L.I. 1652). These procedures require that an Environmental Impact Statement (EIS) be submitted to the EPA for review and be approved to obtain an Environmental Permit, which allows a project to proceed on environmental grounds.

8.1 Summary of the Baseline Information

The topography of the concession area is undulating and mountainous. The land generally rises from 60m above sea level.

The project area is drained by the Nwi River and its tributaries.

8.2 Air Quality

The Project area is entirely rural and ambient air quality is good. Particulate pollution from the seasonal dust laden Harmattan winds occurs principally during the three months of the dry season, from December to February. During this period, it is not exceptional to record a background level of total inhalable dust exceeding the maximum 24 hours peak of 500 mg/m³ quoted by the World Health Organization (WHO).

The project area is entirely rural with relatively good background air quality. Impacts prior to mining was primarily related to smoke from slash/burn agricultural practices, charcoal manufacturing, and fires set naturally or by other human-related activities such as hunting. Background dust impacts were primarily related to vehicular traffic along the Ignite Machinery unpaved road network within the project area and the dust laden harmattan winds occurring annually from December through February.

8.3 Surface Water

Generally, surface water will be diverted around areas of disturbance to avoid impacting water quality. Surface water flows into the Red Volta basin running through the center of the concession, proximal sub-basins are impacted due to the presence of the tailing's storage facility and water storage facility.

8.4 Flora and Fauna

The vegetation is Sudan Savanna consisting of short drought and fire-resistant deciduous trees interspersed with open savanna grassland. Grass is very sparse and in Ignite Machinery areas the land is bare.

Mining activities throughout the life of mine, from exploration and development to post-closure, impact social and environmental systems, including ecosystem services (ES) (Neves et al., 2016) which contribute to human wellbeing (Crossman et al., 2013; MEA, 2005).

As a result, the company will assess, analyze and manage threats to ecosystem included in mining industry performance standards (ICMM, 2020).

8.5 Dust Generation

Vehicular movement on gravel roads may lead to generation of dust within the working communities. Frequent exposure to dust may result in lung related disease. The company will ensure that its vehicles are driven within speed limits in the host communities. During drilling, dust suppression equipment will be used on rigs to minimize exposure of workers to dust. This will be augmented with respiratory protection to be worn as Personal Protective Equipment (PPE).

8.6 Noise

Drilling sites are noisy places, as the use of rigs and heavy machinery generate a lot of noise with potential for hearing damage. To protect workers against noise, the company will evaluate working conditions and noise exposure through risk assessments. Avoiding and reducing exposure can be achieved by applying engineering controls at the noise source or along the noise path to reduce exposures. Regular maintenance of vehicles and machines is also essential to reducing noise. The company will ensure proper use of personal hearing protection amongst noise-exposed employees, while providing necessary health and safety training and maintaining up-to-date health surveillance records.

8.7 UV Exposure

Exploration is often carried out under harsh weather conditions. Exploration workers often spend whole days out in the baking hot sun. Thus, understanding the risk of over-exposure to UV (ultraviolet) radiation in sunlight is essential as it can expose workers to risk of skin cancer or cause serious damage to the eyes if protective glasses are not worn. In the short-term, overexposure to the sun can cause dehydration, headaches and nausea. Workers will be required to use the right PPEs in order to minimize exposure to sunlight.

8.8 Stakeholder Consultations

The Company will engage with community leaders and sometimes wider communities on the scale and anticipated impacts of planned mining and related activities in the communities in which we operate. We commit to design projects that take into consideration the concerns and aspirations of community members; and to explore opportunities and strategies for maximizing benefits of our projects. Our engagement strategies will include grievance mechanisms to allow for peaceful resolve of community grievances before they become risks and escalated conflicts.

8.9 Impact Identification

Project-related air quality impacts include non-point source impacts such as dust generation attributable to mining activities (mainly hauling) and emissions from the mining equipment fleet. Point source emissions include emissions from the trash/medical waste incinerator and emissions from the gold room processing facility. Management of air quality impacts are primarily focused on dust control. Measures including extensive road watering, applying rock running courses to road surfaces, and speed limit controls are utilized to minimize dust. An air quality monitoring program is in place to evaluate respirable dust (PM10) and dust accumulation. Source testing of the point source emissions will be conducted periodically.

8.10 Corporate Social Responsibility Programme

In addition to doing its core business of mining, the company is committed to demonstrating social responsibility towards the communities that host our projects. The scope of our citizenship moves beyond traditional philanthropy to include engagements and critical social interventions that will create shared value for us and the local communities. These will include provision of educational and health facilities, potable water supply, etc.

Consultation with stakeholder communities, traditional and elected leaders, and regulators forms an integral component of the project environmental and socioeconomic impact assessment process.

Stakeholder consultation follows a typical pattern reflecting the stages of evolution of the project knowledge and permitting. Initial consultations will occur to sensitize the community on the project, and gain relevant information for the documentation for the Project Environmental Scoping Report.

In addition to compliance to regulatory requirements, the company will design and commit to a robust Corporate Social Responsibility (CSR) program which cascades across the project life up to mine closure. At the core of the program is our belief to be a responsible mining operator in our host communities by innovating and demonstrating responsibility for enhancing social well-being, protecting the environment, whilst maintaining financial viability of our operations. The CSR program comprises of four key activities, namely; Stakeholder engagement; Employee Engagement; Community development; and Environmental stewardship. Each of these activities are mutually-reinforcing and will be planned and implemented in a way that they extend credible development opportunities for local communities.

8.11 Community Development

The company believes in creating shared value with local communities through successful mine operations. As part of this, the company will work with the District Assembly to prioritize and invest in areas of Education and Skills development, Health infrastructure, Local economic development, as well as Water and Sanitation facilities. In particular, the skills development module will be aligned with the labor requirements of the project to ensure that youth are prioritized for employment along the project life. Added to the operational requirements, the initiatives under the each of the prioritized modules above must align with the medium-term development plan of the District Assembly as a condition of company's support. Ultimately, our investment will help the capacity of the district assembly to deliver on its development mandate. The community development activities that will be considered include the following:

Table 8.1: Priority Areas for IGNITE MACHINERY Ltd. in CSR Activities

Priority sector	Potential activities
Education and skills development	Scholarships and bursaries for tertiary and technical education. Support for technical education will target youth interested in developing their skills in craft and artisanship e.g. dump truck operators, auto-mechanics, welders, carpenters, hospitality stewards, Integrate skills training with company's operational requirements
Health infrastructure and services	Construction of Wards at Regional/district hospitals Clinics/CHPs Compounds in local communities Donation of medical equipment Short-term staff development training in China for health professionals
Local economic development	Sponsorship for agriculture-based initiatives e.g. National farmers' day Training and technical support for women in agro-processing initiatives Crop-based livelihoods activities linked with livelihood restoration programs for Project-affected persons Local Employment Strategy – 100 percent recruitment of unskilled labour from affected communities
Water and Sanitation	Provision of boreholes and small-town water systems Provision of household latrines
Others	Cash donations for community driven projects and initiatives including

CHAPTER 9

9.0 PROJECT COST ESTIMATION

The cost analysis of mining projects base on gold price trends is a key factor in making a decisive decision of the return on investment of the project. Mining is a very capital-intensive venture due to advance methodology for extraction and its environmental consequences. The proposed Amenam commercial mining project is therefore expected to have cost elements. This includes Capital Expenditure and Operating Costs. At this time, the company will be arranging equity and debt funding for the proposed mine. All financial projections were done using an exchange rate of \$1 = GH¢12.00 as of October, 2025 an inflation rate of 10%, 5% royalty and a corporate tax of 35%. The following assumptions were also made in the financial inclusion of the project.

- In estimating the capital costs, it is assumed that basic mining equipment is exempted from import duties as outlined in the minerals code of Ghana.
- In estimating the operating costs, it is assumed that both skilled and unskilled labour is available in Ghana. Therefore, local labour will be used for the project and where training is required the company will ensure that it is given to the suitable candidates.
- The remuneration for employees has been determined based on consultation with other mining companies currently operating in Ghana.

9.1 Projected Cost Estimation

The Capital Expenditure covers the machinery, heavy duty trucks and many other items that the company plans to acquire for the implementation of the mining project. The Capital Expenditure is estimated at \$41,416,251. The estimated costs of the various items are based on price quotations on the Pro forma Invoices obtained from the relevant suppliers and manufacturers. The total Operating Costs of the proposed Amenam project is estimated over the seven (7) year period of the mine.

Other operational expenses include:

Table 9.1 Additional operating cost

Item	Amount (US \$)
Land Compensation (EPA Bond)	10,000,0000.00
Licensing and Permits	1,000,000.00
TOTAL	11,000,000.00

The Summary of the Financial Projections of Amenam proposed mining project is as follows:

1. Capital Costs - **US\$ 41,416,251.0**
2. Operating Cost for First Year - **US\$ 8,295,701.0**
3. Additional Operating Cost - **US\$11,000,000.0**
4. Mine Life Operating Cost (11years) is **US\$ 65,591,032.0**

The price quotations on which the capital costs are based are likely to be increased at the time that the items are to be purchased in case there occurs inflation and change in exchange rates.

9.2 Project Revenue Estimation from Gold Sales

Average gold price of gold as of October,2025 = US\$ 4,343.70/oz. conservatively = 3,000/oz.

However, a base of \$2,500/oz is used for IGNITE MACHINERY Ltd proposed project.

Total ounces of gold in Amenam Concession: 84,967.80oz.

Total Gold sales (Gross Revenue) = 84,967.80oz. x US\$2,500.00 = US\$212,419,500

Hence expected revenue per year

= \$ 212,419,500 ÷ 7yrs

= \$ 30,345,642.85 per year

Table 9.2: Income statement for Ignite Machinery Rentals & Mining Limited

Projected Income	Unit	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Total
Gold Production (ounce)	ounces		12,138.26	12,138.26	12,138.26	12,138.26	12,138.26	12,138.26	12,138.26	84,967.80
Gold Price	US\$/oz	2,500.00	2,500.00	2,500.00	2,500.00	2,500.00	2,500.00	2,500.00	2,500.00	
Revenue	US\$		30,345,642.86	30,345,642.86	30,345,642.86	30,345,642.86	30,345,642.86	30,345,642.86	30,345,642.86	212,419,500.00
Operating Cost			8,295,701.00	8,099,402.00	8,892,045.00	9,892,570.00	9,504,801.00	9,902,531.00	11,003,982.00	65,591,032.00
Gross Profit			22,049,941.86	22,246,240.86	21,453,597.86	20,453,072.86	20,840,841.86	20,443,111.86	19,341,660.86	146,828,468.00
Royalty (1)	0.05		1,517,282.14	1,517,282.14	1,517,282.14	1,517,282.14	1,517,282.14	1,517,282.14	1,517,282.14	10,620,975.00
Pre-Tax Profit			20,532,659.71	20,728,958.71	19,936,315.71	18,935,790.71	19,323,559.71	18,925,829.71	17,824,378.71	136,207,493.00
Corporate Tax	0.35		7,186,430.90	7,255,135.55	6,977,710.50	6,627,526.75	6,763,245.90	6,624,040.40	6,238,532.55	47,672,622.55
Net Profit			13,346,228.81	13,473,823.16	12,958,605.21	12,308,263.96	12,560,313.81	12,301,789.31	11,585,846.16	88,534,870.45
Cumulative Earnings			13,346,228.81	26,820,051.98	39,778,657.19	52,086,921.16	64,647,234.97	76,949,024.29	88,534,870.45	

Table 9.3: Cash flow of proposed Ignite Machinery Rentals & Mining Ltd.

Projected Income	Unit	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Total
Gold Production (ounce)	ounces		12,138.26	12,138.26	12,138.26	12,138.26	12,138.26	12,138.26	12,138.26	84,967.80
Gold Price	US\$/oz	2,500.00	2,500.00	2,500.00	2,500.00	2,500.00	2,500.00	2,500.00	2,500.00	
Revenue	US\$		30,345,642.86	30,345,642.86	30,345,642.86	30,345,642.86	30,345,642.86	30,345,642.86	30,345,642.86	212,419,500.00
Operating Cost			8,295,701.00	8,099,402.00	8,892,045.00	9,892,570.00	9,504,801.00	9,902,531.00	11,003,982.00	65,591,032.00
Gross Profit			22,049,941.86	22,246,240.86	21,453,597.86	20,453,072.86	20,840,841.86	20,443,111.86	19,341,660.86	146,828,468.00
Royalty	5%		1,517,282.14	1,517,282.14	1,517,282.14	1,517,282.14	1,517,282.14	1,517,282.14	1,517,282.14	10,620,975.00
Pre-Tax Profit			20,532,659.71	20,728,958.71	19,936,315.71	18,935,790.71	19,323,559.71	18,925,829.71	17,824,378.71	136,207,493.00
Corporate Tax	35%		7,186,430.90	7,255,135.55	6,977,710.50	6,627,526.75	6,763,245.90	6,624,040.40	6,238,532.55	47,672,622.55
Net Profit			13,346,228.81	13,473,823.16	12,958,605.21	12,308,263.96	12,560,313.81	12,301,789.31	11,585,846.16	88,534,870.45
Cumulative Earnings			13,346,228.81	26,820,051.98	39,778,657.19	52,086,921.16	64,647,234.97	76,949,024.29	88,534,870.45	362,162,988.85
										-
Depreciation			309,398.00	387,098.00	586,793.00	798,345.00	893,784.00	903,869.00	1,302,669.00	27,341,596.00
Capex (Project Cost)		- 41,416,251.00								-
Net Cash Flow		- 41,416,251.00	13,655,626.81	13,860,921.16	13,545,398.21	13,106,608.96	13,454,097.81	13,205,658.31	12,888,515.16	52,300,575.45
DISCOUNT FACTOR	10%	1.00	0.91	0.83	0.75	0.68	0.62	0.56	0.51	
Present Value of NCF		- 41,416,251.00	12,414,330.34	11,454,665.25	10,176,657.68	8,951,813.92	8,353,649.33	7,454,594.12	6,614,385.98	24,003,845.62
IRR		0.26								
NPV		24,003,845.62								

Table 9.4: Sensitivity Analysis (OPEX > 10%)

Projected Income	Unit	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Total
Gold Production (ounce)	ounces		12,138.26	12,138.26	12,138.26	12,138.26	12,138.26	12,138.26	12,138.26	84,967.80
Gold Price	US\$/oz	2,500.00	2,500.00	2,500.00	2,500.00	2,500.00	2,500.00	2,500.00	2,500.00	
Revenue	US\$		30,345,642.86	30,345,642.86	30,345,642.86	30,345,642.86	30,345,642.86	30,345,642.86	30,345,642.86	212,419,500.00
Operating Cost			9,125,271.10	8,909,342.20	9,781,249.50	10,881,827.00	10,455,281.10	10,892,784.10	12,104,380.20	72,150,135.20
Gross Profit			21,220,371.76	21,436,300.66	20,564,393.36	19,463,815.86	19,890,361.76	19,452,858.76	18,241,262.66	140,269,364.80
Royalty	5%		1,061,018.59	1,071,815.03	1,028,219.67	973,190.79	994,518.09	972,642.94	912,063.13	7,013,468.24
Pre-Tax Profit			20,159,353.17	20,364,485.62	19,536,173.69	18,490,625.06	18,895,843.67	18,480,215.82	17,329,199.52	133,255,896.56
Corporate Tax	35%		7,055,773.61	7,127,569.97	6,837,660.79	6,471,718.77	6,613,545.28	6,468,075.54	6,065,219.83	46,639,563.80
Net Profit			13,103,579.56	13,236,915.66	12,698,512.90	12,018,906.29	12,282,298.39	12,012,140.28	11,263,979.69	86,616,332.76
Cumulative Earnings			13,103,579.56	26,340,495.22	39,039,008.11	51,057,914.41	63,340,212.79	75,352,353.07	86,616,332.76	354,849,895.92
										-
Depreciation			309,398.00	387,098.00	586,793.00	798,345.00	893,784.00	903,869.00	1,302,669.00	5,181,956.00
Capex (Project Cost)		- 41,416,251.00								-
Net Cash Flow		- 41,416,251.00	13,412,977.56	13,624,013.66	13,285,305.90	12,817,251.29	13,176,082.39	12,916,009.28	12,566,648.69	50,382,037.76
DISCOUNT FACTOR	10%	1.00	0.91	0.83	0.75	0.68	0.62	0.56	0.51	
Present Value of NCF		- 41,416,251.00	12,193,737.90	11,258,884.89	9,981,250.32	8,754,182.63	8,181,029.55	7,291,087.24	6,449,204.11	22,693,125.64
IRR		0.25								
NPV		22,693,125.64								

Table 9.5: Sensitivity Analysis (Revenue < 10%)

Projected Income	Unit	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Total
Gold Production (ounce)	ounces		12,138.26	12,138.26	12,138.26	12,138.26	12,138.26	12,138.26	12,138.26	84,967.80
Gold Price	US\$/oz	2,500.00	2,500.00	2,500.00	2,500.00	2,500.00	2,500.00	2,500.00	2,500.00	
Revenue	US\$		27,311,078.57	27,311,078.57	27,311,078.57	27,311,078.57	27,311,078.57	27,311,078.57	27,311,078.57	191,177,550.00
Operating Cost			8,295,701.00	8,099,402.00	8,892,045.00	9,892,570.00	9,504,801.00	9,902,531.00	11,003,982.00	65,591,032.00
Gross Profit			19,015,377.57	19,211,676.57	18,419,033.57	17,418,508.57	17,806,277.57	17,408,547.57	16,307,096.57	125,586,518.00
Royalty	5%		950,768.88	960,583.83	920,951.68	870,925.43	890,313.88	870,427.38	815,354.83	6,279,325.90
Pre-Tax Profit			18,064,608.69	18,251,092.74	17,498,081.89	16,547,583.14	16,915,963.69	16,538,120.19	15,491,741.74	119,307,192.10
Corporate Tax	35%		6,322,613.04	6,387,882.46	6,124,328.66	5,791,654.10	5,920,587.29	5,788,342.07	5,422,109.61	41,757,517.24
Net Profit			11,741,995.65	11,863,210.28	11,373,753.23	10,755,929.04	10,995,376.40	10,749,778.13	10,069,632.13	77,549,674.87
Cumulative Earnings			11,741,995.65	23,605,205.93	34,978,959.16	45,734,888.21	56,730,264.61	67,480,042.73	77,549,674.87	317,821,031.16
										-
Depreciation			309,398.00	387,098.00	586,793.00	798,345.00	893,784.00	903,869.00	1,302,669.00	27,341,596.00
Capex (Project Cost)		- 41,416,251.00								-
Net Cash Flow		- 41,416,251.00	12,051,393.65	12,250,308.28	11,960,546.23	11,554,274.04	11,889,160.40	11,653,647.13	11,372,301.13	41,315,379.87
DISCOUNT FACTOR	10%	1.00	0.91	0.83	0.75	0.68	0.62	0.56	0.51	
Present Value of NCF		- 41,416,251.00	10,955,921.97	10,123,654.76	8,985,958.38	7,891,569.17	7,381,979.69	6,578,483.80	5,836,264.94	16,337,581.72
IRR		0.21								
NPV		16,337,581.72								

Table 9.6: Sensitivity Analysis, Ignite Machinery Rentals & Mining Ltd.

Scenario	NPV \$'M'	IRR
Base Case	24,003,846	26%
Operational Cost Increase 10%	22,693,126	25%
Revenue Decrease 10%	16,337,582	21%

CHAPTER 10

10.0 MINE CLOSURE AND REHABILITATION

10.1 Introduction

The Company is required by legislation to implement a land rehabilitation program and prepare a decommissioning plan as part of its overall environmental management strategy. Mine closure and rehabilitation will be done with strict adherence with all laws and regulatory provisions provided by the institutions that are mandated to oversee its successful implementation.

The EPA requires a Reclamation Bond to be posted as part of any mine permitting process. The bond is required to provide financial surety against non-compliance under the approved Closure and Reclamation Plan and is required within six months after the start of operations. Projected reclamation liability for the Amenam project is approximately US\$ 5.24 million.

This compacted amount will be lodged into an escrow account with both the Company and EPA as co-signatories to ensure that the desired reclamation program is achieved.

10.2 General Provisions for Mine Site Rehabilitation and Closure

As has been stated by the company, mining and reclamation of mined out pits shall be done concurrently. The company will liaise with the inspectorate division on monthly update of mining and reclamation progress. The inspectorate due to scheduled visits has the authority to crosscheck and satisfy all required provisions required to be fulfilled with regards to mine closure. The Company will submit a mine closure plan to the Inspectorate Division of EPA for approval before closure initiation kickstart. This plan will include but not limited to surveyed mine layout plan that shows all final workings as prescribed by L.I. 2182, regulation 67.

Furthermore, the Company is of the view that a successful closure could be achieved through progressive closure commitment and planning and in constant consultation with the inspectorate division to be in a position to satisfy the Chief Inspector of Mines that each source of potential pollution and component of the mining project that is to be closed are designed to be stable in the long term.

Mine closure processes consist of several steps including the following:

1. Decommissioning and rehabilitation
2. Relinquishment (issuing of a mine closure certificate)
3. Management of residual and latent risks
4. Mine site reclamation
5. Continuous closure
6. Operating the mine for closure
7. Post closure monitoring and maintenance
8. Engaging stakeholders in the mine closure process

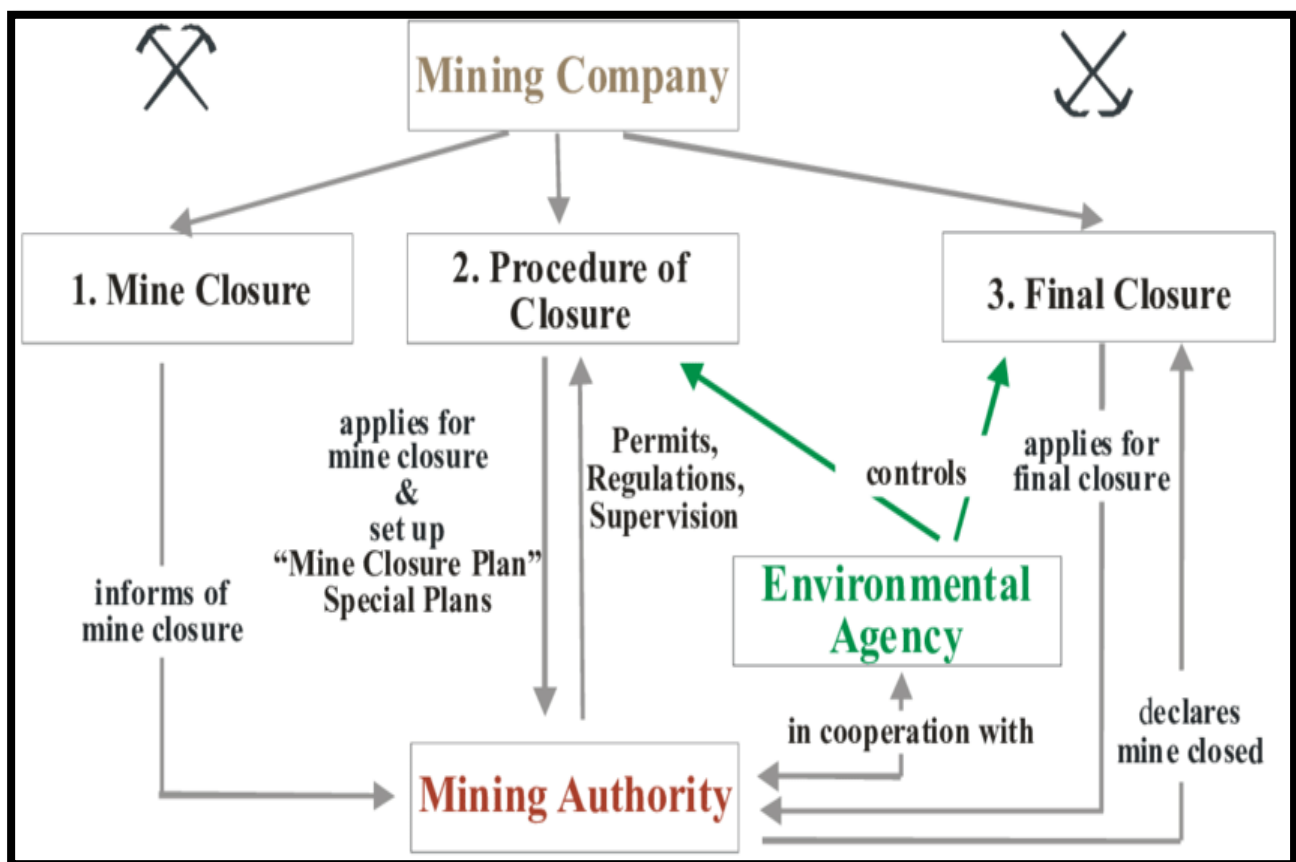


Figure 10.1: Simplified Scheme for A Mine Closure Procedure.

10.3 Reclamation Processes

As already stated, the company is committed to rehabilitation and Mine Closure to ensure that the environment is restored to its primary state. The waterbodies, the air, the trees, cash crops, infrastructure sited on arable lands, dams, tailings dams, ponds and sundries shall all be restored, and the report submitted to the Mine Inspectorate division not later than sixty (60) days before abandonment. Maps for areas that have been mined, partially backfilled, reclaimed, and revegetated shall be prepared monthly to know the deficit and the areas to zero-in on during the mine closure and folding periods. The company is going to ensure that total mine-out figures and maps are well documented, and budgets prepared for the environmental department to embark on seamless backfilling and revegetation during and after the mine life. Consultation with landowners will help to define a map out plan as to how their respective lands shall be put to use. This will inform the environmental unit as the end-use of their land before commencing the restoration.

The site infrastructure including dams, ponds, offices, routes, bridges, culverts, workshops, processing plant; shall be restored and some be marked out as assets for the community with the agreed compensation given the landowners for the loss of their lands.

The Company shall comply with all the Commissions regulations concerning closure of processing plants, removal of hazardous plants and closure of ancillary facilities if we are given the Lease. It is however worth noting that no chemical or chemical precipitates would be used in the gold recovery process.

10.4 Reclamation and Restoration Plan

The Company has drawn up a program to reclaim all areas that will be affected by the mining project, in accordance with Ghana's mining and environmental regulations. The reclamation program is made up of the following components:

- Dewatering
- Backfilling: which involves the use of waste gravel and overburden to re-fill mined out areas
- Top-soiling: Spread of topsoil in preparation for re-vegetation.
- Re-vegetation: The planting of selected tree species over areas that have been back-filled and top soiled.
- Monitoring

The reclamation process will be done in constant consultation of authorizing institutions such as the Minerals Commission, Mines Department, Environmental Protection Agency (EPA) and Water Resources Commission.

Dewatering: It is common knowledge that pits that have been left to the mercy of the weather over time will be filled with either rain or ground water. These pits filled with water will be pumped out under controlled conditions before backfilling begins. The dewatering will be done by high-capacity hydraulic pumps.

Gravel Backfilling: The first stage of the backfilling phase will be to backfill with the washed-out gravel (from the mining activities) from the base and then compacted. This is to try and follow the sequence of the units before they were excavated. The processing plant tailings, which will be made up of washed boulders, cobbles, pebbles and sand will be constantly conveyed to the mined-out areas for backfilling purposes. This operation will be carried out *simultaneously* with mining. By that design, the tipper trucks, which carry the ore to the plant, will also carry washed gravels or tailings on their return to the mining face. This means that suitable machines will have to be placed at the tailings site to load the tipper trucks on their return to the mining face.

The duty of the tailings handling machine will be more than just loading trucks; the machine will be responsible for de-silting the sandpit as well. Hence, an excavator with a somewhat long boom and light to medium weight will be ideal for the job.

Overburden Backfilling: This is the zone that overlies between the gravel layer and the topsoil. For agricultural reasons, this zone usually separates the life supporting topsoil from the subsurface. This zone which may be composed of high percentage of clay will be filled after the gravel filling is completed.

Top Soil Backfilling: The final stage of the backfilling stage will be to fill and spread the top soil from the “topsoil stockpile”. The top soil will be filled after the filling of the overburden.

Re-vegetation: Re-vegetation would begin with the planting of leguminous plant after which local and economic (mixed species of ofram, mahogany, teak and manzonias, all endangered wood species) would be planted over the entire area. Plant species common in the area would be planted to enable the land gain its fertility.

Monitoring: Monitoring of post restoration will be done by respective stakeholders mandated by laws of the industry. In the event of any unsatisfactory result from the reclamation and rehabilitation, Time Concept will be committed to finding lasting solution to such undesirable situation. Much help will be needed from the Agric extension officers stationed at the district assemblies (during the re-vegetation stage); such officers will be part and oversee the final stages of reclamation process as well as the post reclamation in monitoring and evaluation.



Figure 10.2: Mine Closure Overview.

CHAPTER 11

11.0 CONCLUSION

The management of Ignite Machinery Rentals & Mining Ltd., Amenam concession has followed detailed mineral exploration procedure in evaluating its gold deposit for the Amenam gold concession. The company employed bankable due diligence procedures to the mineral resource estimation of its 9.77 sqkm property in the Birim North District of the Eastern Region. The exploration team have been able to acquire sufficient data on which the economic assessment and gold delineation of the concession has been based.

Hardrock mineral resource projections with reference to geological, structural and geophysical data, an area of 766.7k metre squared area is expected to produce a projected tonnage of 1.76Mt, with an average grade of 0.92g/t and 62.36koz gold resource from the Amenam concession.

The evaluation of the exploration data obtained show that alluvial gold gravel beds occur in the valleys and old terraces within the concession. The buried gravel beds have an average thickness of 1.34m and have overburden with average thickness of 3.9m. The estimated reserves of gold deposits in the gravel beds in the concession is approximately 84,967.80oz.

The alluvial mineral deposits can conveniently be exploited through open-pit mining and treatment of the un-consolidated ore-bearing gravels with water-driven washing plant having a capacity of 100m³/h. The company intends to operates one stationary site, and hauling ore the deposit from various mining pits simultaneously (concurrently) from the respective delineated gold zones. The projected mine life is estimated at approximately seven (7) years.

The total revenues that the Company expects to derive from the sales of gold from the Amenam concession is US\$ 212,419,500 as against the initial investment of US\$ 60,711,952. The economic viability of the proposed mining project in the Amenam Gold Concession is considered to be positive obtained from cost benefit analysis conducted. This means that the mining of the alluvial mineral deposits on the Amenam Concession by open pit method could be implemented as it will be an economically profitable venture. The CEO and board of directors of Ignite Machinery Rentals & Mining Ltd., therefore applying for a mining lease to enable them commercialize the alluvial gold mining at Amenam concession by the Hon. minister of the of Ministry of Lands and Natural Resources through the Minerals Commission.

APPENDICES:

Result of pitting program conducted at the Amenam concession

PIT_ID	UTM_EAST	UTM_NORTH	ELEV	Assay_g/m3	MEDIUM	OVERBURDEN THICKNESS(m)	GRAVEL THICKNESS(m)	TOTAL PIT DEPTH	VISIBLE SPECKS	GRID
AMPIT001	716994	708799	182	0.79	gravel	5.2	2.10	7.30	2	Amenam-West
AMPIT002	716694	708832	164	0.44	gravel	1.0	1.20	2.20	3	Amenam-West
AMPIT003	716799	708919	157	1.00	gravel	1.0	1.10	2.10	4	Amenam-West
AMPIT004	717102	708886	166	1.60	gravel	6.1	1.00	7.10	6	Amenam-West
AMPIT005	716952	708994	183	0.10	gravel	7.2	0.90	8.10	3	Amenam-West
AMPIT006	717148	708981	182	0.55	gravel	6.4	0.60	7.00	3	Amenam-West
AMPIT007	717047	709097	163	0.00	gravel	6.3	0.00	0.00	0	Amenam-West
AMPIT008	717195	709145	157	0.53	gravel	5.8	1.10	6.90	3	Amenam-West
AMPIT009	717327	708945	166	0.51	gravel	4.2	1.30	5.50	3	Amenam-West
AMPIT010	717420	709086	183	0.61	gravel	5.2	2.60	7.80	6	Amenam-West
AMPIT011	717197	709255	182	1.13	gravel	6.2	1.20	7.40	3	Amenam-West
AMPIT012	716983	709207	163	0.20	gravel	5.8	1.00	6.80	6	Amenam-West
AMPIT013	717129	709360	157	1.37	gravel	0.4	1.10	1.50	7	Amenam-West
AMPIT014	716870	709273	166	0.20	gravel	1.7	1.80	3.50	2	Amenam-West
AMPIT015	717067	709501	183	0.79	gravel	4.5	1.60	6.10	1	Amenam-West
AMPIT016	716718	709396	182	0.44	gravel	6.3	1.40	7.70	4	Amenam-West
AMPIT017	716842	709698	163	1.30	gravel	0.4	1.20	1.60	2	Amenam-West
AMPIT018	716570	709624	164	1.60	gravel	4.7	1.30	6.00	1	Amenam-West
AMPIT019	716648	709545	160	0.71	gravel	0.8	1.50	2.30	2	Amenam-West
AMPIT020	717085	709610	157	0.91	gravel	4.5	1.70	6.20	4	Amenam-West
AMPIT021	717274	709393	164	1.60	gravel	0.1	0.12	0.22	1	Amenam-West
AMPIT022	717415	709173	157	0.71	gravel	7.2	1.20	8.40	2	Amenam-West
AMPIT023	717601	709322	166	0.15	gravel	5.2	2.00	7.20	2	Amenam-West
AMPIT024	717530	708971	183	0.11	gravel	6.1	0.12	6.22	5	Amenam-West
AMPIT025	717709	709087	157	0.53	gravel	5.2	1.30	6.50	3	Amenam-West
AMPIT026	717581	708807	183	1.00	gravel	6.3	1.00	7.30	6	Amenam-West

AMPIT027	717996	708892	164	0.79	gravel	4.4	1.00	5.40	5	Amenam-West
AMPIT028	718229	708898	157	1.37	gravel	0.8	1.30	2.10	2	Amenam-West
AMPIT029	718417	708889	166	0.54	gravel	4.0	2.00	6.00	5	Amenam-West
AMPIT030	718819	709259	157	0.79	gravel	1.0	1.20	2.20	2	Amenam-West
AMPIT031	718914	709477	164	0.20	gravel	1.0	1.30	2.30	3	Amenam-West
AMPIT032	718605	709308	157	0.40	gravel	6.0	1.50	7.50	2	Amenam-West
AMPIT033	718550	709400	166	1.60	gravel	4.0	2.00	6.00	6	Amenam-West
AMPIT034	718764	709527	183	1.45	gravel	7.0	0.60	7.60	3	Amenam-West
AMPIT035	718503	709530	182	0.00	gravel	4.2	0.30	4.50	2	Amenam-West
AMPIT036	718731	709611	163	1.11	gravel	4.7	0.12	4.82	2	Amenam-West
AMPIT037	718472	709635	157	1.43	gravel	5.3	2.30	7.60	3	Amenam-West
AMPIT038	719130	709266	166	0.56	gravel	5.8	1.70	7.50	2	Amenam-West
AMPIT039	719035	709476	183	3.45	gravel	1.1	0.12	1.22	7	Amenam-West
AMPIT040	719300	709353	182	0.99	gravel	1.4	0.70	2.10	2	Amenam-West
AMPIT041	719247	709533	163	1.45	gravel	6.4	0.40	6.80	3	Amenam-West
AMPIT042	719448	709375	157	0.81	gravel	7.1	0.60	7.70	2	Amenam-West
AMPIT043	719412	709577	166	1.11	gravel	4.9	0.80	5.70	2	Amenam-West
AMPIT044	719624	709368	183	0.30	gravel	3.4	1.90	5.30	2	Amenam-West
AMPIT045	719527	709645	182	1.41	gravel	1.5	1.30	2.80	5	Amenam-West
AMPIT046	719679	709474	163	0.46	gravel	1.6	1.10	2.70	4	Amenam-West
AMPIT047	719814	709636	157	0.54	gravel	1.7	2.00	3.70	6	Amenam-West
AMPIT048	719734	709298	166	0.90	gravel	5.2	1.20	6.40	2	Amenam-West
AMPIT049	720079	709306	183	0.80	gravel	5.3	1.30	6.60	2	Amenam-West
AMPIT050	720070	709434	182	0.50	gravel	1.1	1.10	2.20	3	Amenam-West
AMPIT051	719922	709557	163	0.00	gravel	5.3	1.40	6.70	3	Amenam-West
AMPIT052	720110	709679	157	0.20	gravel	1.6	1.90	3.50	5	Amenam-West
AMPIT053	719997	709919	166	1.10	gravel	5.3	2.00	7.30	1	Amenam-West
AMPIT054	719861	709971	183	1.40	gravel	5.3	1.90	7.20	1	Amenam-West
AMPIT055	720024	710022	182	0.67	gravel	6.1	1.50	7.60	2	Amenam-West
AMPIT056	720203	710074	163	0.90	gravel	1.7	1.60	3.30	2	Amenam-West
AMPIT057	720238	709861	164	1.20	gravel	6.2	1.70	7.90	3	Amenam-West



AMPIT058	720242	709618	160	1.40	gravel	6.4	1.50	7.90	5	Amenam-West
AMPIT059	720423	709822	157	0.10	gravel	1.4	1.80	3.20	2	Amenam-West
AMPIT060	720394	709510	157	1.11	gravel	1.0	0.90	1.90	4	Amenam-West
AMPIT061	720619	709547	166	0.40	gravel	3.9	0.50	4.40	1	Amenam-West
AMPIT062	720383	709352	183	1.15	gravel	6.2	1.20	7.40	4	Amenam-West
AMPIT063	720591	709299	182	4.11	gravel	1.8	1.10	2.90	6	Amenam-West
AMPIT064	720390	709247	163	0.70	gravel	1.4	1.00	2.40	4	Amenam-West
AMPIT065	724465	709613	157	0.00	gravel	1.0	0.90	1.90	0	Amenam-East
AMPIT066	724713	709661	166	0.10	gravel	1.2	0.60	1.80	0	Amenam-East
AMPIT067	724571	709447	183	0.50	gravel	0.5	2.00	2.50	0	Amenam-East
AMPIT068	724927	709554	182	0.00	gravel	5.4	2.40	7.80	0	Amenam-East
AMPIT069	724774	709319	163	0.00	gravel	6.0	1.30	7.30	0	Amenam-East
AMPIT070	725141	709420	157	0.00	gravel	5.1	0.50	5.60	0	Amenam-East
AMPIT071	724962	709104	166	0.00	gravel	6.8	0.12	6.92	0	Amenam-East
AMPIT072	725370	709229	183	0.00	gravel	6.2	1.00	7.20	0	Amenam-East
AMPIT073	725110	708952	182	0.00	gravel	4.5	1.10	5.60	0	Amenam-East
AMPIT074	725655	709031	163	0.50	gravel	4.2	1.80	6.00	0	Amenam-East
AMPIT075	725246	708845	164	0.00	gravel	1.4	1.60	3.00	0	Amenam-East
AMPIT076	725701	708827	160	0.06	gravel	6.1	1.40	7.50	0	Amenam-East
AMPIT077	726222	709647	157	0.20	gravel	5.6	1.20	6.80	3	Amenam-East
AMPIT078	726476	709699	157	0.46	gravel	0.1	0.12	0.22	5	Amenam-East
AMPIT079	726317	709526	166	1.20	gravel	1.4	1.50	2.90	7	Amenam-East
AMPIT080	726529	709609	183	0.40	gravel	5.3	1.70	7.00	1	Amenam-East
AMPIT081	726398	709412	182	1.90	gravel	4.5	1.20	5.70	3	Amenam-East
AMPIT082	726643	709565	163	0.50	gravel	1.4	1.10	2.50	3	Amenam-East
AMPIT083	726535	709335	157	0.90	gravel	4.7	1.40	6.10	5	Amenam-East
AMPIT084	726756	709517	166	1.34	gravel	6.8	1.00	7.80	3	Amenam-East
AMPIT085	726745	709284	183	0.60	gravel	1.0	4.10	5.10	5	Amenam-East
AMPIT086	727040	709370	182	0.00	gravel	0.8	2.10	2.90	4	Amenam-East
AMPIT087	726919	709572	163	0.85	gravel	5.2	2.70	7.90	6	Amenam-East
AMPIT088	727069	709271	157	0.56	gravel	1.0	3.50	4.50	2	Amenam-East



AMPIT089	727135	709545	166	0.00	gravel	0.0	0.00	0.00	0	Amenam-East
AMPIT090	727001	709648	183	0.85	gravel	5.4	3.20	8.60	6	Amenam-East
AMPIT091	727252	709611	182	3.10	gravel	1.4	3.00	4.40	5	Amenam-East
AMPIT092	727151	709692	163	0.73	gravel	1.0	0.90	1.90	3	Amenam-East
AMPIT093	727517	709562	157	1.50	gravel	7.2	0.60	7.80	5	Amenam-East
AMPIT094	727659	709685	166	1.17	gravel	4.5	2.00	6.50	3	Amenam-East
AMPIT095	727638	709468	183	1.40	gravel	4.5	2.40	6.90	4	Amenam-East

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